

Conformal flow and the Riemannian Penrose inequality with minimizing frontiers

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Abstract

We give a detailed conformal-flow proof of the numerical Riemannian Penrose inequality in every dimension $n \geq 3$ for complete asymptotically flat exteriors with nonnegative scalar curvature and full compact frontiers that are outer minimizing and locally perimeter minimizing. The metric extends smoothly through the possibly singular frontier; neither spin nor frontier connectedness is assumed. The proof follows the conformal-flow approach of Bray, Bray–Lee, and Bi–Zhu.

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1 The inequality and the proof

The Riemannian Penrose inequality compares the mass of an asymptotically flat manifold with the total area of its outer minimizing minimal boundary. Its geometric content is that a large minimal enclosure requires a correspondingly large mass. Penrose’s physical argument [31] related an initial trapped area to the mass remaining after gravitational collapse: cosmic censorship, area increase, and settling to a Kerr black hole led to the proposed mass–area bound. For time-symmetric initial data, the energy condition becomes nonnegative scalar curvature and the horizon condition becomes minimality. The resulting Riemannian problem retains the mass–area comparison without requiring an evolution of the spacetime.

In three dimensions, one route from the horizon to infinity uses the Hawking mass [17]. Geroch’s calculation [15] showed its monotonicity along smooth inverse mean curvature flow of two-spheres in nonnegative scalar curvature. Jang and Wald [21, Section 5] derived the Penrose bound assuming a global flow with the required asymptotics. Existence of such a flow was the central difficulty: mean curvature may vanish and the smooth evolution can develop singularities. Before the general proofs, spherical symmetry yielded the inequality [25], and spinorial methods gave mass–area estimates involving an additional normalized Sobolev quantity [18].

Huisken and Ilmanen’s weak inverse mean curvature flow allows jumps across the regions where smooth flow fails [19]. It proves the sharp mass bound for each connected component of an outermost minimal boundary in dimension three. Bray’s conformal flow [8] instead changes the ambient metric while preserving the total horizon area. Positive mass makes the evolving mass nonincreasing, and the limiting comparison yields the sharp bound for the full area, including disconnected boundaries. The harmonic-flat reduction, area-preserving flow, and mass comparison in the present proof descend from this construction.

Bray and Lee extended the conformal method to dimensions below eight [9]. The dimension restriction reflects the use of smooth minimizing hypersurfaces. In dimensions at least eight, a minimizing frontier may be singular, and its singular points affect the flow construction, separation of slices, and the geometric comparison used to decrease mass.

Restricted all-dimensional results were obtained earlier under graphical hypotheses. Lam [22, Corollary 7] proved the inequality for asymptotically flat Euclidean graphs with nonnegative scalar curvature and convex horizon components. De Lima and Girão [11, Theorem 1.8] treated two-sided asymptotically flat quasi-graphs in Euclidean space, with nonnegative scalar curvature and convex horizons in a horizontal hyperplane met orthogonally. These results use additional graphical control of the geometry.

Bi and Zhu [5, Theorem 1.2] prove the all-dimensional inequality together with Schwarzschild rigidity for almost-minimizing Caccioppoli (finite-perimeter) boundaries whose regular part is minimal and which are their own outermost minimizing enclosures, with a $C^{2,\alpha}$ metric extension through the boundary. Their version 2 conformal-flow argument is the immediate framework for this paper. We develop the numerical proof under stronger local minimization and ambient regularity hypotheses: the frontier is locally perimeter minimizing on both sides, the ambient metric is smooth through it, and the scalar curvature has the pointwise decay stated below. Our initially outer minimizing frontier is replaced by its outermost hull without changing its area. The present contribution is a detailed numerical reconstruction for this minimizing-frontier class.

The reconstruction makes explicit the preservation of area before limiting nesting, uniform estimates near regular patches of tangent cones, and complete smooth comparison metrics with positive conformal factors. It uses only the largest limiting frontier radius to reach the numerical conclusion. These reductions explain which geometric and positive-mass inputs suffice without requiring convergence of every frontier component or classification of equality.

1.1 Manifolds, frontiers, and mass

Write $n \geq 3$, $k = n - 1$, $d = n - 2$, and $\omega = \mathcal{H}_\delta^k(S^k)$. We use $O_j(r^{-\gamma})$ for a tensor whose coordinate derivatives of order $\ell \leq j$ are $O(r^{-\gamma-\ell})$. On a Euclidean coordinate end the ADM mass is

$$m_{\text{ADM}}(h) = \frac{1}{2k\omega} \lim_{R \rightarrow \infty} \int_{S_R} (\partial_j h_{ij} - \partial_i h_{jj}) n_\delta^i dA_\delta.$$

The decay and scalar integrability hypotheses below give the usual well-defined mass in this normalization; see [3].

A possibly singular exterior is described by its filled side in a smooth ambient manifold. The exterior contains the coordinate end; its compact frontier Σ is the support of the perimeter measure of the filled side. Its area is the full reduced-boundary perimeter, including every component. The metric extends smoothly through a neighborhood of Σ . The exterior includes its closed frontier. For $p, q \in E$, define the possibly extended length distance

$$d_E(p, q) = \inf \{ \text{Length}_h(\gamma) : \gamma \subset E \text{ is a rectifiable path joining } p \text{ to } q \}, \quad \inf \emptyset = +\infty.$$

Completeness means that each finite-distance equivalence class $\{q \in E : d_E(p, q) < \infty\}$ is complete for the restricted metric. This does not identify E with the length completion of its open exterior or assert finite-length access to every singular frontier point. Every point and component of the full frontier remains included in E , with its full perimeter counted.

The frontier is *outer minimizing* if every finite-perimeter enclosure of its filled side with compact additional support has at least its perimeter. It is *locally perimeter minimizing* if

compactly supported changes in sufficiently small neighborhoods, including changes on either side, cannot reduce perimeter. The latter condition is stronger than weak stationarity. It is the local comparison needed in the flow construction. At free regular points it gives the minimal-hypersurface equation.

Theorem 1.1 (Riemannian Penrose inequality with minimizing frontiers). *Let (E, h) be a connected complete n -dimensional exterior, including its compact frontier Σ in the enclosing-set sense above. Suppose that h is smooth on E and through Σ , and that*

1. $R_h \geq 0$, $R_h \in L^1(E, dV_h)$, and $R_h = O(r^{-n-\beta})$ for some $\beta > 0$ on the end;
2. outside a compact set there is exactly one Euclidean coordinate end with $h - \delta = O_2(r^{-\tau})$, where $\tau > d/2$;
3. the full frontier is outer minimizing and locally perimeter minimizing as the boundary of the filled side. Its regular part is a smooth embedded minimal hypersurface, and its singular set has Hausdorff dimension at most $n - 8$.

Then

$$m_{\text{ADM}}(h) \geq \frac{1}{2} \left(\frac{\mathcal{H}_h^k(\Sigma)}{\omega} \right)^{d/k}. \quad (1.1)$$

All boundary components enter the displayed area. Neither spin nor connectedness of the frontier is assumed. The zero-area case is included: the density lower bound makes a nonempty locally minimizing frontier have positive perimeter, so zero area means an empty frontier and the conclusion is nonnegative mass.

For $3 \leq n \leq 7$, the frontier is smooth and the numerical assertion follows from Bray–Lee’s published Theorem 1.4 [9]. Its numerical part has no spin assumption; spin belongs to the separate equality assertion there. The scalar decay required in that theorem holds here. The rest of this paper develops the singular argument for $n \geq 8$, including the reduction from harmonic-flat ends to the stated weaker asymptotics.

The constant is illustrated by the Schwarzschild–Tangherlini metric [35]

$$h_m = \left(1 + \frac{m}{2r^d} \right)^{4/d} \delta, \quad r \geq (m/2)^{1/d}, \quad m > 0.$$

Its inner sphere is minimal, its mass is m , and its area is $\omega(2m)^{k/d}$. Thus it realizes the numerical constant in Equation (1.1).

The enclosure lemmas in Section 2 identify full enclosing area with the area of a minimizing frontier and detach that frontier from a strict inner barrier. Their geometric conclusions are used independently in boundary-graph deformations [28]. Corollary 2.4 combines them with Theorem 1.1 to give a Penrose bound for smooth strict inner barriers. This passage from enclosing area to a minimizing frontier also supplies the numerical comparison in [30] and the smooth nonnegative-mass reduction in [29].

1.2 Four quantities that control the proof

First assume the initial metric $g_0 = h$ has a harmonic-flat end, meaning $g_0 = H^{4/d}\delta$ there with $H > 0$ Euclidean harmonic and $H \rightarrow 1$. Extend g_0 smoothly a short distance behind its frontier and truncate on that buried side at a smooth compact boundary. This gives an ambient exterior

M and an initial filled region K_0 containing the buried boundary; the original end exterior is $M \setminus K_0$. The extension needs no scalar-curvature sign behind the original frontier.

The flow has metrics $g_t = u_t^{4/d} g_0$ and filled regions K_t with frontiers Σ_t and end exteriors $E_t = M \setminus K_t$. At infinity $u_t \rightarrow e^{-t}$. If y denotes the original end coordinate, we measure mass in the normalized coordinate $x = e^{-2t/d} y$, in which the metric tends to δ at spatial infinity for each fixed t . Four quantities organize the proof:

$$A = P_{g_t}(K_t), \quad m(t) = m_{\text{ADM}}(g_t), \quad \mu(t) = m(t) - c_t, \quad R_t \quad (\text{outer radius}).$$

For the radius, represent the compact core by the ball of radius $R_0 e^{-2t/d}$ when the original chart starts at $|y| = R_0$; R_t is the maximum radius of the union of that ball and the frontier portion in the normalized chart. The capacity $c_t = \mathfrak{c}(\Sigma_t, g_t)$ in the deficit $\mu(t)$ is the normalized energy of the harmonic function w_t equal to one at the conductor and zero at infinity:

$$\mathfrak{c}(\Sigma_t, g_t) = \frac{1}{d\omega} \int_{E_t} |dw_t|_{g_t}^2 dV_{g_t}.$$

The flow keeps A fixed and satisfies $m' = -2\mu$ almost everywhere. The mass–capacity comparison gives $\mu \geq 0$; a one-sided time estimate then gives $\mu(t) \rightarrow 0$. The limiting harmonic potential identifies the largest limiting frontier radius, and enclosing spheres give the sharp area inequality.

The constructions establishing these facts have a necessary order. Section 2 states the independent enclosure results; their proofs are collected in Section 13, where they enter the final approximation argument. In Sections 3–4, the discrete flow supplies uniform density and potential estimates. We prove preservation of the limiting minimum area before deriving nesting of the limiting outermost hulls. This follows the order in Bi–Zhu version 2, Lemmas 2.21 and 2.23 and Corollary 2.22 [5].

For mass–capacity comparison we need a smooth boundary separated from the singular frontier. Sections 5–6 use a single median of the distance between frontiers to normalize all regular patches and prove this separation. The method descends from the singular maximum principles of Simon and Ilmanen [33, 20], as developed in [36, Section 4], and adapts Bi–Zhu’s relative forcing and normalization [5, Proposition 2.34 and Lemma 2.35]. Section 7.2 then smooths on the exterior side while retaining the curvature sign, following the distance–offset strategy of [4].

Before applying positive mass, Sections 8 and 11 construct complete smooth comparison metrics. Brendle–Wang’s theorem [10, Corollary 1.6] requires strictly positive scalar curvature and an all-order end expansion. From it we obtain nonnegative mass on harmonic-flat ends and mass–capacity comparison by doubling and collar repair [26]. A one-end reduction then permits the quantitative estimate of Lee [23]. Appendix A proves a local weighted-minimizer estimate using [27] to substantiate singular-set control in that positive-mass method. The smooth positive-mass theorem itself remains an independent input.

Finally, Section 12 identifies the limiting outer radius and completes the harmonic-flat argument. Section 14 first creates a strict smooth inner barrier and then approximates the general end, controlling both mass and every enclosing area. Thus the approximation does not require a theorem preserving a singular minimal frontier.

Mean-curvature convention. At an inner frontier, ν points toward the asymptotic end and $H_\nu = \text{div}_\Sigma \nu$. The outward normal of the exterior region is $-\nu$. Mean-curvature signs below are always stated using one of these explicit normals.

2 Full enclosures and smooth obstacles

A smooth inner boundary need not minimize the area of its enclosures. The three lemmas in this section replace it by a full minimizing frontier, retaining every component and every portion of contact with the original boundary. A strictly negative inner mean curvature then makes the minimizing frontier detach from that boundary. These are geometric results for a fixed smooth metric; they require neither nonnegative scalar curvature nor a mass hypothesis.

We state the results here for later reference. Their proofs are given together in Section 13, immediately before their use in the passage from harmonic-flat to general asymptotically flat ends. The local obstacle regularity is the same input that underlies minimizing hulls in Huisken–Ilmanen’s inverse mean curvature flow [19].

2.1 The full area and its minimizing enclosure

Let (X, g) be a smooth connected Riemannian manifold of dimension $n \geq 3$, complete with its nonempty compact smooth boundary B included. Assume that X has one coordinate end with compact complement and that, in its coordinates,

$$g_{ij} - \delta_{ij} = O_2(r^{-q}) \quad \text{for some } q > 0. \quad (2.1)$$

Only the eventual strict mean convexity of its coordinate spheres is needed for the three enclosure lemmas below. In particular, scalar curvature and ADM mass play no role in those lemmas.

A *full smooth enclosing cut* is the entire intrinsic manifold boundary $\partial_{\text{man}} D$ of a connected smooth domain $D \subset X$ that is closed in X , contains the distant coordinate end, and has compact boundary. Its intrinsic interior must lie in $\text{int } X$. All boundary components and every portion coincident with B are counted. Thus $D = X$ is admissible and its cut is B . Define

$$a_g(B) = \inf_D \text{Area}_g(\partial_{\text{man}} D). \quad (2.2)$$

Attach a compact smooth filling O behind B and extend g smoothly across B , obtaining a connected boundaryless manifold M with $M \setminus O = \text{int } X$. Such a filling exists: take a second copy of a compact truncation of X , reverse the collar direction along B , and cap its spherical end boundary with a ball. This construction does not require an orientation. The extension is auxiliary; it will not change any area in (2.2). Since the attached part is compact, the extended metric can and will be chosen complete.

Let $\mathcal{A}$ be the bounded finite-perimeter subsets E of M that contain O up to a volume-null set. Sets agreeing almost everywhere are identified. If $\partial^* E$ denotes the reduced boundary and ν_E its measure-theoretic outward unit normal, set

$$P_g(E) = |D\mathbf{1}_E|_g(M) = \mathcal{H}_g^{n-1}(\partial^* E). \quad (2.3)$$

This is perimeter in the extended manifold. In particular $P_g(O) = \text{Area}_g(B)$. Relative perimeter in $\text{int } X$ would instead assign zero to O and would lose all contact area. We use the standard compactness, approximation, trace, and Gauss–Green results for finite-perimeter sets [1, 24].

Lemma 2.1 (The full enclosure minimum). *Under the preceding hypotheses,*

$$0 < a_g(B) = \min_{E \in \mathcal{A}} P_g(E) \leq \text{Area}_g(B). \quad (2.4)$$

All minimizers are supported in one compact subset of M . The family of minimizers is compact for $L^1(M)$ convergence, and that convergence also entails convergence of their full perimeters.

Each minimizer has a closed representative E whose frontier is $T = \text{supp } |D\mathbf{1}_E|_g$. Its exterior $M \setminus E$ is connected and contains the distant end. There are full smooth enclosing cuts in $\text{int } X$ with areas tending to $P_g(E)$. When $3 \leq n \leq 7$, these cuts can be chosen to converge to T in C^1 . Every minimizer is outer minimizing: if $F \supset E$ is bounded and has finite perimeter, then $P_g(F) \geq P_g(E)$.

The proof in Section 13.1 identifies the perimeter minimum with the infimum over full smooth cuts. Counting contact with the original boundary is essential to this identification.

2.2 Regularity at a smooth obstacle

The next local statement supplies the regularity used in the enclosure lemma. It concerns constrained perimeter minimization alone, independently of global existence or connectedness.

Lemma 2.2 (Smooth-obstacle regularity). *Let M^n carry a smooth metric, let O have smooth boundary, and let E locally minimize perimeter among sets containing O . On a relatively compact region meeting ∂O , assume that ∂O has bounded smooth geometry. Then E has a closed representative whose topological frontier is $T = \text{supp } |D\mathbf{1}_E|_g$. Its regular part has multiplicity one, and its singular set $\mathcal{S}$ satisfies*

$$\mathcal{S} = \emptyset \quad (n \leq 7), \quad \dim_{\text{H}} \mathcal{S} \leq n - 8 \quad (n \geq 8).$$

The regular part outside ∂O is smooth and minimal. Every contact point in $T \cap \partial O$ is regular, with local graph in $C^{1,\alpha} \cap W^{2,p}$ for every $0 < \alpha < 1$ and every $1 < p < \infty$. In dimensions $3 \leq n \leq 7$ the entire frontier is $C^{1,1}$, with local graph functions in $W^{2,\infty}$. Moreover, $\mathcal{H}_g^{n-1}(\mathcal{S}) = 0$ and $\mathcal{H}_g^{n-1}(T) = P_g(E)$ wherever the total perimeter is finite.

The proof is given in Section 13.2.

2.3 Detachment from a strict inner barrier

Contact with the obstacle is possible when its mean curvature is zero. Strictly negative mean curvature, with the normal pointing toward the end, instead forces every minimizing enclosure to contain a fixed exterior collar. The resulting frontier is a locally minimizing boundary in a smooth ambient neighborhood.

Lemma 2.3 (Uniform detachment from a strict inner barrier). *In the setting of Lemma 2.1, suppose that $H_B = \text{div}_B \nu_B < 0$ on every component of B , with ν_B pointing into X . There is $\varepsilon > 0$, depending only on a fixed collar of (B, g) , such that every minimizing filled set contains the exterior distance collar $0 \leq t < \varepsilon$ in its interior. Consequently its frontier T is compact, nonempty, and disjoint from B , and is locally perimeter minimizing in the smooth ambient metric. It is stationary as a whole boundary varifold, smooth and minimal outside a compact singular set of dimension at most $n - 8$. If $3 \leq n \leq 7$, T is a smooth minimal full cut.*

The closed exterior $D = \overline{M \setminus E}$ is connected, has the same single coordinate end, and is a complete closed subset for the ambient distance. Every sequence that is Cauchy for lengths of paths constrained to D has a limit in D in the same sense, including when the limit lies on the frontier. Its frontier is outer minimizing with all components counted, and $\mathcal{H}_g^{n-1}(T) = a_g(B)$.

Section 13.3 proves the collar inclusion and the completeness assertions. In Section 14, this detached frontier will be the boundary to which we apply the harmonic-flat Penrose inequality.

Combining the enclosure lemmas with Theorem 1.1 also gives a mass bound for a smooth inner boundary that is strictly mean concave toward the end, without assuming that the boundary itself minimizes area.

Corollary 2.4 (The inequality for a strict inner barrier). *Let (X, g) be a smooth connected n -dimensional exterior, $n \geq 3$, complete including its nonempty compact smooth boundary B , with one Euclidean coordinate end and compact end complement. Suppose that*

$$g - \delta = O_2(r^{-\tau}), \quad \tau > \frac{n-2}{2}, \quad R_g \geq 0, \quad R_g \in L^1(X, dV_g), \quad R_g = O(r^{-n-\beta}) \quad (\beta > 0).$$

If $H_B < 0$ for the normal pointing into X , then

$$m_{\text{ADM}}(g) \geq \frac{1}{2} \left(\frac{a_g(B)}{\omega} \right)^{(n-2)/(n-1)}.$$

Proof. Lemma 2.1 supplies a minimizing enclosure of full perimeter $a_g(B)$. By Lemma 2.3, its frontier is detached from B , locally perimeter minimizing, and outer minimizing; its closed end exterior is connected and complete. The restricted metric retains the scalar-curvature and end hypotheses, and is smooth through the frontier. The regularity and area assertions of the same lemmas therefore permit application of Theorem 1.1 to this exterior, giving the displayed bound. $\square$

3 The initial enclosure, buried boundary, and flow notation

We construct the conformal flow for $n \geq 8$, following Bray and Bi-Zhu [8, 5]. The flow keeps the full frontier area fixed while changing the exterior metric. We first specify its ambient manifold and filled side, including the connectedness needed for the harmonic potentials.

Extend the original smooth metric a short distance behind its compact frontier, and truncate there at a smooth compact boundary B . Denote the resulting smooth ambient exterior by (M, g_0) and its original filled side by J . The set J contains a fixed collar of B . Smooth boundary charts, the compact core, and the complete asymptotically flat tail make M complete with B included. The metric on the buried side need not have nonnegative scalar curvature.

Use the canonical closed representative of J , whose frontier is the full perimeter support. Its open complement is connected. Indeed, there is one component containing the coordinate end, and any other component U is bounded with boundary in the original frontier. That frontier has finite $(n-1)$ -measure and a singular part of zero $(n-1)$ -measure, so the finite-boundary-measure criterion gives finite perimeter for U . On regular boundary charts its normal is opposite to the filled-side normal. Filling U therefore gives

$$P_{g_0}(J \cup U) = P_{g_0}(J) - P_{g_0}(U).$$

The nonempty bounded open set U has positive perimeter, contradicting outer minimization of J . This proves connectedness of the open end exterior. The perimeter facts used here are local BV facts [1, 24].

Local perimeter minimality of the original frontier and its compactness give one radius for all sufficiently small local comparisons. Shrink that radius so the variation balls also avoid the buried

collar. Balls missing the frontier have zero local perimeter. We may replace J by its outermost minimizing enclosure K_0 : outer minimization preserves its area, and the lattice comparison below preserves its local minimality. Write $E_0 = M \setminus K_0$. The replacement has connected open exterior by the same filling argument. Scalar nonnegativity holds on E_0 , and later it will be used on $E_t \subset E_0$. The auxiliary physical boundary remains buried throughout the flow. With $k = n - 1$ and $d = n - 2$, the flow constructed below has $E_t \subset E_0$ and defining equations

$$u_t = 1 + \int_0^t v_s ds, \quad g_t = u_t^{4/d} g_0, \quad \Delta_{g_0} v_t = 0 \text{ in } E_t, \quad v_t = 0 \text{ on } K_t, \quad v_t \longrightarrow -e^{-t}.$$

Consequently $v_t = 0$ and $u_t = 1$ on the same fixed collar of B . The g_0 -capacitary potential is $1 + e^t v_t$; the g_t -capacitary potential, denoted below by w_t , is

$$w_t = 1 + v_t/u_t \quad \text{on } E_t, \quad w_t = 1 \quad \text{on } K_t.$$

These two potentials must be distinguished. Both are constant on that collar and have zero physical-boundary flux. Capacity test functions are compactly supported relative to M , smooth up to B , and equal to one near K_t ; they are not required to vanish on B . Equivalently, one uses the Dirichlet problem on E_t and extends by one to K_t . Every regular level below one excludes the buried collar, so the capacity-flux calculation has only its level boundary and its sphere at infinity. No mean-curvature condition at B enters.

The smooth reference metric remains g_0 . The working metric g_t is smooth on the open exterior, where u_t is g_0 -harmonic and $R_{g_t} = u_t^{-4/d} R_{g_0} \geq 0$. Across a singular frontier one uses the comparison and Hölder estimates of Lemma 4.1, followed by the simultaneous one-sided regularity and relative forcing estimates of Section 5. The perimeter-plus-volume forcing problem uses filled-set competitors containing K_t ; they may meet its frontier. Corollary 6.2 proves detachment including at singular points. Once its new frontier is strictly detached, it has a neighborhood compactly contained in this smooth exterior. Its one-sided smoothing and the smooth mass-capacity argument therefore take place entirely outside the original frontier and omit B altogether.

4 Construction and preserved area of the flow

Lemma 4.1 (Construction and area of the conformal flow). *Let K_0 be the initial outermost minimizing enclosure just described, and put*

$$E_0 = M \setminus K_0, \quad \Sigma_0 = \partial K_0, \quad A_0 = P_{g_0}(K_0).$$

Here and below perimeter is relative to $\text{int } M$ and counts the whole frontier. There are increasing closed filled sets $K_t \supset K_0$, with connected end exteriors $E_t = M \setminus K_t$, and positive continuous functions u_t such that

$$\begin{aligned} g_t &= u_t^{4/d} g_0, & u_t &= 1 + \int_0^t v_s ds, \\ \Delta_{g_0} v_t &= 0 \quad \text{in } E_t, & v_t &= 0 \quad \text{on } K_t, & v_t &\longrightarrow -e^{-t} \quad \text{at infinity.} \end{aligned} \tag{4.1}$$

For every t we have $e^{-t} \leq u_t \leq 1$. Each K_t locally minimizes perimeter in the continuous metric g_t against all variations in balls below the initial comparison radius, including variations across K_0 . The set K_t is the outermost g_t -minimizing enclosure of K_0 and satisfies

$$P_{g_t}(K_t) = A_0. \tag{4.2}$$

On each finite time interval the conductors have common compact support, the factors have a common positive lower bound and spatial Hölder modulus, and the frontiers have uniform two-phase density and almost-minimizing estimates. In particular there are $\alpha_T > 0$, $C_T < \infty$, and $r_T > 0$ such that

$$P_{g_0}(K_t; B_r(x)) \leq P_{g_0}(L; B_r(x)) + C_T r^{k+\alpha_T} \quad (4.3)$$

whenever $0 \leq t \leq T$, $r < r_T$, and $L \triangle K_t \subseteq B_r(x)$.

The discrete method is due to Bray [8]; its singular-frontier form follows [5, Section 2.1]. We follow the order of [5, Lemmas 2.21 and 2.23 and Corollary 2.22]: area preservation precedes nesting of the limiting outermost hulls. Freezing on a discrete conductor by itself does not give freezing on a potentially larger limiting conductor.

The proof has three stages. A static weighted enclosure problem supplies compactness and boundary estimates for a single conductor. We apply those estimates to the discrete flow and prove that its area loss tends to zero. Finally, we identify the minimum area of each limiting metric; that equality gives nesting and the integral evolution.

4.1 Weighted enclosures and their escape potentials

For a continuous positive weight a , write

$$P_a(L; U) = \int_{U \cap \partial^* L} a \, d\mathcal{H}_{g_0}^k.$$

The next lemma requires only continuity of a for its density and potential estimates. Its last assertion records the stronger comparison available when the weight is Hölder continuous.

Lemma 4.2 (A weighted enclosure and its potential). *Every compact finite-perimeter obstacle has a unique outermost minimizing enclosure for any continuous perimeter weight bounded above and below by positive constants.*

For the quantitative conclusions, let K_0 be a compact finite-perimeter filled set containing a collar of B , locally perimeter minimizing in g_0 with comparison radius $r_0 > 0$. Put $A_0 = P_{g_0}(K_0)$, and let a be continuous with $c \leq a \leq 1$ and $a = 1$ on K_0 , where $c > 0$. Every P_a -minimizing enclosure K has weighted perimeter at most A_0 and is contained in a ball $B_{R_(c)}$ depending only on c , A_0 , the initial support, and the reference geometry.*

Every such enclosure locally minimizes P_a against unconstrained variations in balls of radius less than r_0 . Its canonical closed representative has connected open end exterior and uniform two-phase volume and perimeter density bounds. The unique escape potential q_K , harmonic in that exterior, zero on K , and tending to one at infinity, is continuous and has a Hölder modulus depending on the same data. It satisfies $0 < q_K < 1$ in the exterior.

If in addition $\text{osc}_{B_r} a \leq C_a r^\alpha$ for $0 < \alpha < 1$, then, for local competitors L ,

$$P_{g_0}(K; B_r) \leq P_{g_0}(L; B_r) + C r^{k+\alpha},$$

where C also depends on C_a .

Proof. Confinement and existence. First minimize P_a among enclosures of K_0 , either globally or with an outer constraint $L \subset B_Q$. The competitor K_0 bounds the minimizing perimeter by A_0 . Choose R_0 containing K_0 and the compact core, and choose $\gamma \geq 1$ such that $\gamma^{-2}\delta \leq g_0 \leq \gamma^2\delta$ in the coordinate end. Put

$$a_- = c\gamma^{-k}, \quad a_+ = \gamma^k.$$

For almost every $R > R_0$, truncation by B_R is admissible and cancellation of the common interior perimeter gives

$$P_\delta(L; \{|x| > R\}) \leq \frac{a_+}{a_-} \mathcal{H}_\delta^k(L \cap S_R). \quad (4.4)$$

For the constrained problem this is used when $R < Q$; beyond Q the tail is empty. Set $V(R) = |L \cap \{|x| > R\}|_\delta$. Euclidean isoperimetry applied to the cut tail, including its spherical boundary, and the coarea formula imply

$$V(R)^{k/n} \leq C_{\text{iso}} \left(1 + \frac{a_+}{a_-}\right) [-V'(R)].$$

At a regular cutting radius $R_1 \in [R_0, R_0 + 1]$, a second use of Euclidean isoperimetry gives

$$V(R_1) \leq V_* := \left[C_{\text{iso}} \left(a_-^{-1} A_0 + \mathcal{H}_\delta^k(S_{R_0+1}) \right) \right]^{n/k}.$$

Indeed, the original end perimeter is at most $a_-^{-1} A_0$ and the added cutting boundary is a portion of that sphere. While $V > 0$,

$$\frac{d}{dR} V(R)^{1/n} \leq - \left[n C_{\text{iso}} \left(1 + \frac{a_+}{a_-}\right) \right]^{-1}.$$

Consequently the tail vanishes by the radius

$$R_*(c) = R_0 + 1 + n C_{\text{iso}} \left(1 + \frac{a_+}{a_-}\right) V_*^{1/n}. \quad (4.5)$$

We use the canonical closed representative $L = \text{supp}(\chi_L dV_{g_0})$; zero tail volume therefore means actual containment in $B_{R_*(c)}$. No density estimate, connectedness assertion, or potential estimate was needed.

Minimize first in each bounded exhaustion domain B_Q . BV compactness and weighted lower semicontinuity give a constrained minimizer. The radius in (4.5) is independent of Q , so a sequence with $Q \rightarrow \infty$ has a limit on one fixed support ball. Every fixed compact competitor is admissible eventually; the limit is thus a global minimizer. All global minimizers have the same support bound. Choose one of maximal background volume by compactness once more. Submodularity and minimality show that the union and intersection of any two minimizers are again minimizers. The maximal-volume one must therefore contain every other minimizer, first almost everywhere and then canonically. This is the unique outermost hull. The same argument constructs the minimizing enclosure of any compact finite-perimeter obstacle J for any continuous weight bounded above and below by positive constants: use $P_a(J)$ as the competitor bound and choose R_0 to contain J . Equality of the weight to one on J is unnecessary for this individual existence assertion.

If the prescribed initial enclosure J was replaced by its outermost minimizing hull K_0 , the replacement also preserves local minimality, not only area. For $L \triangle K_0 \in U$ in an initial variation ball, apply the local minimality of J to $L \cap J$ and the enclosure minimality of K_0 to $L \cup J$. Submodularity gives

$$\begin{aligned} P_{g_0}(K_0; U) &\leq P_{g_0}(L \cup J; U) \\ &\leq P_{g_0}(L; U) + P_{g_0}(J; U) - P_{g_0}(L \cap J; U) \\ &\leq P_{g_0}(L; U). \end{aligned}$$

Thus the same initial comparison radius applies to K_0 .

Removal of the initial obstacle. Let K be any minimizing enclosure. For a local competitor $L \triangle K \subseteq B_r(x)$ with $r < r_0$, initial local minimality and $a = 1$ on K_0 give

$$P_a(K_0; B_r(x)) \leq P_a(L \cap K_0; B_r(x)).$$

Global enclosure minimality applies to $L \cup K_0$. Combining it with the local submodularity inequality yields

$$P_a(K; B_r(x)) \leq P_a(L; B_r(x)). \quad (4.6)$$

Thus K is a local background quasiminimizer with factor $Q = c^{-1}$. The comparison includes balls crossing the initial obstacle and uses its local minimality against removals. Cancelling unchanged perimeter also localizes the comparison to smaller variation domains.

Two-phase density before boundary regularity. Choose uniform small background coordinate balls away from a smaller buried collar. Smoothness on the compact core and the differentiated asymptotic-flatness bounds on the end give uniform local absolute and relative isoperimetric constants, sphere-area bounds, and scaled C^{1,α_0} bounds for the divergence-form coefficients of Δ_{g_0} , for some $\alpha_0 \in (0, 1)$. No uniform bounds for derivatives of all orders are required. Shrink their radius below r_0 . Balls missing ∂K_0 have zero initial local perimeter and cause no exception. We may use geodesic balls in the following slicing argument by localization of the exact weighted comparison.

Let K be one of the hulls and $x \in \text{supp } |D\chi_K|$. Put

$$m(r) = \text{Vol}_{g_0}(K \cap B_r(x)), \quad e(r) = \text{Vol}_{g_0}(B_r(x) \setminus K).$$

Both numbers are positive for every $r > 0$, since otherwise the perimeter measure would vanish near x . Delete or fill $B_r(x)$, first comparing in a slightly larger ball, and then decrease the comparison radius. BV slicing and quasiminimality give, for almost every r ,

$$P_{g_0}(K; B_r(x)) \leq Qm'(r), \quad P_{g_0}(K; B_r(x)) \leq Qe'(r).$$

If I is the local absolute isoperimetric constant, the perimeter bound and isoperimetric inequality give

$$m(r)^{k/n} \leq I(P_{g_0}(K; B_r(x)) + m'(r)) \leq I(Q + 1)m'(r).$$

Integrating the resulting inequality for $m^{1/n}$ from its zero limit at radius zero, and doing the same for the exterior phase, gives

$$\min\{m(r), e(r)\} \geq [nI(Q + 1)]^{-n} r^n.$$

Also $m'(r) + e'(r) = \text{Area}_{g_0}(\partial B_r(x)) \leq Sr^k$, so adding the two comparison inequalities gives $P_{g_0}(K; B_r(x)) \leq QSr^k/2$. Perimeter-measure monotonicity extends the upper bound to every small radius; relative isoperimetry and the two phase volumes give the lower bound. Thus, with constants depending only on the stated data,

$$\begin{aligned} \min\{\text{Vol}_{g_0}(K \cap B_r(x)), \text{Vol}_{g_0}(B_r(x) \setminus K)\} &\geq c_* r^n, \\ c_* r^k &\leq P_{g_0}(K; B_r(x)) \leq C_* r^k \quad (x \in \partial K, \ 0 < r < r_*). \end{aligned} \quad (4.7)$$

For the canonical representative $K = \text{supp}(\chi_K dV_{g_0})$, these estimates identify ∂K with $\text{supp } |D\chi_K|$. They hold at every frontier point, not only almost every reduced-boundary point. Null-set additions or artificial isolated points are not part of the representative.

A bounded complementary component of a minimizing hull can be filled, removing its strictly positive perimeter. This contradicts minimality; equivalently one may use the finite-perimeter component decomposition. The canonical representatives have no measure-zero cracks producing artificial interfaces. Since the manifold has only one end and the conductor is compact, the remaining exterior is its connected end component.

The weak capacitary problem. The harmonic function is constructed before any continuity at the frontier is assumed. For an individual compact conductor K , let $\mathcal{D}^{1,2}(M)$ be the completion, in the gradient norm, of smooth functions compactly supported relative to M . Such functions may be nonzero on the physical boundary. The asymptotically flat Sobolev inequality, for $n \geq 3$, embeds this space in $L^{2n/d}$. Minimize

$$\int_M |\nabla W|_{g_0}^2 dV_{g_0}$$

over its closed convex subset consisting of functions whose quasi-continuous representative is at least one quasi-everywhere on K . Here quasi-everywhere means outside a set of Newtonian capacity zero. A smooth function equal to one near K makes the class nonempty. The Hilbert-space direct method gives a unique minimizer W_K ; truncation gives $0 \leq W_K \leq 1$ and $W_K = 1$ quasi-everywhere on K .

Variations of both signs supported in $M \setminus K$ prove harmonicity there. Nonnegative variations supported away from the physical boundary remain admissible, and give

$$\int_M \langle \nabla W_K, \nabla \phi \rangle_{g_0} dV_{g_0} \geq 0 \quad (\phi \geq 0).$$

Consequently the escape potential $q_K = 1 - W_K$, extended by zero on K , is locally H^1 , satisfies $0 \leq q_K \leq 1$, and obeys

$$\Delta_{g_0} q_K \geq 0 \quad \text{weakly away from the physical boundary,} \quad \Delta_{g_0} q_K = 0 \quad \text{in } M \setminus K. \quad (4.8)$$

Sobolev integrability of W_K and interior estimates on end annuli imply $W_K \rightarrow 0$ uniformly at infinity. A radial supersolution $r^{-\beta}$, $0 < \beta < d$, and interior gradient estimates give, after increasing a radius depending on K ,

$$W_K = O(r^{-\beta}), \quad |\nabla W_K| = O(r^{-1-\beta}).$$

No uniform support radius is assumed in this individual estimate. Thus $q_K \rightarrow 1$ at infinity. The homogeneous space is important: W_K need not belong to unweighted L^2 in dimensions three and four. Since K contains a collar of the physical boundary, $W_K = 1$ and $q_K = 0$ there, and the boundary contributes no test or flux term to this exterior problem.

The exterior-indicator boundary contraction. Fix $x \in \partial K$ and work in uniform background coordinates, using Euclidean coordinate radii in this paragraph. By (4.7) and metric comparability, $|K \cap B_r(x)| \geq d_* r^n$. Choose $\kappa \in (0, 1/2)$ so small that $|B_{\kappa r}(x)| \leq d_* r^n / 2$. Coarea supplies a radius $\rho \in [\kappa r, r]$ at which the Sobolev traces exist and

$$\mathcal{H}_\delta^k(K \cap \partial B_\rho(x)) \geq d'_* r^k.$$

At this radius the zero extension of q_K has zero trace almost everywhere on the conductor portion of the sphere. Set

$$\chi = \mathbf{1}_{(M \setminus K) \cap \partial B_\rho(x)}, \quad M(r) = \operatorname{ess\,sup}_{B_r(x)} q_K.$$

Let H be the g_0 -harmonic replacement in the full ball with boundary trace q_K . Weak subharmonicity in (4.8) and the weak maximum principle give $q_K \leq H$. If P_ρ is the Poisson kernel, then

$$H(y) \leq M(r) \int_{\partial B_\rho(x)} P_\rho(y, \xi) \chi(\xi) dS_\delta(\xi).$$

The datum is the exterior indicator χ , not $1 - \chi$: only the former dominates the possibly positive trace of q_K . For the fixed smooth background operator, scaled smooth-ball Green-function estimates give

$$P_\rho(y, \xi) \geq c_P \rho^{1-n} \quad (y \in B_{\rho/2}(x), \xi \in \partial B_\rho(x)).$$

The constant is uniform under the local ellipticity and coefficient bounds already fixed; the kernel integrates to one. Its missing mass on the conductor portion of the sphere is therefore at least a uniform $\delta_* > 0$. Since $B_{\kappa r/2}(x) \subset B_{\rho/2}(x)$, it follows that

$$M(\kappa r/2) \leq (1 - \delta_*) M(r).$$

This comparison does not assume continuity at the frontier: the harmonic replacement uses an actual Sobolev trace at an eligible radius, whereas the indicator is used only in the Poisson integral and is not asserted to belong to $H^{1/2}$.

Iteration, starting from $q_K \leq 1$, gives some $\alpha \in (0, 1)$ for which

$$\operatorname{ess\,sup}_{B_r(x)} q_K \leq C_*(r/r_*)^\alpha \quad (0 < r \leq r_*).$$

For example, decrease to an exponent smaller than one and no larger than $\log(1 - \delta_*)/\log(\kappa/2)$. Interior smoothness now supplies a continuous representative equal to zero at every frontier point. The strong maximum principle gives $0 < q_K < 1$ on the connected exterior. Uniqueness among continuous exterior solutions with these boundary and end values follows by applying the maximum principle on expanding truncated exteriors.

For completeness the same estimate gives a full spatial Hölder modulus, not just decay at boundary points. If two points a distance ℓ apart have one within 2ℓ of K , both values are bounded by $C_* \ell^\alpha$. Otherwise a local ball joining them stays in the exterior. At distance ϱ from K , interior gradient estimates and the boundary decay give $|\nabla q_K| \leq C_* \varrho^{\alpha-1}$ when $\varrho < r_*/4$; at larger distance fixed-radius interior estimates suffice. Integration between the two points again bounds their difference by $C_* \ell^\alpha$. Boundedness handles larger separations. Hence the zero extensions satisfy

$$\|q_K\|_{C^{0,\alpha}(M, g_0)} \leq C_*. \quad (4.9)$$

The constants depend on c , the background geometry, and the initial comparison radius. No derivatives of the weight enter this estimate.

The comparison for a Hölder weight. Suppose $\operatorname{osc}_{B_r} a \leq C_a r^\alpha$. If $P_{g_0}(L; B_r) \geq P_{g_0}(K; B_r)$ the conclusion is immediate. Otherwise the density upper bound gives both perimeters at most

$C_* r^k$. The exact weighted comparison yields

$$\begin{aligned} P_{g_0}(K; B_r) &\leq \frac{\sup_{B_r} a}{\inf_{B_r} a} P_{g_0}(L; B_r) \\ &\leq P_{g_0}(L; B_r) + Cr^\alpha P_{g_0}(L; B_r) \\ &\leq P_{g_0}(L; B_r) + Cr^{k+\alpha}. \end{aligned}$$

This proves the final assertion. $\square$

4.2 The discrete flow and its area loss

We now apply the static lemma successively. All enclosure minimizations allow compact finite-perimeter supersets, including sets with bounded complementary cavities, so union and intersection remain admissible. Put $p = 2k/d$, the exponent converting the conformal factor to its perimeter weight.

Fix $0 < \epsilon < 1/2$, set $u_0^\epsilon = 1$, and let $K_0^\epsilon = K_0$. Inductively, put

$$h_j^\epsilon = (u_j^\epsilon)^{4/d} g_0, \quad a_j^\epsilon = (1 - \epsilon)^j.$$

For $j \geq 1$, let K_j^ϵ be the outermost h_j^ϵ -minimizing enclosure of K_{j-1}^ϵ . Let v_j^ϵ be the unique harmonic function on $M \setminus K_j^\epsilon$ with zero boundary value and end value $-a_j^\epsilon$, extended by zero to K_j^ϵ . Define

$$\begin{aligned} u_{j+1}^\epsilon &= u_j^\epsilon + \epsilon v_j^\epsilon, \\ u_t^\epsilon &= u_j^\epsilon + (t - j\epsilon) v_j^\epsilon, \\ K_t^\epsilon &= K_j^\epsilon, \quad v_t^\epsilon = v_j^\epsilon. \end{aligned} \tag{4.10}$$

The last two lines apply when $j\epsilon \leq t < (j+1)\epsilon$; use the new hull and velocity at a grid endpoint. Thus $u_t^\epsilon = 1 + \int_0^t v_s^\epsilon ds$ exactly. The maximum principle gives

$$-a_j^\epsilon \leq v_j^\epsilon \leq 0, \quad (1 - \epsilon)^{j+1} \leq u_t^\epsilon \leq 1 \quad (j\epsilon \leq t < (j+1)\epsilon).$$

More precisely, if $\theta = t - j\epsilon \in [0, \epsilon]$, then concavity of $\log(1 - \theta)$ gives

$$u_t^\epsilon \geq (1 - \epsilon)^j (1 - \theta) \geq (1 - \epsilon)^{t/\epsilon} \geq 4^{-t}.$$

Consequently $u_t^\epsilon \geq 4^{-T}$ for $0 \leq t \leq T$. In particular every finite interval has mesh-independent metric ellipticity constants. The end values are exactly

$$u_j^\epsilon \longrightarrow a_j^\epsilon, \quad u_t^\epsilon \longrightarrow a_j^\epsilon (1 - (t - j\epsilon)).$$

The discrete conductors are nested by construction. Comparison on $M \setminus K_{j+1}^\epsilon$, including the change of the constant at infinity, gives

$$v_j^\epsilon \leq v_{j+1}^\epsilon \leq 0. \tag{4.11}$$

No limiting nesting is used here.

Lemma 4.3 (Uniform estimates for the discrete flow). *For every finite T , the preceding induction defines the flow on $[0, T]$. Every grid conductor K_j^ϵ is the outermost h_j^ϵ -minimizing enclosure of the fixed K_0 . The conductors are nested and have common compact support. Their velocities, and the factors u_t^ϵ , have a common spatial Hölder modulus; u_t^ϵ is also uniformly Lipschitz in time. The frontiers have uniform density and background almost-minimizing estimates. Their grid areas $A_j^\epsilon = P_{h_j^\epsilon}(K_j^\epsilon)$ are nonincreasing, and the area is frozen between grid times.*

Proof. The fixed-obstacle minimizing property. Every K_j^ϵ is also the outermost h_j^ϵ -minimizing enclosure of the fixed K_0 . Indeed, suppose this holds at step $j-1$, and let $L \supset K_0$ be any compact finite-perimeter competitor. The two consecutive metrics agree on K_{j-1}^ϵ , because v_{j-1}^ϵ vanishes there. The boundary of $L \cap K_{j-1}^\epsilon$ is supported in that same closed set. Hence

$$\begin{aligned} P_{h_j^\epsilon}(K_{j-1}^\epsilon) &= P_{h_{j-1}^\epsilon}(K_{j-1}^\epsilon) \\ &\leq P_{h_{j-1}^\epsilon}(L \cap K_{j-1}^\epsilon) = P_{h_j^\epsilon}(L \cap K_{j-1}^\epsilon). \end{aligned}$$

Perimeter submodularity therefore gives

$$P_{h_j^\epsilon}(L \cup K_{j-1}^\epsilon) \leq P_{h_j^\epsilon}(L).$$

The set on the left is an admissible competitor for K_j^ϵ , which proves global minimality. If L is any global minimizer, the same inequalities show that its union with K_{j-1}^ϵ is a constrained minimizer. Outermostness of K_j^ϵ then gives $L \subset K_j^\epsilon$. The induction starts at the actual outermost initial hull, not at an arbitrary almost-minimizing frontier.

Set $A_j^\epsilon = P_{h_j^\epsilon}(K_j^\epsilon)$. Freezing on the previous conductor and its admissibility at the next step imply

$$0 \leq A_j^\epsilon \leq A_{j-1}^\epsilon \leq A_0.$$

The metric at an intermediate time need not make the frozen K_j^ϵ globally minimizing. Its area nevertheless equals A_j^ϵ , since $v_j^\epsilon = 0$ on that frontier; comparison with an arbitrary competitor always uses h_j^ϵ . On a fixed finite interval h_j^ϵ and g_t^ϵ differ uniformly by $O_T(\epsilon)$ for $j\epsilon \leq t < (j+1)\epsilon$.

Closing the induction and its uniform estimates. At step j , the constructed continuous positive factor defines the weighted hull problem. The fixed-obstacle minimizing property just proved lets us apply Lemma 4.2 with $a = (u_j^\epsilon)^p$ and $c = 4^{-pT}$. It gives the density estimates and the continuous potential

$$v_j^\epsilon = -a_j^\epsilon q_{K_j^\epsilon}, \quad \|q_{K_j^\epsilon}\|_{C^{0,\alpha}} + \|v_j^\epsilon\|_{C^{0,\alpha}} \leq C_T.$$

In particular v_j^ϵ vanishes on the whole canonical conductor. Thus the next factor is continuous and agrees with u_j^ϵ on K_j^ϵ , including its perimeter traces. This closes the induction underlying the freezing and fixed-obstacle comparisons.

The exact discrete integral identity gives a uniform spatial Hölder bound for u_t^ϵ on $[0, T]$, while $|v_j^\epsilon| \leq 1$ gives the time Lipschitz bound. Hence $\text{osc}_{B_r}(u_j^\epsilon)^p \leq C_T r^\alpha$. The final assertion of the static lemma now gives the additive almost-minimizing estimate with error $C_T r^{k+\alpha}$.

Uniform finite-time support. The global grid-minimality induction permits application of (4.5) with the fixed obstacle K_0 , the fixed area bound A_0 , and $c = 4^{-pT}$. Hence, enlarging the resulting radius if necessary,

$$K_t^\epsilon \subset B_{R_T}, \quad \text{Vol}_{g_0}(K_t^\epsilon) \leq V_T \quad (0 \leq t \leq T). \quad (4.12)$$

It follows directly from the static truncation comparison and is independent of the number or shape of the conductor components. □

Proposition 4.4 (Vanishing discrete area loss). *For every finite T , the areas of the discrete frontiers tend uniformly to A_0 for $0 \leq t \leq T$ as $\epsilon \downarrow 0$.*

Proof. Global minimality in the preceding grid metric, with K_j^ϵ as competitor, gives

$$\begin{aligned}
0 \leq A_{j-1}^\epsilon - A_j^\epsilon &\leq \int_{\partial^* K_j^\epsilon} [(u_{j-1}^\epsilon)^p - (u_j^\epsilon)^p] d\mathcal{H}_{g_0}^k \\
&= \int_{\partial^* K_j^\epsilon} \left[1 - \left(1 + \epsilon \frac{v_{j-1}^\epsilon}{u_{j-1}^\epsilon} \right)^p \right] d\mathcal{H}_{h_{j-1}^\epsilon}^k \\
&\leq -p\epsilon \int_{\partial^* K_j^\epsilon} \frac{v_{j-1}^\epsilon}{u_{j-1}^\epsilon} d\mathcal{H}_{h_{j-1}^\epsilon}^k \\
&\leq C_T \epsilon A_0 \sup_{\partial K_j^\epsilon} |v_{j-1}^\epsilon|.
\end{aligned}$$

The third line follows from Bernoulli's inequality, with $v_{j-1}^\epsilon \leq 0$.

Separate the grid transitions into those for which

$$\text{Vol}_{g_0}(K_j^\epsilon \setminus K_{j-1}^\epsilon) > V_T \sqrt{\epsilon}$$

and the remaining transitions. Discrete nesting and (4.12) bound the number of the first kind by $\epsilon^{-1/2}$, so their total area loss is $O_T(\sqrt{\epsilon})$. For a transition of the second kind, the two phase-density inequalities imply

$$d_{\mathcal{H}}(\partial K_j^\epsilon, \partial K_{j-1}^\epsilon) \leq C_T (V_T \sqrt{\epsilon})^{1/n}$$

once ϵ is small. Indeed, a boundary point separated from the other frontier by a ball of radius ρ contributes at least $c_T \rho^n$ to the swept volume: use the filled phase at a point of the new frontier and the exterior phase at a point of the old one. The old velocity vanishes on the old frontier; its Hölder modulus therefore bounds its supremum on the new frontier by $C_T \epsilon^{\alpha/(2n)}$. Summing the ϵ -weighted losses over at most $T/\epsilon + 1$ transitions gives

$$0 \leq A_0 - A_j^\epsilon \leq C_T (\sqrt{\epsilon} + \epsilon^{\alpha/(2n)}) \longrightarrow 0 \quad (4.13)$$

uniformly for $j\epsilon \leq T$. The same assertion holds at intermediate times because the current frontier has frozen area. Only discrete nesting has entered this calculation. $\square$

4.3 The limiting minimum area, nesting, and integral evolution

The discrete areas now converge to A_0 . We first show that A_0 is the minimum enclosure area in every limiting metric. This fact will identify the selected outermost hulls and their velocities.

Proof of Lemma 4.1. Compactness and the limiting minimum area. The functions u_t^ϵ are uniformly Lipschitz in time and uniformly Hölder in space. Choose a mesh sequence tending to zero for which they converge locally uniformly, jointly in space and time, to a positive u_t , and put $g_t = u_t^{4/d} g_0$. The support bound also controls the end normalization. Put $\beta = d/2$ and enlarge R_T until $\Delta_{g_0} r^{-\beta} < 0$ for $r \geq R_T$. The function $1 + v_t^\epsilon/a_j^\epsilon$ is harmonic there, lies between zero and one on the inner sphere, and tends to zero at infinity. The maximum principle gives

$$0 \leq v_j^\epsilon + a_j^\epsilon \leq a_j^\epsilon (R_T/r)^\beta.$$

Writing $a_\epsilon(t) = a_j^\epsilon(1 - t + j\epsilon)$ on the current step and integrating yields

$$|u_t^\epsilon - a_\epsilon(t)| \leq T(R_T/r)^\beta \quad (0 \leq t \leq T, \ r \geq R_T).$$

The constants $a_\epsilon(t)$ converge uniformly to e^{-t} on $[0, T]$. These tail bounds turn local uniform convergence of the factors into uniform convergence on $[0, T] \times M$ and retain the constant $-e^{-t}$ for every fixed-time limit of the velocities. Thus the compactness passage preserves the end normalization.

We claim that, for every t , the minimum g_t -perimeter among enclosures of K_0 equals A_0 . Fix a time t and a bounded finite-perimeter enclosure $L \supset K_0$. Grid minimality at $j = \lfloor t/\epsilon \rfloor$, the uniform convergence of the lagged metric, and (4.13) imply

$$P_{g_t}(L) \geq A_0. \quad (4.14)$$

All discrete velocities vanish on K_0 , so $u_t = 1$ there and $P_{g_t}(K_0) = A_0$. Hence A_0 is exactly the minimum enclosure perimeter for g_t .

Common support and the background perimeter bound give a subsequential BV limit $\widehat{K}_t$ of the grid conductors. Lower semicontinuity in the uniformly converging positive weights gives $P_{g_t}(\widehat{K}_t) \leq A_0$. Since this is itself an enclosure of K_0 , (4.14) gives the reverse inequality. Thus every such limit is a g_t -minimizer of perimeter A_0 . This also proves perimeter convergence. The density estimates then upgrade convergence of the canonical sets to Hausdorff convergence of their frontiers.

Define K_t to be the outermost g_t -minimizing enclosure of K_0 . It has perimeter A_0 and contains every $\widehat{K}_t$. Lemma 4.2 bounds its support by a finite-time common radius and gives exact local weighted minimality, since $u_t = 1$ on K_0 . It also supplies the uniform density, local area, and background almost-minimizing estimates. A bounded component of its exterior could be filled with a strict decrease in perimeter; thus no such component occurs. The unique asymptotically flat end accounts for the remaining exterior component. These observations also apply to every $\widehat{K}_t$.

Nesting of the limiting outermost hulls. For $s > t$, the monotonicity of the discrete conformal factors passes to the limit, so $u_s \leq u_t$. The already established minimum areas give

$$A_0 \leq P_{g_s}(K_t) \leq P_{g_t}(K_t) = A_0.$$

It follows that $u_s = u_t$ almost everywhere on the perimeter of K_t . Continuity of the factors and the positive perimeter density at every frontier point improve this to

$$u_s = u_t \quad \text{on } \partial K_t. \quad (4.15)$$

At this stage no freezing throughout K_t has been asserted.

Take any joint subsequential limit $(\widehat{K}_s, \widehat{v}_s)$ of the grid conductors and velocities at time s . The boundary Hölder estimate, interior harmonic compactness, and the fixed-sphere tail comparison identify $\widehat{v}_s$ as the harmonic function with zero extension to $\widehat{K}_s$ and end value $-e^{-s}$. It is strictly negative on its connected end exterior. On the other hand, the discrete integral identity and (4.11) give

$$u_t^\epsilon - u_s^\epsilon = - \int_t^s v_\tau^\epsilon d\tau \geq (s - t)(-v_s^\epsilon).$$

Passing to this subsequence yields

$$u_t - u_s \geq (s - t)(-\widehat{v}_s) > 0 \quad \text{on } M \setminus \widehat{K}_s.$$

Together with (4.15), this implies $\partial K_t \subset \widehat{K}_s$. The connected exterior of $\widehat{K}_s$ agrees with the exterior of K_t near infinity and cannot cross ∂K_t , so it is contained in that exterior. Therefore

$$K_t \subset \widehat{K}_s \subset K_s \quad (t < s). \quad (4.16)$$

This proves nesting using only the discrete integral identity and the previously proved area equality.

Identification of the integral flow. On each finite interval the function $V(t) = \text{Vol}_{g_0}(K_t)$ is bounded and nondecreasing. Its positive left jumps occur at only countably many times. Let $\mathcal{J}$ be the union of these exceptional sets over integer time intervals, together with $\{0\}$. If $t \notin \mathcal{J}$, every fixed-time grid limit satisfies, by (4.16),

$$\bigcup_{r < t} K_r \subset \widehat{K}_t \subset K_t.$$

The union may be taken over rational $r < t$, and its volume is the left limit of V . At a nonjump time this equals $V(t)$. The canonical identity $K_t = \text{supp}(\chi_{K_t} dV_{g_0})$ therefore gives

$$\overline{\bigcup_{r < t} K_r} = K_t :$$

a point of K_t outside that closure would have a neighborhood of positive K_t -volume disjoint from the union, contradicting equality of volumes. Since $\widehat{K}_t$ is closed, it equals K_t .

Thus at every nonexceptional time all subsequential grid limits agree. Compactness proves convergence of the entire chosen sequence in L^1 and in Hausdorff distance of canonical sets and frontiers. The latter assertion follows directly from the two phase-density estimates: a boundary point remaining away from the limiting boundary would have a ball wholly in one limiting phase, contrary to the corresponding positive discrete phase volume and L^1 convergence. The reverse direction uses the two positive limiting phase volumes. Common support prevents boundary points from escaping in this argument. The boundary Hölder estimates, interior harmonic compactness, uniqueness, and uniform tail comparison then give

$$\|v_t^\epsilon - v_t\|_{C^0(M)} \longrightarrow 0 \quad (t \notin \mathcal{J}),$$

where v_t is the harmonic function on E_t , zero on K_t and tending to $-e^{-t}$. The static density and capacity arguments construct this selected potential at exceptional times as well. Nesting and the maximum principle make v_t nondecreasing in time, so it is measurable. Dominated convergence in the exact discrete integral identity gives

$$u_t = 1 + \int_0^t v_s ds.$$

In fact the convergence of these integrals is uniform in space and on $[0, T]$, since

$$\sup_{0 \leq t \leq T} \left\| \int_0^t (v_s^\epsilon - v_s) ds \right\|_{C^0(M)} \leq \int_0^T \|v_s^\epsilon - v_s\|_{C^0(M)} ds \longrightarrow 0.$$

The scalar norm is measurable by taking the supremum over a countable dense spatial set; it is bounded by two, so the displayed almost-everywhere convergence justifies the last dominated-convergence step. The same exceptional set therefore works for every spatial point. The possible

failure of grid velocities to converge to the selected v_t at a jump time has no effect on this integral. Now, and only now, nesting and the integral identity give $u_s = u_t$ throughout K_t for $s \geq t$, since each later velocity vanishes there.

The end normalization gives $v_s \geq -e^{-s}$, so the same identity improves the finite-time lower bound to

$$u_t \geq 1 - \int_0^t e^{-s} ds = e^{-t}.$$

It also shows that u_t is g_0 -harmonic in E_t : each earlier velocity is harmonic there, and interior estimates justify integration on compact subsets of that exterior. All conductors contain the fixed collar of the physical boundary, so $v_t = 0$ and $u_t = 1$ on that collar. No boundary condition or mean-curvature assumption at the buried boundary is added. The conformal transformation of the scalar Laplacian gives

$$\Delta_{g_t} \left(\frac{v_t}{u_t} \right) = u_t^{-(n+2)/d} \left(\Delta_{g_0} v_t - \frac{v_t}{u_t} \Delta_{g_0} u_t \right) = 0 \quad \text{in } E_t.$$

Consequently $w_t = 1 + v_t/u_t$ is the g_t -harmonic capacitary potential: its boundary value is one and its end value is zero. The inequalities $-e^{-t} \leq v_t \leq 0$ and $u_t \geq e^{-t}$ give $0 \leq w_t \leq 1$ directly. Its constant extension over K_t equals one on the same buried collar.

Finally choose the convergent mesh subsequence diagonally over $T = 1, 2, \dots$. The true hull at each time is determined by its limiting metric, so the constructions on overlapping finite intervals agree. This gives the flow for all $t \geq 0$. Equation (4.2) already holds at every time by the minimizing-value argument, including jump times. The local weighted comparison and its Hölder-weight conversion give (4.3) uniformly on each finite interval, as asserted. $\square$

5 Regular patches and the size of the conformal forcing

Our next objective is to separate two distinct flow frontiers, including their singular points. At a hypothetical contact, rescaling makes both frontiers approach the same minimizing cone. Its regular patches are the places where their separation can be described by a scalar graph equation. Before using that equation, we need estimates for the whole intervening time family, and we must measure the harmonic forcing relative to its own size. A strict Hopf derivative at each fixed scale would not by itself survive normalization.

Fix $T < \infty$, and write $k = n - 1$ and $d = n - 2$. We use the flow already constructed: K_t is its closed filled side, $E_t = M \setminus K_t$ its connected end exterior, and $\Sigma_t = \text{supp } |D\chi_{K_t}|$ its full perimeter support. The normal always points from the filled side into this exterior. In the smooth reference metric g_0 , the frontiers have uniform local perimeter and two-phase density bounds and an almost-minimizing error $\beta r^{k+\mu}$ with $\mu > 0$. We also have

$$g_t = u_t^{4/d} g_0, \quad 0 < c_T \leq u_t \leq 1, \quad |v_t| \leq 1, \quad u_t = 1 + \int_0^t v_\sigma d\sigma.$$

The factors have a common background Hölder modulus. The sets are nested, each v_t is harmonic in E_t and zero on K_t , and each u_t is harmonic in E_t . The initial frontier is locally perimeter minimizing in g_0 . All neighborhoods below avoid the buried physical boundary.

The rescaled almost-minimizing errors tend to zero. Compactness, perimeter convergence, and small-excess regularity therefore give multiplicity-one minimizing boundary tangent cones and single $C^{1,\theta}$ graphs on their compact regular patches [34, Chapter 7]. We develop the uniform

estimates needed on those patches, following the regularity strategy of [6, Section 2.3] and the relative-forcing analysis of [5, Proposition 2.34 and Lemma 2.35].

Lemma 5.1 (Simultaneous regular neighborhoods). *Fix a regular point p of one Σ_t and a coordinate plane parallel to its tangent plane there. Given a sufficiently small slope bound, there are nested coordinate neighborhoods $U \Subset V$ of p such that every frontier meeting U is a single graph in V over that plane, with the prescribed slope bound and a uniform $C^{1,\theta}$ estimate. On smaller neighborhoods, every fixed finite order of one-sided regularity of the graphs, harmonic velocities, and conformal factors is uniform for $0 \leq t \leq T$. These conclusions also hold uniformly for the rescaled time families on each fixed compact regular patch of a limiting cone.*

Proof. The proof has two parts. First, ordering prevents a second frontier from developing excess beside a flat regular patch. Then the harmonic equation and the minimal graph equation improve the resulting common graphical bounds together.

A common graphical neighborhood. We use the following compactness consequence of the power almost-minimizing estimate. A sequence with uniform local perimeter and two-phase density bounds has a locally L^1 -convergent subsequence with locally convergent perimeter measures. If the limit has a regular point, the boundaries converge as single $C^{1,\theta}$ graphs on a smaller neighborhood of that point, with uniform bounds. Strict perimeter convergence excludes additional sheets; L^1 convergence alone would not suffice. These are the almost-minimizer compactness and flat-boundary regularity conclusions used in [5, Theorem 2.27].

Fix the reference regular point $p \in \Sigma_t$ and choose coordinates in which its tangent plane is horizontal. Consider any sequence of frontiers Σ_{σ_j} with points $p_j \in \Sigma_{\sigma_j}$ tending to p . After a subsequence, the filled sets all lie on the same ordered side of K_t . The compactness just stated gives a limiting filled set K_* , and two-phase density puts p on its perimeter support. Ordering passes to the limit. Every tangent filled set of K_* at p is therefore ordered with the tangent half-space of K_t ; its multiplicity-one minimizing boundary cone is supported in a closed half-space.

Such a cone is a plane. Indeed, the normal coordinate is nonnegative on the cone and satisfies $\Delta_{\text{link}} x_\perp = -(k-1)x_\perp$ on its spherical link. Integration forces that coordinate to vanish. Local area growth and the singular-set codimension at least seven supply cutoffs with vanishing tangential L^1 gradient, so the integration includes the singular set. Multiplicity one and the two-phase density bounds give the nonempty plane with density one. Density-one regularity thus makes ∂K_* a single $C^{1,\theta}$ graph near p , tangent to the reference sheet. Regular-patch convergence gives the same fixed graphical neighborhood and a uniform bound for Σ_{σ_j} when j is large.

Here is the uniformity conclusion. Fix a small slope threshold and choose a corresponding bound for the scaled $C^{1,\theta}$ graph norm in the flat-boundary regularity theorem. If no neighborhoods $U \Subset V$ worked for the whole time family, choose radii $r_j \downarrow 0$ and frontiers meeting $B_{r_j^2}(p)$ at points p_j that fail this graph bound on the radius- r_j cylinder centered at p_j . The preceding subsequence instead converges on a fixed graphical neighborhood of p . Its slopes at p_j tend to the horizontal slope, and their oscillation on a radius- r_j cylinder tends to zero under the uniform $C^{1,\theta}$ bound. A fixed shrink of the cylinders allows their centers to move from p to p_j . Restricting the fixed graphs then contradicts the chosen failures. This proves the common graphical neighborhood.

For a rescaled family, repeat this contradiction in a fixed regular cone chart. The reference graphs converge to its smooth sheet, the rescaled smooth background metrics have uniform bounds, and the power errors tend to zero. The same ordered-limit and density-one argument gives fixed inner and outer graphical neighborhoods, uniformly in the rescaling index. A finite cover gives the

assertion on a compact regular cone patch. None of these arguments uses separation for frontiers in different conformal metrics.

One-sided derivatives and the first graph equation. Choose three nested neighborhoods $V_0 \Subset V_1 \Subset V_2$ within this common graphical region. For any time σ , if Σ_σ meets V_1 , its uniform $C^{1,\theta}$ graph and the zero Dirichlet trace of v_σ give one-sided boundary estimates

$$\|v_\sigma\|_{C^{1,\alpha}(\overline{E_\sigma \cap V_0})} \leq C, \quad 0 < \alpha < \theta.$$

Here and below norms on graph domains use their uniformly controlled charts. If Σ_σ misses V_1 , its distance from V_0 is bounded below. On each relevant component v_σ is then harmonic on a whole neighborhood or identically zero, and interior estimates give the same bound. It would not suffice merely to know that the frontier misses V_0 . Nesting gives $E_t \subset E_\sigma$ for $\sigma \leq t$, so all these exterior estimates can be integrated on the common later exterior.

For the first stationarity argument, flatten the graphical boundary and reflect the negative exterior function v_σ oddly. Its zero trace implies that tangential first derivatives vanish at the graph, and odd reflection matches the normal first derivative. This gives a uniformly controlled $C^{1,\alpha}$ extension, nonnegative on the interior side, which dominates the original zero extension. Where a frontier is absent, keep the harmonic or zero function. On a smaller common neighborhood, fixed finite charts and a partition of unity preserve agreement on the exterior and domination on the interior. Integrating these extensions gives

$$\tilde{u}_t \geq u_t, \quad \tilde{u}_t = u_t \quad \text{on } \overline{E_t}, \quad \|\tilde{u}_t\|_{C^{1,\alpha}} \leq C.$$

Shrinking the neighborhood preserves a uniform positive lower bound. Thus $\tilde{g}_t = \tilde{u}_t^{4/d} g_0$, $d = n - 2$, is a uniformly controlled $C^{1,\alpha}$ extension of the one-sided metric.

These integrals require only measurable, not continuous, time dependence. The original harmonic velocities are pointwise monotone in time, interior derivatives are limits of measurable difference quotients, and nested graph heights are measurable. Reflection in the graphs and the finite chart constructions preserve this measurability. Equivalently, one can first integrate the derivatives directly on the common later exterior by dominated convergence and use extensions only for the graph equation.

By Lemma 4.1, each K_t is locally perimeter minimizing in g_t , including for variations across K_0 . Since $\tilde{g}_t \geq g_t$ and the two area functionals agree on Σ_t , it is also locally minimizing in $\tilde{g}_t$. The graph is stationary in this extension metric. The minimal graph equation and uniform elliptic regularity improve the graphs to $C^{2,\alpha}$ on smaller disks. The constants depend on the graph and metric bounds. The common local minimizing comparison makes the equation valid on the whole regular graph disk, including where that disk meets the initial conductor. At time zero the metric is the smooth reference metric and the same local minimizing comparison holds. Thus the estimate is uniform for the whole time family, including zero.

Closing the bootstrap. We now have a common $C^{2,\alpha}$ graph bound, including at time zero. Boundary estimates on these improved domains give uniform $C^{2,\alpha}$ estimates for all harmonic velocities. Integrate them on each common later exterior to improve all conformal factors, and apply the minimal graph equation again. Repetition on nested neighborhoods gives every prescribed finite order. At these later stages use a bounded extension operator for uniformly regular graph domains. Higher-order extensions need not dominate the metric: stationarity is already established, and agreement of the boundary value and first derivatives gives the same

minimal graph equation. In particular, no uniform positive Hopf lower bound is needed to control the sign of higher-order extensions.

For each fixed order only finitely many nested neighborhoods are used. On a compact regular cone patch the rescaled smooth background metrics have uniform bounds of every such order, while the lower bound for u_t and the bound for v_t are unchanged. All the estimates are therefore uniform in the blow-up index. Together with the original C^1 graphical convergence, interpolation gives convergence in every lower fixed differentiability order. In particular the one-sided working metrics converge to their constant contact-point values with the derivatives needed in the Jacobi expansion. $\square$

The preceding lemma controls derivatives absolutely. Separation also requires a relative estimate: the error made by transferring a normal derivative between nearby graphs must be small compared with that derivative, even when the harmonic function itself tends to zero.

Lemma 5.2 (Relative harmonic derivatives). *Consider uniformly $C^{2,\alpha}$ graphical domains in a fixed cylinder, with uniformly elliptic $C^{2,\alpha}$ metric coefficients. Let $z_i > 0$ be harmonic on the exterior side, with zero boundary trace, and let A_i be interior points whose distances from the graph and the lateral boundary are bounded below by a fixed positive fraction of the cylinder radius. These are the interior corkscrew points used in the estimate. Put $a_i = z_i(A_i)$. On nested central cylinders,*

$$\|z_i\|_{C^{2,\alpha}} \leq Ca_i, \quad ca_i \leq \partial_\nu z_i \leq Ca_i$$

on the original boundary, where ν points into the exterior. If a second graph lies in the closure of that exterior and converges in C^1 to the first, then on the second graph the derivative in its corresponding exterior-pointing unit normal also lies between ca_i and Ca_i , with adjusted fixed constants, for large i . All assertions are relative to a_i ; no positive lower bound for a_i is required.

Proof. Normalize z_i by a_i . Interior Harnack chains and the boundary Carleson estimate bound this normalized function on a smaller cylinder. Boundary Schauder estimates then give the stated $C^{2,\alpha}$ bound. An interior Harnack chain supplies a fixed positive comparison value on a sphere in a uniform interior tangent ball. The tangent-ball barrier gives the positive lower bound for the normal derivative on the central boundary patch; the upper bound follows from the derivative estimate. These are the standard local boundary estimates for uniformly regular elliptic Dirichlet problems; see [16].

Join corresponding points of the two graphs by the common graph coordinate segment, which lies on the exterior side of the first graph. The Hessian bound and convergence of the normals show that the error in transferring the derivative is at most

$$Ca_i(\text{height difference} + \text{normal difference}).$$

For large i this is less than half the first positive lower bound. The same factor a_i multiplies the bound and its error, even if $a_i \rightarrow 0$. This proves the relative assertion. $\square$

Applying the relative estimate to two flow times. Fix $0 \leq s < t \leq T$, and suppose a simultaneous blow-up at a contact point has a common regular cone patch. We apply the lemma to the positive functions whose integral measures the change from u_s to u_t . Nesting traps every intermediate filled set between the endpoints. Uniform almost-minimizing compactness and small-excess regularity give uniform single-sheet $C^{1,\theta}$ graphs for all intermediate times: an offending sequence of times would still have the squeezed multiplicity-one set and perimeter limit,

contradicting small-excess regularity. In a common graph direction the intermediate heights lie between the endpoint heights. They converge uniformly in C^0 and, by interpolation with the uniform graphical bound, in C^1 on a smaller patch. Nesting alone would not imply the latter statement. Lemma 5.1 supplies the higher-order bounds for the whole family.

The harmonic functions used in Lemma 5.2 must be

$$z_\sigma = -\frac{v_\sigma}{u_s} \quad \text{on } E_\sigma, \quad s \leq \sigma \leq t. \quad (5.1)$$

Both numerator and denominator are g_0 -harmonic on that domain, and direct calculation gives

$$\Delta_{g_s} \left(\frac{f}{u_s} \right) = u_s^{-(n+2)/d} \left(\Delta_{g_0} f - \frac{f}{u_s} \Delta_{g_0} u_s \right).$$

Thus z_σ is positive and g_s -harmonic on its connected exterior and has zero trace on its own frontier. The denominator is fixed at time s , not at time σ .

Choose a common corkscrew point A_i in the later exterior, above all intermediate rescaled graphs. Put $a_{i,\sigma} = z_\sigma(A_i) > 0$. The preceding lemma and the uniform family bounds give relative estimates independent of i and σ on the fixed patch. Transfer each normal derivative to the later graph. The error is

$$Ca_{i,\sigma}(\text{height difference} + \text{normal difference}) = o(1)a_{i,\sigma},$$

uniformly in σ . Hence that derivative is between $ca_{i,\sigma}$ and $Ca_{i,\sigma}$. Set

$$\lambda_i = \int_s^t a_{i,\sigma} d\sigma > 0.$$

The normalization is measurable and bounded in time. Since its denominator is fixed, the exact identity is

$$1 - \frac{u_t}{u_s} = \int_s^t z_\sigma d\sigma. \quad (5.2)$$

The relative derivative estimate gives an integrable majorant for exterior difference quotients at the later graph. Dominated convergence therefore permits differentiation there and proves

$$-C\lambda_i \leq \partial_{\nu_t}^{g_s}(u_t/u_s) \leq -c\lambda_i. \quad (5.3)$$

The ratio u_t/u_s is uniformly positive, so the extra positive factor in the conformal mean-curvature formula preserves these bounds. Only exterior traces are differentiated; no derivative across the zero extension of a velocity is taken, and no time derivative at a jump time is required.

The same estimate applies to the other deformation used below: a fixed harmonic conformal factor u , normalized to have boundary value one. Its positive harmonic function is $z = 1 - u$. The strict inequality $u < 1$ on the connected exterior is essential. Lemma 5.2 then transfers its derivative from the original frontier to the nearby candidate graph. All constants may depend on the fixed regular cone patch, finite time interval, and fixed perturbation parameter. No uniform estimate through a cone singularity or as the perturbation parameter vanishes is asserted.

6 Singular separation from a single distance median

The next proposition excludes singular contact for an ordered pair whose regular parts do not meet. The applications below will verify that regular comparison applies. The idea is to rescale

at a possible singular contact and divide every graph separation by the same number: the median distance from the earlier support to the later one. This produces one positive supersolution on the regular part of the common tangent cone. Suitable record scales make its infimum decay to zero near the vertex. After a bounded truncation and removal of the singular set by capacity cutoffs, the minimizing-hypersurface mean-value inequality contradicts that decay.

This follows the singular-separation approach of [6, Proposition 2.36], using the relative forcing, half-volume normalization, record scales, and truncation developed in [5, Proposition 2.34 and Lemma 2.35]. We keep the median independent of the regular-patch exhaustion. That distinction is essential: a positive Jacobi supersolution alone is possible on a singular cone—for example, r^{-2} on the seven-dimensional Simons cone. The contradiction will use the additional small-ball infimum estimate.

Proposition 6.1 (Separation of the ordered supports). *Let $K \subset K'$ be canonical closed filled sets in a smooth n -manifold, $n \geq 8$, with multiplicity-one perimeter supports Σ and Σ' . Put $k = n - 1$. On a neighborhood of their possible contact set, let g_{ref} be a smooth metric and assume that there are $\Lambda < \infty$, $\alpha > 0$, and $r_0 > 0$ such that, for $E = K, K'$,*

$$P_{g_{\text{ref}}}(E; B_r(x)) \leq P_{g_{\text{ref}}}(L; B_r(x)) + \Lambda r^{k+\alpha} \quad \text{if } L \Delta E \Subset B_r(x), \quad 0 < r < r_0.$$

The balls lie in that neighborhood. Assume that every intersection point is singular for both supports.

Let $g = a^{4/(n-2)} g_{\text{ref}}$, with $a > 0$ continuous, be the metric used to measure graph heights. At a possible contact p , choose coordinates with $g(p) = \delta$ and rescale by $\eta_i(y) = p + r_i y$, $r_i \downarrow 0$. The power comparison gives simultaneous minimizing-cone subsequences with local perimeter convergence; the two limiting cones coincide, as shown below. For every such subsequence with common cone C , assume the following on each compact regular patch, after shrinking a larger regular patch:

1. *The rescaled supports converge as single smooth graphs, and extensions of $g_i = r_i^{-2} \eta_i^* g$ converge to δ in C^2 . The graph and metric derivatives needed below have uniform local bounds.*
2. *The later support is the g_i -normal graph of a function $\zeta_i > 0$ over the earlier support, and*

$$L_i \zeta_i = F_i \leq 0, \quad L_i \longrightarrow J_C := \Delta_C + |A_C|^2. \quad (6.1)$$

In fixed regular-patch coordinates, the second-order operators are uniformly elliptic, their coefficients have uniform $C^{0,\theta}$ bounds for some $\theta > 0$, and converge locally to those of J_C .

3. *On a slightly larger patch there are numbers $\lambda_i > 0$ and constants $c, C > 0$, independent of i , such that*

$$-C\lambda_i \leq F_i \leq 0, \quad F_i \leq -c\lambda_i \quad \text{on an inner ball.}$$

The numbers and constants may depend on the fixed patch, and λ_i may tend to zero.

Then $\Sigma \cap \Sigma' = \emptyset$.

Proof. Suppose the supports meet at the origin, and put $k = n - 1 \geq 7$.

A common cone with connected regular part. The power almost-minimizing comparison gives local strict perimeter compactness and almost-monotonicity of density. Hence every simultaneous blow-up has a subsequence converging to multiplicity-one minimizing boundary cones for $g_{\text{ref}}(p)$. They remain nested. Since $g(p)$ is a constant scalar multiple of $g_{\text{ref}}(p)$, both cones are stationary also in the normalized Euclidean metric. Their supports are connected, because radial segments join all points to the vertex; their singular sets have dimension at most $k - 7$. The same-metric strong maximum principle of [36, Theorem 1.1 and Remark (3)] identifies the two supports, since both contain the vertex. Multiplicity one identifies their perimeter measures. This works for every simultaneous blow-up, not only the sequence selected below.

The regular part of the common cone C is connected. To see this, restrict its varifold to a regular component. Its stationarity extends across the singular set by the local area bound and cutoffs with vanishing tangential L^1 gradient, since $\mathcal{H}^{k-1}(\text{Sing } C) = 0$. Each regular component is dilation invariant: the radial dilation orbit is a path in that component. Its closure therefore contains the vertex. Distinct component closures intersect only in the singular set, because a regular point has a connected smooth-sheet neighborhood. This would give intersecting stationary integral varifolds whose intersection has zero $\mathcal{H}^{k-1}$ measure, contrary to [36, Theorem 3.2 and the proof of Theorem 1.1, pp. 8–9]. These are same-metric statements and do not assume the different-metric conclusion being proved.

A median chosen before the regular patches. Choose a fixed smooth coordinate chart normalized so that the earlier working metric at the contact point is Euclidean. Let $D(y) = \text{dist}_{\mathbb{R}^n}(y, \Sigma')$, let μ_r denote the perimeter measure of $\Sigma \cap B_r$, and define the fixed median

$$q(r) = \inf \left\{ a \geq 0 : \mu_r(\{D \leq a\}) \geq \frac{1}{2} \mu_r(B_r) \right\}. \quad (6.2)$$

Atoms are allowed. Both $\{D \leq q(r)\}$ and $\{D \geq q(r)\}$ have at least half the measure. The contact set has zero perimeter measure, since there is no regular contact and the singular sets have zero perimeter measure. Thus $q(r) > 0$ for small positive r . Also $q(r) \leq r$, since the origin belongs to Σ' .

In fact $q(r)/r \rightarrow 0$. If not, choose a contradicting sequence and then a simultaneous minimizing-cone subsequence. Compactness and the density bounds give local Hausdorff convergence of both rescaled supports to the same cone. On B_1 , distance to the second support therefore tends uniformly to zero, contradicting the assumed lower bound for the rescaled medians.

Put $f(r) = q(r)/r$. Given any $R_i \downarrow 0$, choose $0 < r_i \leq R_i$ such that

$$f(r_i) \geq \frac{1}{2} \sup_{0 < s \leq R_i} f(s).$$

The supremum is finite and positive; neither continuity of the median nor attainment of the supremum is required. For every $0 < \rho \leq 1$, this choice gives

$$\frac{q(\rho r_i)}{q(r_i)} \leq 2\rho. \quad (6.3)$$

Pass to a simultaneous cone subsequence at these radii, and put

$$a_i = \frac{q(r_i)}{r_i} > 0, \quad \Sigma_i = r_i^{-1} \Sigma, \quad \Sigma'_i = r_i^{-1} \Sigma'.$$

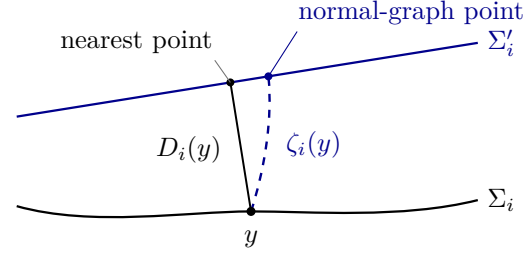
Every graph separation on every patch will be divided by this same a_i .

One median for the whole ball

$$\begin{aligned}
D_i(y) &= \text{dist}_{\mathbb{R}^n}(y, \Sigma'_i) \\
a_i &= \frac{q(r_i)}{r_i} \\
\mu_i(\{D_i \leq a_i\}) &\geq \frac{1}{2}\mu_i(B_1) \\
\mu_i(\{D_i \geq a_i\}) &\geq \frac{1}{2}\mu_i(B_1) \\
&\downarrow \\
W_i &= \frac{\zeta_i}{a_i}
\end{aligned}$$

on every compact regular patch

Two measurements on a regular patch



$$D_i = (1 + o(1))\zeta_i \text{ on compact regular cores}$$

Figure 1: A single distance median and its use on regular patches. The curves are schematic regular cross-sections of the rescaled ordered supports; no graph through the singular set is depicted. The straight segment measures the ambient Euclidean nearest-support distance $D_i(y)$, whereas the dashed segment is the normal exponential segment of length $\zeta_i(y)$ in the rescaled earlier working metric g_i . For $T_i(x) = x/r_i$, the measure $\mu_i = r_i^{-k}(T_i)_\# \mu_{r_i}$ is the rescaling of the same perimeter measure used to define $q(r_i)$, on all of $\Sigma_i \cap B_1$. The number $a_i = q(r_i)/r_i$ is fixed using that whole measure before regular patches are chosen. Both median inequalities allow atoms, and every patch uses the same a_i . The displayed relative comparison is uniform on each compact regular core as $i \rightarrow \infty$.

From ambient distance to normal height. The record scales and the normalizing numbers a_i are now fixed. We next relate their ambient-distance definition to the graph variable in (6.1). On a compact regular core of C , choose a slightly larger regular neighborhood. For large i , Σ'_i is the g_i -normal exponential graph of $\zeta_i > 0$ over Σ_i , where g_i is the rescaled earlier working metric. Their relative slopes tend to zero. For a point in the smaller core, its nearest point on Σ'_i lies in the larger neighborhood: the local normal candidate has distance tending to zero, while the complement of that neighborhood stays a fixed positive distance away. The Lipschitz-graph distance estimate in coordinates flattening Σ_i gives

$$\text{dist}_{\mathbb{R}^n}(y, \Sigma'_i) = (1 + o(1))\zeta_i(y) \quad (6.4)$$

uniformly on the smaller core. The normal exponential segment has Euclidean length $(1 + o(1))\zeta_i$ and direction approaching the Euclidean normal, because the contact metric was normalized and the rescaled metrics converge there in C^1 . Thus the same relative Lipschitz-graph estimate applies to its endpoint. This remains a relative estimate even when ζ_i is much smaller than the blow-up error; an additive comparison with the cone would not suffice. Figure 1 distinguishes these two measurements and the use of one median on all regular patches.

Choose a compact regular core inside $C \cap B_1$ whose omitted area is less than one eighth of its total area, and enlarge it slightly while keeping away from the singular set and sphere. Such cores exist because those two sets have zero k -measure on the cone. Graphical perimeter convergence makes the omitted area less than one quarter for large i . Each median set therefore retains positive measure on this fixed core.

Local bounds with the same normalization on every patch. Set $W_i = \zeta_i/a_i$. The lower median set and (6.4) provide a point with $W_i \leq 2$ in one of finitely many regular core patches. Choose that patch along a subsequence. Weak Harnack for the nonnegative supersolution

$L_i W_i \leq 0$ gives a uniform local L^p bound for some $p > 0$, before any bound for F_i/a_i has been proved. To handle the zero-order term, write

$$L_i = A_i^{ab} \partial_{ab} + B_i^a \partial_a + c_i$$

and replace c_i by $\min\{c_i, 0\}$, obtaining $\widehat{L}_i$. Since $W_i \geq 0$, we still have $\widehat{L}_i W_i \leq 0$. The usual weak Harnack inequality with nonpositive zero-order coefficient applies with uniform constants on every fixed patch.

An L^p bound on an overlap of fixed positive measure supplies a bounded-value point for the next patch. Weak Harnack there propagates the bound. Connectedness of $\text{Reg } C$ gives finite chains from the anchor patch to every fixed compact regular set. Consequently

$$\|W_i\|_{L^p(K)} \leq C_K \quad \text{for every } K \Subset \text{Reg } C. \quad (6.5)$$

The constants may grow with K . No uniform Harnack chain through the singular set has been asserted.

Controlling the forcing before passing to a limit. The local L^p bounds have been obtained without dividing a right-side estimate by the possibly small number a_i . They now let us control that quotient. In a ball where $F_i \leq -c\lambda_i$, solve

$$\widehat{L}_i \phi_i = -1, \quad \phi_i|_{\partial B} = 0.$$

Uniform ellipticity, coefficient bounds, and the maximum principle give $\phi_i \geq c_0 > 0$ on a smaller ball. Since $\widehat{L}_i \zeta_i \leq F_i$ and $\zeta_i \geq 0$ on the boundary, comparison gives

$$\zeta_i \geq c\lambda_i \phi_i.$$

Combining this with (6.5) yields $\lambda_i/a_i \leq C_K$. The assumed upper bound for $|F_i|$ and a finite cover now give

$$\|F_i/a_i\|_{L^\infty(K)} \leq C_K. \quad (6.6)$$

The order is important: first weak Harnack, then the positive Dirichlet comparison, and only then bounded-right-side elliptic estimates.

The weak Harnack exponent need not exceed one. The local boundedness estimate for the now bounded-right-side equation bounds the supremum on an inner ball by the L^2 norm on a larger ball and the right-side bound. Interpolate the L^2 norm between the bounded L^p norm and the larger-ball supremum, apply Young's inequality, and iterate on nested radii to absorb that supremum. For each i the preliminary finite suprema exist by local regularity. This gives uniform local L^∞ bounds; interior elliptic estimates then give uniform $W^{2,q}$ bounds for every finite q . One countable exhaustion and one diagonal subsequence give a single function W , with local C^1 convergence on all of $\text{Reg } C$, satisfying

$$W \geq 0, \quad W \in W_{\text{loc}}^{2,q}(\text{Reg } C) \quad (q < \infty), \quad (\Delta_C + |A_C|^2)W \leq 0. \quad (6.7)$$

The upper median set also retains positive measure on the fixed regular core. Equation (6.4) gives points there with $W_i \geq 1/2$. Local uniform convergence makes W nonzero. Since $\Delta_C W \leq -|A_C|^2 W \leq 0$, the strong maximum principle and connectedness make $W > 0$ throughout the regular part. No separately normalized patch limits are used.

The infimum near the vertex. We have obtained one positive limit on the entire regular part. The record-scale choice now gives information on its behavior near the vertex without changing its normalization or taking new patchwise limits.

Fix $0 < \rho \leq 1$. The lower median set in $\Sigma_i \cap B_\rho$ has half its area and, by (6.3), normalized ambient distance at most 2ρ . Choose a compact regular core in $C \cap B_\rho$ omitting less than one quarter of its area. Perimeter convergence and (6.4) give a point in this core with $W_i \leq 3\rho$ for large i . The already chosen local uniform convergence applies to this core. Thus the same limit satisfies

$$\inf_{\text{Reg } C \cap B_\rho} W \leq 3\rho. \quad (6.8)$$

It is enough to use a countable sequence of $\rho \downarrow 0$. The constant 3 does not depend on the core or on ρ : for each fixed radius the relative error in (6.4) tends to zero before this limit is taken. Therefore the infimum of W in every ball about the vertex is zero. Its essential infimum there is also zero, since every small regular value has a neighborhood of positive perimeter measure with comparably small values.

A bounded supersolution across the singular set. To apply a mean-value inequality on the whole minimizing cone, we need a Sobolev supersolution across its singular set. First discard the nonnegative Jacobi potential, obtaining $\Delta_C W \leq 0$, and then set $h = \min\{W, 1\}$. Approximation by increasing concave truncations shows that h is a bounded positive weak *Laplace* supersolution on the regular part. It need not be a Jacobi supersolution. For a regular-part cutoff η , use the nonnegative test $\eta^2(1 - h)$ to obtain

$$\int \eta^2 |\nabla_C h|^2 \leq 2 \int \eta(1 - h) |\nabla_C h| |\nabla_C \eta|,$$

and hence

$$\int \eta^2 |\nabla_C h|^2 \leq 4 \int |\nabla_C \eta|^2. \quad (6.9)$$

The singular set has zero tangential two-capacity on compact regions. Indeed, cover the relevant compact singular set by finitely many balls $B_{\rho_j}(x_j)$, with centers on that set, maximum radius tending to zero, and $\sum_j \rho_j^{k-2} \rightarrow 0$. This is possible because its dimension is at most $k - 7$. Let θ_j vanish on the ball, equal one off its double, and satisfy $|D\theta_j| \leq C/\rho_j$. Put $\chi = \min_j \theta_j$. Almost everywhere its gradient is one of the active gradients, so local area growth gives

$$\int_C |\nabla_C \chi|^2 \leq \sum_j \int_{C \cap B_{2\rho_j}(x_j)} |\nabla_C \theta_j|^2 \leq C \sum_j \rho_j^{k-2} \rightarrow 0. \quad (6.10)$$

There is no overlap-count factor. These cutoffs vanish near the whole relevant singular set and converge to one on the regular part.

Use $\eta = \psi\chi$ in (6.9), where ψ is a fixed compact spatial cutoff. Fatou's lemma gives locally finite Dirichlet energy for h across the singular set. Moreover, $\psi h \chi \rightarrow \psi h$ in the local surface $W^{1,2}$ norm: the term $h \nabla_C \chi$ tends to zero by boundedness and (6.10), while the omitted $\nabla_C h$ term tends to zero by absolute continuity of its now finite energy. Thus h genuinely extends as a local Sobolev function on the whole minimizing boundary; its values on the measure-zero singular set are immaterial.

For a nonnegative ambient smooth compactly supported test function φ , insert $\varphi\chi$ into the regular-part supersolution inequality. The cutoff error is bounded by

$$\|\varphi\|_\infty \|\nabla_C h\|_{L^2(\text{supp } \varphi)} \|\nabla_C \chi\|_2 \rightarrow 0.$$

The other term converges by dominated convergence. Hence

$$\int_C \langle \nabla_C h, \nabla_C \varphi \rangle \geq 0$$

for tests meeting the singular set as well.

The mean-value contradiction. The bounded function h is now a positive weak Laplace supersolution on the whole minimizing boundary, with locally finite energy and zero essential infimum in every ball about the vertex. These are the conditions needed for the minimizing-boundary weak Harnack theorem of Bombieri–Giusti [7]; the precise form needed here is stated in [12, Theorem 3.1, p. 11]. For an area-minimizing hypersurface in B_R and a positive weak Laplace supersolution, a normalized L^p mean on B_r is bounded by its infimum on B_r when $r \leq \beta R$ and $0 < p < k/(k-2)$. The constants depend only on dimension and p . This is an extrinsic estimate on the minimizing boundary, not an assertion that regular-patch Harnack chains have uniform constants through cone singularities.

Apply it to $h + \epsilon$ on the cone, with fixed concentric balls and, for example, $p = 1$. Choose the smaller ball to contain a regular open subset on which $h > 0$; the cone and the same h are defined on any required larger ball. Its conclusion is

$$\frac{1}{\mathcal{H}^k(C \cap B_r)} \int_{C \cap B_r} (h + \epsilon) \leq C \operatorname{ess\,inf}_{C \cap B_r} (h + \epsilon).$$

The essential infimum of h is zero by (6.8). Sending $\epsilon \downarrow 0$ would make the positive integral on the left zero, a contradiction. This proves separation. No growth selection at infinity or limiting truncation level is needed. $\square$

6.1 The graph equation for the two deformations

It remains to verify the separation proposition for the flow and for the prescribed-curvature enclosure. Both use the same graph equation; the difference is a small zero-order term in the latter problem. We derive the equation using the end-pointing normal throughout.

Normalize the earlier working metric g at the contact point to be Euclidean, write g_i for its rescaling at radius r_i , and use g_i -unit normals and g_i -normal exponential graphs throughout. For this calculation assume uniform $C^{3,\alpha}$ bounds for the extended metrics and graphs on a larger regular cone patch. Lemma 5.1 supplies them for flow slices; the candidate construction below will supply them in the second application. Continuity of the original metric gives uniform convergence to its contact-point value. Apply bounded graph-domain extension operators to its difference from that constant metric; these preserve uniform convergence. Interpolation then gives $g_i \rightarrow \delta$ in C^2 on a smaller patch. The graph bounds similarly give the C^2 convergence needed below. A mere $C^{1,\alpha}$ metric bound would not justify convergence of the Ricci term.

The linearization of $H = \operatorname{div} \nu$ at the earlier minimal graph is $-J_i$, where

$$J_i = \Delta_{\Sigma_i, g_i} + |A_i|_{g_i}^2 + \operatorname{Ric}_{g_i}(\nu_i, \nu_i).$$

Integrating the derivative of the mean-curvature graph operator from height zero to ζ_i writes its nonlinear remainder as

$$Q_i \zeta_i = Q_i^{ab} \partial_{ab} \zeta_i + Q_i^a \partial_a \zeta_i + Q_i^0 \zeta_i,$$

with coefficients tending locally to zero. Let the later metric be $g'_i = U_i^{4/d} g_i$, and let its prescribed end-oriented curvature be $f_i(y) = r_i f(\eta_i(y))$, where $\eta_i(y) = p + r_i y$. The conformal formula gives the exact equation

$$(J_i - Q_i)\zeta_i = F_i - U_i^{2/d} f_i, \quad F_i = \frac{2k}{d} U_i^{-1} \partial_{\nu'_i}^{g_i} U_i. \quad (6.11)$$

For flow slices $f = 0$ and U_i is the rescaled ratio u_t/u_s . Equations (5.1)–(5.3) give the required two-sided relative bound for F_i .

6.2 Separation of flow slices

Fix $0 \leq s < t$. We prove that the two closed supports are disjoint. Suppose $p \in \Sigma_s \cap \Sigma_t$ is regular for at least one frontier. The simultaneous regular-neighborhood lemma makes both graphs regular there. Nesting puts p on every intermediate frontier: an interior point of an intermediate conductor would be interior to K_t . These ordered graphs have a common tangent plane and the same end-oriented normal. Their velocities vanish at p , so $u_t(p) = u_s(p)$.

For each $\sigma \in [s, t]$, the function $z_\sigma = -v_\sigma/u_s$ is positive and g_s -harmonic on its own exterior E_σ , vanishes at p , and has strictly positive derivative in the common g_s -unit normal direction. Uniform boundary C^1 bounds and the positive lower bound for u_s give an integrable bound for these measurable derivatives. The uniform C^2 common-tangent graph charts provide a single short outward normal segment in E_t , hence in every E_σ ; all exterior difference quotients can be taken there. Thus, with $q = u_t/u_s$, Equation (5.2) gives

$$\partial_\nu q(p) = - \int_s^t \partial_\nu z_\sigma(p) d\sigma < 0.$$

Extend the later metric from its exterior as a C^1 metric. Its first jet at p is determined by that one-sided metric, and the earlier metric has the matching one-sided jet. The conformal mean-curvature formula at this point gives

$$H_{\Sigma_s}^{g_t}(p) = q(p)^{-2/d} \left(H_{\Sigma_s}^{g_s}(p) + \frac{2k}{d} q(p)^{-1} \partial_\nu q(p) \right) < 0.$$

The earlier curvature is zero, including at $s = 0$. The later graph has zero curvature in the same extension metric. Since $K_s \subset K_t$, the tangent graph comparison gives $H_{\Sigma_t}^{g_t}(p) \leq H_{\Sigma_s}^{g_t}(p) < 0$, a contradiction. Only the common exterior first jets were used; the ratio need not be constant on the later frontier or harmonic throughout E_s . Thus regular contact is impossible. At any remaining contact, the power almost-minimizing estimates and density bounds give the common minimizing-cone limits used in Proposition 6.1. The regular-patch bootstrap, relative harmonic derivative estimate, and graph equation verify its remaining hypotheses. The proposition excludes singular contact as well, proving $\Sigma_s \cap \Sigma_t = \emptyset$.

6.3 The prescribed-curvature enclosure before detachment

We now build the enclosing frontier needed for the mass–capacity comparison. The construction applies both to a flow slice and to the original smooth ambient metric in the approximation argument. In either case, the initial frontier is locally perimeter minimizing; its enclosing candidate must be understood before detachment is known.

Let K be a compact canonical closed filled set with nonempty frontier Σ and connected open end exterior E . Work in either of the following settings:

1. $K = K_s$ is a flow conductor and $h = b^2 g_s$ for a fixed constant $b > 0$;
2. h is smooth through Σ and on E , and K is locally perimeter minimizing in h on a neighborhood of Σ . The exterior has one coordinate end and compact complement, with $h_{ab} = a_0^2 \delta_{ab} + O_2(r^{-\tau_0})$ for constants $a_0, \tau_0 > 0$.

In the second case extend h smoothly a short distance behind the frontier and, if needed, use the buried-boundary construction already described. No outermost property of K is required. In both settings K is locally minimizing in its continuous metric h : for a flow slice this is Lemma 4.1, and in the smooth setting it is the stated hypothesis. The density and minimizing cone conclusions follow from the same local comparison.

Let u be a positive h -harmonic function on E , continuous up to Σ , with $u = 1$ there, $u < 1$ in E , and $u \rightarrow \ell \in (0, 1)$ at infinity. Extend it by one on K and set $h' = u^{4/d} h$. Assume on the coordinate end that $h'_{ab} = a^2 \delta_{ab} + O_2(r^{-\tau})$, where $a, \tau > 0$. This differentiated decay will confine the candidate inside a fixed coordinate sphere.

The extended perturbation has the Hölder modulus needed for the weighted perimeter comparison. To see this directly in the flow case, the conformal Laplacian identity makes

$$z = u_s(1 - u)$$

a bounded positive g_0 -harmonic function on E_s , continuous with zero frontier value and end value $c = e^{-s}(1 - \ell) > 0$. Let q_{K_s} be the escape potential for g_0 , constructed in the proof of Lemma 4.1. Bounded continuous Dirichlet uniqueness gives $z = cq_{K_s}$. The already proved Hölder modulus of q_{K_s} and the positive Hölder factor u_s therefore give

$$u = 1 - \frac{cq_{K_s}}{u_s}$$

with a Hölder extension across Σ . For a smooth initial metric, use the same background capacity construction with h in place of g_0 . Local minimizing density gives its boundary contraction estimate, and uniqueness gives $1 - u = (1 - \ell)q_K$. Thus h' is a positive Hölder metric in both cases, before the candidate is constructed.

Choose a smooth ambient function h_0 that is zero on K , positive on E , and integrable in h' . A locally finite partition of unity in E , with coefficients decreasing sufficiently rapidly, makes its zero extension smooth and its integral finite. Its first derivative vanishes along Σ ; in particular, on compact sets, $|h_0(x)| \leq C \operatorname{dist}(x, \Sigma)$.

For $\delta > 0$ chosen below, the candidate K' minimizes

$$\mathcal{F}(L) = P_{h'}(L) + \delta \int_{L \setminus K} h_0 dV_{h'}$$

among all compact finite-perimeter supersets of K in the ambient manifold. The added region $L \setminus K$ need not stay a positive distance from the original frontier. The parameters, forcing, and its derivatives are fixed here. To fix the sign, a normal variation $\varphi\nu$ of the candidate frontier has first variation

$$D\mathcal{F}[\varphi\nu] = \int_{\partial K'} (H_\nu + \delta h_0) \varphi dA_{h'}.$$

Its prescribed end-oriented curvature is therefore $H = -\delta h_0$. The function h_0 vanishes on the original filled side; its value on the candidate frontier has not been prescribed to be zero.

Existence, compact support, and a connected end exterior. On the asymptotically flat end, h' is smooth and approaches a positive constant multiple of the Euclidean metric, with the corresponding differentiated decay. Choose R_* enclosing K so that every coordinate sphere beyond R_* has strictly positive outward mean curvature in h' . This choice precedes that of δ . First minimize $\mathcal{F}$ over all finite-perimeter supersets of K in a coordinate truncation B_Q , including the compact core, with $Q > R_*$. The competitor K gives a perimeter bound. BV compactness, lower semicontinuity for the positive continuous perimeter weight, and L^1 convergence of the bounded volume density on B_Q give a minimizer. If X is the outward h' -unit normal to the coordinate spheres, the divergence formula gives, for almost every $r > R_*$,

$$P_{h'}(L \cap B_r) \leq P_{h'}(L) - \int_{L \setminus B_r} \operatorname{div}_{h'} X \, dV_{h'}.$$

The divergence is positive. Clipping also decreases the nonnegative volume term, strictly decreasing $\mathcal{F}$ whenever the removed tail has positive volume. The minimizer is therefore supported in B_{R_*} , independently of Q and δ . Clipping an arbitrary compact competitor into this same ball proves global minimality, with no artificial outer constraint remaining.

To ensure a filled enclosure, choose $\delta > 0$ sufficiently small after fixing h' and R_* . Compact-support isoperimetry gives

$$\operatorname{Vol}_{h'}(U)^{k/n} \leq C_{\text{iso}} P_{h'}(U)$$

for sets $U \subset B_{R_*}$ disjoint from the buried collar. Such sets have zero trace on a slightly larger smooth truncation, so this is the absolute inequality. Require

$$\delta \|h_0\|_{L^\infty(B_{R_*})} C_{\text{iso}} \operatorname{Vol}_{h'}(B_{R_*})^{1/n} < 1.$$

In a larger truncation B_{Q_2} , decompose $B_{Q_2} \setminus L$ into its finite-perimeter components. Exactly one contains the connected outer annulus; let U be the union of the others. Perimeter additivity, with cancellation of the outer sphere, shows that filling U removes its perimeter. Consequently

$$\mathcal{F}(L \cup U) - \mathcal{F}(L) = -P_{h'}(U) + \delta \int_U h_0 \, dV_{h'} < 0$$

unless U has zero volume. This contradicts minimality. The selected minimizer is thus filled and has connected end exterior, while retaining minimality against all finite-perimeter enclosing competitors. No positive lower bound on δ is needed.

Local variations across the original frontier. The existence argument has produced a global enclosing minimizer. To obtain its graph equation near possible contact, we remove the enclosure constraint for local variations. Let $L \triangle K'$ be compactly supported in a sufficiently small variation ball. Since $h' = h$ on K , the local minimizing property of K gives

$$P_{h'}(L \cap K) \geq P_{h'}(K), \quad P_{h'}(L \cup K) \leq P_{h'}(L),$$

where the second inequality is perimeter submodularity and all comparisons are localized to that ball. This argument uses only local minimality of the given K ; it does not replace it by an outermost hull. The volume term is modular under union and intersection, and its density is zero on K . In particular its value on $L \cup K$ equals its value on L . Constrained minimality of K' against $L \cup K$ therefore gives $\mathcal{F}(K') \leq \mathcal{F}(L)$. This proves local unconstrained minimality. Its comparison with the smooth reference metric yields the uniform almost-minimizing estimate from

the Hölder weight, with an additional error at most Cr^n from the bounded volume forcing. The canonical density representative removes null cracks; for these almost-minimizing sets, absence of bounded complementary BV components therefore gives the connected open end exterior asserted above.

Uniform graph estimates before separation. At a hypothetical common tangent-cone patch, the local comparison just proved and one-sided small-excess regularity give uniform $C^{1,\theta}$ candidate graphs. For a flow conductor, the original graph and its one-sided metric have the all-order bounds in Lemma 5.1. In the smooth initial setting, the rescaled smooth metric has uniform bounds and the minimal graph equation supplies the same regular-patch estimates. Harmonic boundary estimates for u , whose original graphical boundary value is constant, give all fixed finite orders on smaller patches. The metric h' consequently has controlled extensions from the original exterior.

For the first graph equation, extend $1 - u$ oddly through the original regular graph. It is negative on the filled side, so the corresponding $C^{1,\alpha}$ extension of u is at least one there and dominates its original extension. In the smooth initial setting multiply this factor into the smooth metric h . In the flow setting, first use the dominating extension of b^2g_s from Lemma 5.1, and multiply the two extended conformal factors. After shrinking the patch the resulting metric is positive, dominates h' , and agrees with it on the original exterior, hence on the candidate graph. The forcing density is unchanged where h_0 is nonzero, because that region lies in the original exterior; on K it is zero in both metrics. Thus K' remains a local minimizer of the same perimeter-plus-volume functional in this regular extension metric. Its weak Euler equation is exactly

$$H_\nu^{h'} = -\delta h_0.$$

The uniformly elliptic prescribed-mean-curvature graph equation improves the candidate to $C^{2,\alpha}$. The original metric and harmonic perturbation have the higher-order coefficient extensions already established, so elliptic bootstrapping gives every fixed higher-order candidate estimate on smaller patches. After the first equation is established, these extensions need only match the one-sided boundary jets; domination is no longer required. Under a radius- r_i blow-up the forcing is $r_i(-\delta h_0)(r_i y)$, with uniform local coefficient bounds of every fixed order. The candidate's estimates therefore precede and do not assume singular separation.

These estimates already exclude contact at a regular point. At contact with the original graph, $h_0 = 0$, whereas the strict harmonic conformal derivative makes the original graph have strictly negative end-oriented mean curvature in h' . The candidate equation has zero curvature there. Ordered graph comparison forbids contact. If only one frontier was initially known to be regular there, the one-sided regular-neighborhood argument first makes the other regular. This excludes regular contact only; singular contact is the separate issue addressed next.

The candidate graph equation and singular detachment. Apply Equation (6.11) with earlier metric $g = h$, later metric $h' = u^{4/d}h$, and prescribed curvature $f = -\delta h_0$. Since f is smooth and zero on K , it satisfies $|f(x)| \leq C \operatorname{dist}(x, \Sigma)$. The original distance from a point at rescaled graph height ζ_i to Σ is at most $Cr_i\zeta_i$, and therefore

$$|U_i^{2/d}f_i| \leq Cr_i^2\zeta_i.$$

Put $b_i = U_i^{2/d}f_i/\zeta_i$ and absorb this term into the operator:

$$L_i = J_i - Q_i + b_i, \quad |b_i| \leq Cr_i^2, \quad L_i\zeta_i = F_i \leq 0.$$

The coefficient also has a uniform Hölder bound. Indeed, f vanishes on the earlier graph, so the fundamental theorem of calculus along the normal graph segment expresses f_i/ζ_i as r_i^2 times an average of first derivatives of f . The smooth forcing and the uniform graph and metric bounds control that average in the fixed regular-patch coordinates. Thus $b_i \rightarrow 0$ and $L_i \rightarrow J_C$ with the coefficient control required by Proposition 6.1.

The sign of f is not needed in this absorption. The strictly negative right side instead comes from the harmonic conformal factor: Lemma 5.2, applied to $1 - u > 0$, gives the patchwise two-sided relative bounds for F_i . All parameters have remained fixed during this argument.

Corollary 6.2 (Detachment of slices and perturbed enclosures). *For any fixed $0 \leq s < t$, the conformal-flow frontiers Σ_s and Σ_t are disjoint as closed perimeter supports. For either the flow or smooth initial exterior in Section 6.3, let u satisfy the harmonic, boundary, strictness, and end hypotheses stated there. For the forcing and sufficiently small fixed $\delta > 0$ constructed there, the perimeter-plus-volume minimizer is disjoint from the original frontier, including its singular set. Each fixed pair of these compact supports has positive distance.*

Proof. The flow assertion was proved in Section 6.2. For the perturbed enclosure, the preceding regular comparison excludes regular contact. The power almost-minimizing estimate, the regular-patch metric and graph bounds, and the displayed forcing equation verify Proposition 6.1, which excludes the remaining singular contact. Compactness of each pair of supports then gives positive distance. The constants and the resulting distance may depend on the fixed time pair or perturbation parameter. $\square$

7 Capacity and one-sided smoothing

The mass decrease in conformal flow is controlled by the difference between ADM mass and capacity. To show that this difference is nonnegative, we first move a possibly singular frontier toward the asymptotic end and smooth it with a definite mean-curvature sign. The capacity does not decrease under this operation. We can then apply a smooth mass-capacity inequality, proved below from positive mass.

Throughout this section $n \geq 3$, $\omega = |S^{n-1}|$, and ν denotes the normal from a compact removed region toward the asymptotically flat end. For a frontier Σ we use $H_\nu = \operatorname{div}_\Sigma \nu$. For a smooth compact conductor with exterior (E, h) , normalize capacity by

$$\mathfrak{c}(\Sigma, h) = \frac{1}{(n-2)\omega} \int_E |dw|_h^2 dV_h, \quad \Delta_h w = 0, \quad w|_\Sigma = 1, \quad w \rightarrow 0. \quad (7.1)$$

Equivalently, this is the normalized infimum of Dirichlet energy among smooth compactly supported trial functions equal to one in a neighborhood of the conductor. The same variational definition applies to a nonsmooth compact conductor. If one conductor contains another, its admissible class is smaller, so its capacity is no smaller.

7.1 A harmonic perturbation and its detached enclosure

We first construct a boundary with a strict curvature sign; the next subsection will smooth it. Let (E, g) be a flow-slice exterior or a smooth-ambient initial exterior as in Theorem 1.1, and assume its frontier is nonempty. Let w be its capacitary potential and put $v = w - 1$. The strong

maximum principle gives $-1 < v < 0$ in the open exterior, while $v = 0$ on the frontier. For $0 < \varepsilon < 1$ set

$$u_\varepsilon = \frac{1 + \varepsilon v}{1 - \varepsilon}, \quad \tilde{g}_\varepsilon = u_\varepsilon^{4/(n-2)} g.$$

Harmonicity gives $R_{\tilde{g}_\varepsilon} = u_\varepsilon^{-4/(n-2)} R_g \geq 0$ on the exterior. The perturbing factor tends to one at infinity, preserving the end normalization of g . To apply the detachment corollary, whose harmonic factor has boundary value one, express the same metric as

$$g_\varepsilon^{\text{base}} = (1 - \varepsilon)^{-4/(n-2)} g, \quad \hat{u}_\varepsilon = 1 + \varepsilon v, \quad \tilde{g}_\varepsilon = \hat{u}_\varepsilon^{4/(n-2)} g_\varepsilon^{\text{base}}.$$

Then $\hat{u}_\varepsilon = 1$ on the frontier and $1 - \varepsilon < \hat{u}_\varepsilon < 1$ in E . If $g_\varepsilon^{\text{base}} = \kappa^2 g$, then

$$\Delta_{g_\varepsilon^{\text{base}}} = \kappa^{-2} \Delta_g, \quad \nu_{g_\varepsilon^{\text{base}}} = \kappa^{-1} \nu_g, \quad H^{g_\varepsilon^{\text{base}}} = \kappa^{-1} H^g.$$

Constant scaling preserves harmonicity, minimality and the local almost-minimizing hypotheses, with changed constants. It also preserves the signs and relative bounds of normal derivatives. At regular boundary points the Hopf derivative of $\hat{u}_\varepsilon$ in the direction ν is strictly negative, so the conformal mean-curvature formula makes the original frontier a strict inner barrier in $\tilde{g}_\varepsilon$.

Apply the prescribed-curvature construction of Section 6.3 to this fixed perturbation, using its smooth forcing h_0 , zero on the original filled side and positive in its exterior. With the sufficiently small fixed volume parameter $\delta > 0$, Corollary 6.2 separates the candidate from the full original frontier. Denote its connected end exterior by E' . In the perturbed metric $\tilde{g}_\varepsilon$, a variation of its filled side by $\varphi\nu$ has first variation

$$D\mathcal{F}[\varphi\nu] = \int_{\partial E'} (H_\nu + \delta h_0) \varphi dA, \quad (7.2)$$

and therefore its regular frontier satisfies

$$H_\nu = -\delta h_0 \leq -\gamma < 0. \quad (7.3)$$

Here $\gamma > 0$ exists because the entire detached frontier is a compact subset of the original exterior, where $h_0 > 0$. The perturbed metric is smooth on its neighborhood. All perturbation and forcing parameters remain fixed through separation and smoothing; no curvature or separation bound uniform in ε is needed.

Strictness matters: if $u \equiv 1$, the forcing is zero, and the original frontier is outer minimizing and minimal, the original exterior remains a minimizing competitor. Separation cannot follow from $u \leq 1$ alone.

Remark 7.1 (Orientation and the literature). On the Euclidean exterior $r \geq a$, our convention gives

$$H_\nu = (n-1)/a > 0, \quad m_{\text{ADM}} = 0, \quad w = (a/r)^{n-2}, \quad \mathfrak{c} = a^{n-2} > 0.$$

The smooth mass-capacity inequality therefore requires the nonpositive sign with this normal. This example distinguishes the statement here from the positive- H_ν inequality in [6, Lemma 3.11], which is version 1. Version 2 uses the outward normal of the exterior in its Lemma 3.10, hence the opposite normal at the inner frontier. Its Corollary 2.37 includes the strict harmonic-factor hypothesis, and its Lemma 3.7 includes the base-metric normalization [5].

7.2 Smooth frontiers with the required sign

The distance-offset construction in [4, Section 3 and Proposition 2.6], arXiv:2605.12403v2, supplies the geometric strategy for the next proposition. The conclusion needed for capacity is conductor inclusion and a mean-curvature bound. We prove precisely this conclusion using Rifford's minimum-family theorem locally and a final smoothing of transverse graphs. Boundary-area convergence is not needed.

Proposition 7.2 (One-sided smoothing in a shrinking collar). *Let Ω be an open subset of a smooth Riemannian manifold with compact full boundary Σ . Assume that the metric is smooth on an ambient neighborhood of Σ . Let $\Lambda \subset \Sigma$ consist of the points touched by a geodesic ball contained in Ω . Suppose that Λ is a smooth embedded hypersurface, locally bounds Ω on its accessible side, and has mean curvature at least $\gamma > 0$ for the normal pointing out of Ω .*

For every $\varepsilon > 0$, there is an open set $\Omega_\varepsilon \subset \Omega$ with smooth compact boundary such that

$$\Omega \setminus \Omega_\varepsilon \subset \{x \in \Omega : \text{dist}(x, \Sigma) < \varepsilon\}, \quad H_{\text{out}, \Omega_\varepsilon} \geq \gamma - \varepsilon.$$

Proof. If Σ is empty, take $\Omega_\varepsilon = \Omega$. Otherwise we first choose a regular distance level, then take a short offset of its superlevel, and finally smooth the resulting $C^{1,1}$ hypersurface. The offset controls curvature; the final smoothing loses an arbitrarily small amount of that bound.

Write $\rho(x) = \text{dist}(x, \Sigma)$ on Ω . All distances and geodesics below are taken in the ambient metric. Choose a relatively compact ambient neighborhood of Σ and a smaller compact safety neighborhood. Fix $r_* > 0$ much smaller than the distance separating their boundaries and the injectivity radii on the larger compact neighborhood. Short paths used in the proof then stay in the region of controlled smooth geometry, and the squared-distance functions used for pairs less than $4r_*$ apart are smooth. No completeness of an auxiliary ambient extension is needed.

Regular distance levels. If $0 < \rho(x) < r_*$, every nearest point of Σ to x belongs to Λ . Indeed, a shortest segment from x to the full boundary stays in Ω until its endpoint, and a shorter terminal subsegment supplies an interior tangent ball there. The nearest-point set is compact. It therefore lies in the interior of a compact smooth hypersurface patch $Q \subset \Lambda$ with smooth boundary. Choose Q sufficiently close to that set that $\text{dist}(y, q)^2$ is smooth for $q \in Q$ and y near x .

Because Λ is embedded, there is an ambient open set $\mathcal{V}$ containing the nearest-point set with $\mathcal{V} \cap \Lambda \subset \text{int}_\Lambda Q$. After shrinking the neighborhood of x , every nearest point to each such y lies in $\mathcal{V}$. Otherwise, compactness of $\Sigma \setminus \mathcal{V}$ would give a nearest point of x outside $\mathcal{V}$. These nearby nearest points belong to Λ by the preceding tangent-ball argument, and hence lie in the interior of Q . Thus

$$\rho(y)^2 = \min_{q \in Q} \text{dist}(y, q)^2$$

on an open neighborhood of x . Apply [32, Theorem 3 and the compact-parameter-with-boundary extension in the proof of Theorem 1]. A countable family of these neighborhoods covers the small distance collar, so the critical values of ρ^2 form a null set. On each compact interval of positive distances, the square-root map is Lipschitz and $\partial_C(\rho^2) = 2\rho \partial_C \rho$, where ∂_C denotes the Clarke differential. Thus the positive critical values of ρ also form a null set. Choose $s_i \downarrow 0$ such that

$$0 \notin \partial_C \rho(x) \quad \text{whenever } \rho(x) = s_i.$$

The compact parameter patch is the reason this theorem applies even though the accessible locus Λ need not be closed.

Positive reach of the superlevels. The distance function is locally semiconcave away from Σ . To obtain smooth upper supports, take a minimizing segment from Σ to a point of the collar and retain a short terminal piece. Distance from its earlier endpoint, plus the length of the preceding piece, is a smooth upper support near the terminal point. The local Hessian comparison estimate gives a uniform upper Hessian bound on a sufficiently small fixed ball. Consequently, in a geodesic coordinate ball,

$$\rho(y) \leq \rho(x) + \langle p, \log_x y \rangle + C \operatorname{dist}(x, y)^2 \quad (p \in \partial_C \rho(x)). \quad (7.4)$$

Here the superdifferential of the semiconcave function agrees with its Clarke differential.

We include the regular-superlevel implication underlying the positive-reach criterion associated with [2]. At a point of a regular level, let $\mathcal{K} = \partial_C \rho(x)$. This is a compact convex set not containing zero. Separation gives a direction v_0 such that $\langle p, v_0 \rangle > 0$ for every $p \in \mathcal{K}$. The directional derivative of a semiconcave function is $\min_{p \in \mathcal{K}} \langle p, v \rangle$. Every direction for which this minimum is positive therefore enters the superlevel. Separation of convex cones implies that the polar of its contingent tangent cone, and in particular each proximal outward normal, is contained in $-\operatorname{cone}(\mathcal{K})$. Since $\mathcal{K}$ is convex, each unit proximal outward normal can be written $\mathbf{n} = -p/|p|$ with $p \in \mathcal{K}$.

Compactness of the level and upper semicontinuity of the superdifferential give a uniform lower bound $|p| \geq \eta > 0$ there. For a nearby point y of the superlevel, Equation (7.4) yields

$$\langle \mathbf{n}, \log_x y \rangle \leq \frac{C}{\eta} \operatorname{dist}(x, y)^2.$$

This gives uniqueness of sufficiently nearby nearest-point projection. Indeed, suppose that z had two distinct nearest points x, y , at the same distance a . In a sufficiently small bounded-geometry ball, the Taylor expansion of squared distance from z gives

$$0 = \operatorname{dist}(z, y)^2 - \operatorname{dist}(z, x)^2 \geq -2a \langle \mathbf{n}, \log_x y \rangle + \frac{1}{2} \operatorname{dist}(x, y)^2.$$

For $a < \eta/(4C)$ this contradicts the preceding bound. Enlarging the constants slightly covers the curvature terms. Compactness excludes distant distinct nearest points when a is small and makes this projection radius uniform. Thus

$$F_i = \{x \in \Omega : \rho(x) \geq s_i\}$$

has positive reach. It is closed in the ambient manifold because Σ is the full boundary and $s_i > 0$, and its boundary is compact. The same separating direction v_0 shows that its outward normal cone is pointed: it cannot contain opposite nonzero normals. Only the regular-superlevel implication has been used.

Offsets and their curvature. Fix a regular value $s = s_i$ and put $F = F_i$. For $0 < \tau < \operatorname{reach}(F)$, sufficiently small compared with s and the ambient injectivity scale, define

$$O_\tau = \{x : \operatorname{dist}(x, F) < \tau\}, \quad \Gamma_\tau = \partial O_\tau.$$

The closure of O_τ lies in Ω , since $\rho \geq s - \tau > 0$ there. Moreover, $\Omega \setminus O_\tau$ lies in the s -collar of Σ .

Each point of Γ_τ has a unique footpoint $q \in F$ and can be written $p = \exp_q(\tau v)$, where v is a unit outward normal to F . The same ray realizes distance to F for every positive parameter

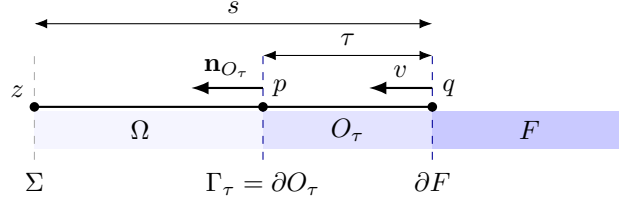


Figure 2: The offset construction along a minimizing segment in the singleton-normal case. First take $F = \{\rho \geq s\}$, then enlarge it by $\tau < s$ to O_τ ; the gap to Σ on this segment is $s - \tau$. The outward normals point toward Σ , hence equal $-\nu$ in the exterior application. This local schematic asserts no tubular coordinates at singular points.

below $\min\{\text{reach}(F), r_*\}$. These facts follow from uniqueness of projection, first variation and continuation of a minimizing normal segment; they are also the nearest-point and normal-cone facts used in [4, Section 3.1]. No differentiability of the projection is required.

The normal ray supplies tangent balls on both sides of Γ_τ . The interior supporting radius is bounded below when τ stays away from zero. For the exterior supporting ball, fix $0 < R < \min\{\text{reach}(F), r_*\}$ first and take $\tau < R/2$. Continuing the normal ray to R gives an exterior supporting radius $R - \tau \geq R/2$. Local graph comparison with these spheres gives a compact $C^{1,1}$ hypersurface. For this fixed F , the principal curvatures have a uniform lower bound as $\tau \downarrow 0$; their upper bound may grow like τ^{-1} .

The offset is now a $C^{1,1}$ hypersurface lying strictly inside Ω . To finish its construction we must transfer the positive curvature of accessible boundary points to this offset. We do so by comparison with smooth interior test domains. Let a smooth region $D \subset O_\tau$ touch Γ_τ from inside at $p = \exp_q(\tau v)$. Its outward normal at contact is the parallel translate of v along the normal ray.

Suppose first that the unit normal cone of F at q is a singleton. Take a shortest segment from q to Σ . Its initial direction is an outward normal to F : its negative is the gradient of a smooth distance upper support, and the superlevel inequality places the initial direction in the polar tangent cone. It must therefore equal v . Its endpoint $z \in \Sigma$ is accessible and smooth. Figure 2 shows the order of the two offsets along this segment. Enlarging D by distance $s - \tau$ touches Σ at z . The enlargement stays on the Ω side, since every point of D has distance at least $s - \tau$ from Σ .

Let $C_R \geq 0$ bound the negative part of the ambient Ricci curvature on the fixed neighborhood. Along the touching geodesic, the normal Riccati equation gives

$$H(\text{enlarged } D \text{ at } z) \leq H(D \text{ at } p) + C_R(s - \tau).$$

First-contact comparison with the accessible part of Σ gives $H(\text{enlarged } D \text{ at } z) \geq \gamma$.

Here is the endpoint argument if the enlargement is focal at z . Localize D near p and shrink it strictly away from p , preserving the touching segment, so that p is the unique terminal footpoint. In tangent coordinates with outward normal in the positive vertical direction, replace its boundary graph u by $u - a|x|^2$, where $a > 0$ can be arbitrarily small. This shrinks the domain and increases its outward second fundamental form at contact by $2a$ times the tangent metric. The boundary index form of the original minimizing normal segment is nonnegative; the perturbation adds $2a|J(0)|^2$ for each variation field J . Variations with $J(0) = 0$ have positive index because the segment is shorter than the ambient conjugacy scale. Thus the perturbed segment is nonfocal through its endpoint. Apply the Riccati comparison to the perturbed domain and let $a \downarrow 0$. Its

mean curvature at contact tends to that of D , proving also in this case that

$$H(D \text{ at } p) \geq \gamma - C_R s.$$

Suppose next that the normal cone contains a direction not parallel to v . Convexity supplies a nontrivial one-sided angular arc through v . Exponentiating this arc at radius τ produces a curve in Γ_τ . The test boundary has the same first derivative at contact. Second-order comparison along the arc, even when it is one-sided, therefore gives a principal curvature at least $1/(2\tau)$ for small τ . The exterior supporting ball gives a lower bound $-C_F$ for all the other principal curvatures, independent of τ for this fixed F . Hence

$$H(D \text{ at } p) \geq \frac{1}{2\tau} - (n-2)C_F.$$

This exceeds any fixed lower bound when τ is sufficiently small. The pointedness proved above excludes the remaining apparent possibility of opposite normals without an angular arc.

Choose the regular value $s < \varepsilon$ sufficiently small that $C_R s < \varepsilon/4$. Then choose τ sufficiently small compared with s , the reach of this fixed F , and its curvature constants. The resulting $C^{1,1}$ hypersurface has outward mean curvature at least $\gamma - \varepsilon/4$ in the barrier sense. At almost every twice-differentiability point of a $C^{1,1}$ graph, quadratic test functions give the same inequality for its classical almost-everywhere mean curvature. The order of these choices is important: neither the reach nor C_F is asserted to remain uniform as $s \downarrow 0$.

Smooth graph approximation. We use the following direct fact. A compact cooriented embedded $C^{1,1}$ hypersurface with almost-everywhere outward mean curvature at least c has smooth embedded approximations converging in C^1 whose mean curvature is at least $c - o(1)$.

To prove it, choose a smooth ambient vector field V uniformly transverse to the hypersurface, by smoothly approximating its continuous unit normal near the compact hypersurface. The C^1 inverse function theorem and compactness give a fixed flow tube for V , in which each sufficiently short flow line meets the hypersurface exactly once. Local graph convolution and a partition of unity give a smooth C^1 approximation Γ_0 sufficiently close to lie in this tube and remain uniformly transverse to V . Projection along these fixed flow lines is a global diffeomorphism. The original hypersurface is therefore the flow graph of a $C^{1,1}$ function u over Γ_0 . This fixed transverse flow avoids any assumption about the normal injectivity radius of an arbitrary C^1 approximation.

In local coordinates its mean-curvature operator has the form

$$H(u) = A^{ab}(x, u, Du) \partial_{ab} u + B(x, u, Du),$$

where the coefficients are smooth on a compact set of allowed graph jets. The sign of A depends on the graph convention and is immaterial for the following averaging argument. Choose a finite smooth atlas with subordinate partition of unity χ_α and smooth u using normalized nonnegative convolution kernels $\rho_{\alpha,\zeta}$. The functions u_ζ converge to u in C^1 and have uniformly bounded second derivatives. Their second derivatives equal the partition-weighted averages of $D^2 u$ plus a uniformly vanishing term. Indeed, differentiating the partition introduces differences $u_\zeta - u$ and $Du_\zeta - Du$, which tend uniformly to zero, while the sums of the derivatives of the partition vanish.

Because $D^2 u$ is uniformly bounded, the coefficient replacement inside those averages,

$$A(x, u_\zeta, Du_\zeta) \longmapsto A(y, u(y), Du(y)),$$

gives a uniformly vanishing commutator. The lower-order coefficients converge uniformly as well. Coordinate changes and connection terms are included in B ; their additional smoothing errors are controlled by the same C^1 convergence. Consequently

$$H(u_\zeta)(x) = \sum_{\alpha} \chi_{\alpha}(x) \int \rho_{\alpha,\zeta}(x,y) H(u)(y) dy + o(1) \geq c - o(1)$$

uniformly in x , proving the asserted approximation.

Apply this construction to the fixed hypersurface Γ_τ . Its positive distance from Σ ensures that a sufficiently small C^1 perturbation remains inside Ω . Choose the region on the same side of the smoothed boundary that contains the deep part of Ω . It differs from O_τ only in a tubular neighborhood of Γ_τ . Choose the width of this neighborhood smaller than both its distance from Σ and $\varepsilon - s$. The removed collar is then contained in the ε -collar of Σ . Finally choose ζ so small that the mean-curvature loss is less than $\varepsilon/2$. The resulting region is Ω_ε , and its mean curvature is at least $\gamma - \varepsilon$. This proves the proposition. $\square$

For the almost-minimizing frontiers used here, the accessible locus is exactly the regular part. At a point touched by a ball contained in the exterior, a tangent cone lies on one side of a plane. The half-space property for minimizing boundaries and small-excess regularity make the point regular. Conversely, a regular point has a sufficiently small supporting ball on the chosen side. Thus the smooth embeddedness and local one-sided hypotheses of Proposition 7.2 hold. The proposition is applied to the full compact frontier. An application to only selected boundary components would also require a positive distance from the remaining components; compactness of their union alone does not imply that separation.

Application to a detached frontier. Let E' be a detached minimizing exterior from the signed variational construction, with $H_\nu \leq -\gamma$ on its regular frontier. Apply the proposition to $\Omega = E'$. Its outward normal at the inner frontier is $-\nu$, so $H_{\text{out},\Omega} = -H_\nu \geq \gamma$. With $\varepsilon = j^{-1}$, write $\Omega_j = \Omega_{j-1}$. Since $\Omega_j \subset \Omega$, the new frontier lies toward the asymptotically flat end. With the end-oriented normal it satisfies

$$H_\nu(\partial\Omega_j) \leq -\gamma + j^{-1} < 0 \tag{7.5}$$

for large j . Its compact removed region contains the original removed region. In the variational definition of capacity this makes the admissible class smaller, and hence

$$\mathfrak{c}(\partial\Omega_j, h) \geq \mathfrak{c}(\Sigma, h). \tag{7.6}$$

If Ω_j has bounded components, retain only the component containing the asymptotic end. This selects whole components of its smooth compact boundary, preserves the curvature sign, and enlarges the conductor further. The same capacity comparison therefore holds for the resulting connected smooth exterior. The argument uses only the shrinking collar, the curvature sign and conductor inclusion; no boundary-area convergence is needed.

This smooth exterior is complete with its boundary included. For each fixed ε , the harmonic perturbation is uniformly comparable with the starting metric, and the retained closed exterior lies a positive distance from the original frontier. The finite-length concatenation argument in Lemma 2.3 therefore gives completeness for its intrinsic length distance. These are the smooth boundary and completeness properties needed in the mass-capacity lemma below.

8 Positive mass and the mass–capacity inequality

The preceding construction reduces the singular-frontier argument to a smooth mass–capacity inequality. We prove that inequality from a positive mass theorem, independently of Theorem 1.1. The conformal-flow argument at this stage has a harmonically flat end, and that is the class we need. A harmonically flat end has metric $U^{4/(n-2)}\delta$, with $U > 0$ Euclidean harmonic and $U \rightarrow 1$. We use the independently stated smooth positive mass result [10, Corollary 1.6]. Its actual hypotheses are strict positive scalar curvature and an all-order end expansion

$$g = (1 + \alpha r^{2-n})\delta + O_j(r^{2-n-\sigma}), \quad \sigma > 0, \quad (8.1)$$

for every fixed j . The complete smooth manifold in that corollary and its governing definitions has no orientability assumption. In dimensions at least four other ends are permitted. Its mass is $(n-1)\alpha$, whereas our ADM mass is $(n-2)\alpha/2$; their signs agree. The following argument uses it only for $n \geq 8$; the lower-dimensional numerical positive mass theorem and [9] provide the classical alternative.

The dimension-descent proof of the cited theorem uses a singular-set covering estimate for weighted perimeter minimizers with a volume forcing. Appendix A proves the needed estimate for that class, using [27, Theorem 1.3]; its role in [10, Definition 3.19, Lemma 3.20, and Theorem 3.34] is specified there. With that input accounted for, the remaining steps of the mass–capacity proof concern smooth metrics: first pass from strict to nonnegative scalar curvature on harmonic ends, then double the smooth exterior and repair its seam.

8.1 Nonnegative mass on harmonic ends

Lemma 8.1. *A smooth complete boundaryless manifold of dimension $n \geq 8$ with finitely many ends, all harmonically flat, and nonnegative scalar curvature has nonnegative ADM mass at each end.*

Proof. Work on the connected component containing the chosen end. Choose a positive smooth function ρ , equal to $r^{-n-\beta}$ on each distant end, with $0 < \beta < 1$. The global Sobolev inequality on a manifold with finitely many Euclidean ends, followed by exhaustion, gives the positive solution of

$$-\Delta v = \rho, \quad v \rightarrow 0 \quad \text{on every end.}$$

More explicitly, $\rho \in L^{2n/(n+2)}$, so the energy completion of compactly supported functions gives a weak solution by the coercive Dirichlet form. Positivity follows by testing the negative part. The local elliptic estimates and end barriers give a bounded smooth solution; the homogeneous decaying maximum principle gives uniqueness. On each harmonically flat end, Uv satisfies a Euclidean Poisson equation with right side $-U^{(n+2)/(n-2)}\rho$. The Newton potential, split into inner, comparable, and outer annuli, gives

$$v = Ar^{2-n} + O_j(r^{2-n-\beta'}) \quad \text{for } 0 < \beta' < \beta.$$

Equivalently, subtract the Newton kernel’s monopole and use the finite β' moment of the right-hand side. The contribution of the finite inner boundary is a decaying harmonic function with the same type of expansion. Scaled interior estimates give all fixed orders of derivatives. Thus $(1+tv)^{4/(n-2)}g$ satisfies Equation (8.1), is complete, and has strictly positive scalar curvature for every $t > 0$, since

$$-\frac{4(n-1)}{n-2}\Delta(1+tv) + R_g(1+tv) > 0.$$

Apply the cited positive mass theorem and let $t \downarrow 0$. The ADM mass changes by $2tA$ at the chosen end, which proves the claim. $\square$

8.2 Doubling and conformal repair

We can now turn the capacity of a smooth boundary into a mass decrease on a complete manifold. The doubling must keep its second end complete while positive mass is applied; a positive parameter will do this, and it can be sent to zero afterward.

The following doubling, seam smoothing, and conformal repair have their three-dimensional antecedent in [8, Definition 17 and Theorem 9, pp. 206–211]. The normalization there gives the same three-dimensional capacity bound. The higher-dimensional positive-mass input is instead the smooth theorem and the reduction just established.

Lemma 8.2 (Smooth mass–capacity inequality). *Let (N, h) , of dimension $n \geq 8$, be a smooth connected exterior, complete including its compact smooth boundary, with exactly one end and compact end complement. Suppose the end is harmonically flat and the scalar curvature is nonnegative. Write Γ for its boundary. If $H_\nu \leq 0$ for the normal toward the end, then $m_{\text{ADM}}(h) \geq \mathfrak{c}(\Gamma, h)$.*

Proof. If Γ is empty, its capacity is zero and the assertion is Lemma 8.1. Assume henceforth that Γ is nonempty. Let $\phi = 1 - w$, where w is the potential in Equation (7.1), and write $c = \mathfrak{c}(\Gamma, h)$. On the doubled manifold define $\psi = \phi$ on the first copy and $\psi = -\phi$ on the second. In signed normal distance across the seam, ψ and its first derivative agree: its value is zero and both normal derivatives equal $\partial_\nu \phi$. For $0 < a < 1$, put

$$V_a = \frac{1 + a + (1 - a)\psi}{2}, \quad h_a = V_a^{4/(n-2)} \bar{h},$$

where $\bar{h}$ is the doubled corner metric. The factor is bounded below by a , tends to one at the first end and to a at the second, and is weakly harmonic. Consequently h_a is complete, has two harmonically flat ends after constant rescaling at the second, and has nonnegative scalar curvature away from the seam.

Choose the common transverse normal from the second copy to the first. The two mean curvatures of $\bar{h}$ with this normal are $-H_\nu$ and H_ν . Their jump in the corner condition is therefore $-2H_\nu \geq 0$. The normal derivative of V_a is continuous, so its contribution cancels in the conformal jump formula; the jump for h_a has the same sign. We now explain the analytic smoothing step to avoid extending the dimensional or topological statement of a corner theorem without justification.

Apply the local Gaussian-collar construction of [26, Section 3, Proposition 3.1 and Equation (36)]. Its tensor convolution uses the coorientation of this collar and is independent of the topology or number of ends outside it. It gives uniformly comparable C^2 metrics, equal to h_a off a shrinking collar, whose negative scalar curvature has a uniform bound. For each fixed collar parameter, approximate this metric in C^2 by a smooth metric, keeping it unchanged outside a slightly larger collar. Scalar curvature is continuous in the metric's C^2 jets, so choosing the approximation sufficiently close preserves the negative-scalar bound up to a fixed enlargement. Denote these smooth metrics by $h_{a,\delta}$. Their negative scalar curvature is supported in a collar whose volume tends to zero. Choose a smooth $q_\delta \geq (R_{h_{a,\delta}})_-$, uniformly bounded and supported in a slightly larger shrinking collar. Solve

$$\Delta_{h_{a,\delta}} z_\delta + \frac{n-2}{4(n-1)} q_\delta z_\delta = 0, \quad z_\delta \longrightarrow 1 \quad \text{at both ends.}$$

Here is the required existence and mass estimate on this two-ended manifold. Put $a_n = 4(n-1)/(n-2)$, $2^* = 2n/(n-2)$ and $p = 2n/(n+2)$. A fixed complete metric with these ends satisfies $\|f\|_{2^*}^2 \leq S_a \int |df|^2$: combine the Euclidean end inequalities with a compact-core Poincaré estimate having trace on one fixed end annulus. Uniform metric comparison gives the same inequality for $h_{a,\delta}$, uniformly in small δ . Since $\|q_\delta\|_{n/2} \rightarrow 0$, the form

$$B_\delta(f, f) = a_n \int |df|^2 - \int q_\delta f^2 \geq \frac{a_n}{2} \int |df|^2$$

is coercive on the homogeneous energy completion of compactly supported smooth functions. Solving $B_\delta(u_\delta, f) = \int q_\delta f$ gives

$$\|u_\delta\|_{2^*} + \|du_\delta\|_2 \leq C_a \|q_\delta\|_p.$$

Testing the negative part gives $u_\delta \geq 0$. Local elliptic regularity and the harmonic exterior equation give a smooth solution $z_\delta = 1 + u_\delta \geq 1$, tending to one on both ends; the end limits also follow from $u_\delta \in L^{2^*}$. The corrected smooth metric $z_\delta^{4/(n-2)} h_{a,\delta}$ has nonnegative scalar curvature. It is still harmonically flat at both ends. In separately normalized end coordinates write $u_\delta = A_{i,\delta} r_i^{2-n} + O_j(r_i^{1-n})$; the coefficients are nonnegative because $u_\delta \geq 0$. Flux integration over both ends gives

$$a_n(n-2)\omega \sum_{i=1}^2 A_{i,\delta} = \int q_\delta z_\delta dV \leq \|q_\delta\|_1 + C_a \|q_\delta\|_p^2 = o(1).$$

Thus each coefficient tends to zero, and each end mass converges to that of h_a . Lemma 8.1 proves nonnegativity of this limiting mass. This argument is local at the seam and accommodates both ends; it does not invoke the one-ended global statement of [26, Theorem 1] outside its stated class.

On the retained end,

$$\phi = 1 - cr^{2-n} + O_j(r^{1-n}), \quad V_a = 1 - \frac{1}{2}(1-a)cr^{2-n} + O_j(r^{1-n}),$$

so the conformal ADM formula gives

$$0 \leq m_{\text{ADM}}(h_a) = m_{\text{ADM}}(h) - (1-a)c.$$

Let $a \downarrow 0$. Both ends remained complete for every application of the smooth positive mass theorem; no theorem about a compactified point was used. $\square$

The signed variational construction, one-sided smoothing and Lemma 8.2 now give the smooth comparison needed for a singular time slice: an enclosing smooth conductor has at least the original capacity, while its end mass is unchanged by moving the frontier. Section 9 applies this comparison to a small harmonic perturbation and removes that perturbation. The same positive two-ended construction will provide the complete inputs used in Section 11.

9 Mass and capacity along the flow

The smooth comparison applies to every flow slice after a small harmonic perturbation. Together with the end expansion, it gives the mass decrease that drives the remaining argument.

Proposition 9.1 (Mass and capacity of the conformal flow). *For the conformal flow of Lemma 4.1 in dimension $n \geq 8$, with nonempty initial frontier and harmonically flat initial end, let $m(t)$ be its ADM mass in normalized coordinates and let $c_t = \mathfrak{c}(\Sigma_t, g_t)$. Then*

$$m(t) \geq c_t \geq 0, \quad \mu(t) := m(t) - c_t \geq 0.$$

The mass is locally absolutely continuous and satisfies $m'(t) = -2\mu(t)$ almost everywhere.

Proof. Fix a slice with normalized metric h , and write m, c, w for its mass, capacity, and capacity potential in normalized end coordinates. For $\lambda > 0$ small set $p = 1 + \lambda w$ and $h_\lambda = p^{4/d} h$. Conformal harmonicity and the end expansions give

$$w_{\Sigma, \lambda} = \frac{(1 + \lambda)w}{p}, \quad c_{\Sigma, \lambda} = (1 + \lambda)c, \quad m_\lambda = m + 2\lambda c.$$

This is the earlier separating perturbation with $\lambda = \varepsilon/(1 - \varepsilon)$. Its detached smooth enclosure Γ has $H_\nu < 0$. Capacity monotonicity and Lemma 8.2, applied to its smooth exterior, give

$$m + 2\lambda c = m_\lambda \geq c_\Gamma \geq (1 + \lambda)c.$$

Sending $\lambda \downarrow 0$ proves $m \geq c$ on the original time slice.

For the evolution formula, write the initial end as $g_0 = H^{4/d} \delta$, where $H > 0$, $\Delta_\delta H = 0$, and $H \rightarrow 1$. The products Hu_t and Hv_t are Euclidean harmonic on the end. Expanding them there and dividing by H gives, for any fixed $0 < \beta < 1$,

$$u_t = e^{-t} + B(t)r^{-d} + O_1(r^{-d-\beta}), \quad v_t = -e^{-t} + b(t)r^{-d} + O_1(r^{-d-\beta}), \quad B' = b, \quad B(0) = 0.$$

Flux identification and normalization of the end coordinates give

$$c_t = e^{-t}(B + b), \quad m(t) = e^{-2t}m(0) + 2e^{-t}B(t), \quad \mu(t) = m(t) - c_t.$$

Since B is locally absolutely continuous, so is m , and differentiation gives $m' = -2\mu$ almost everywhere. $\square$

A restarted flow at time s has conformal and evolution factors u_{s+t}/u_s and v_{s+t}/u_s . The denominator stays fixed at the restart time. The capacity potential has finite homogeneous energy in every dimension; its unweighted L^2 tail is also finite when $n \geq 5$, since it is controlled by $\int_R^\infty r^{3-n} dr$.

The normalized end factors. Let $\Phi_t(x) = e^{2t/d}x$ map the normalized coordinate x to the original coordinate. The normalized flow metric is $G_t = \Phi_t^* g_t$ on this end. Transport the flow and velocity factors by

$$U_t = e^t u_t \circ \Phi_t, \quad V_t = e^t v_t \circ \Phi_t.$$

The normalized background metric and its factor are

$$h_t = H \circ \Phi_t, \quad \bar{G}_t = e^{-4t/d} \Phi_t^* g_0 = h_t^{4/d} \delta.$$

The velocity V_t is $\bar{G}_t$ -harmonic on the portion of the normalized exterior in this end chart. Define the total factors and the auxiliary relative factor as follows:

role	relative factor	Euclidean-harmonic factor
flow metric	U_t	$F_t = h_t U_t$
velocity	V_t	$Z_t = h_t V_t$
auxiliary metric	$W_t = (U_t - V_t)/2$	$J_t = h_t W_t$

Then $G_t = F_t^{4/d} \delta$, and

$$J_t = \frac{F_t - Z_t}{2} = F_t(1 - (w_t \circ \Phi_t)/2), \quad \tilde{G}_t = J_t^{4/d} \delta.$$

The total factors are harmonic outside a sphere enclosing the current frontier and compact core. Their constant terms are 1, -1 , 1, respectively; the masses of the flow and auxiliary metrics are $m(t)$ and $\mu(t)$. Section 11 constructs complete smooth metrics whose end factors approximate J_t , allowing a quantitative positive-mass estimate to control it.

The flow calculation is first made for harmonically flat ends. Section 14 then proves the passage to the general asymptotically flat metrics used here. It first introduces a strict smooth boundary, proves a Dirichlet conformal approximation with mass convergence, and takes a detached minimizing enclosure. The resulting metric comparison controls all enclosing areas.

10 Mass decay and the enclosure radius

Area preservation and Proposition 9.1 reduce the argument to the decay of $\mu(t) = m(t) - c_t$ and the location of the frontier. The next estimates provide a fixed harmonic tail in normalized coordinates, on which the quantitative positive-mass theorem will apply. Put $d = n - 2$ and $q = 2(n - 1)/d$, and retain $\omega = |S^{n-1}|$ for the Euclidean area of the unit sphere.

Proposition 10.1 (Vanishing deficit and bounded normalized radius). *For the flow in Proposition 9.1, $\mu(t) \rightarrow 0$ as $t \rightarrow \infty$. There is a fixed $R_{\max}$ such that*

$$\Sigma_t \subset B_{R_{\max} e^{2t/d}} \quad (t \geq 0). \quad (10.1)$$

Here the radius is measured in the original end coordinates, with the compact core included in the ball. Equivalently, every normalized frontier lies in $B_{R_{\max}}$.

Proof. **The integrated mass identity.** Since $B' = b$ and $B(0) = 0$,

$$\frac{d}{dt}(e^{-t}B(t)) = e^{-t}(b(t) - B(t)).$$

Consequently

$$\int_0^t \mu(s) ds = \frac{1}{2}(1 - e^{-2t})m(0) - e^{-t}B(t) = \frac{m(0) - m(t)}{2}.$$

This supplies integrability of the nonnegative μ . Set $a = m - 2\mu = 2c_t - m$. Nesting and the maximum principle make $e^t v_t$ nondecreasing: the later potential is zero on the larger filled set and both have limit -1 at infinity. Its end coefficient $e^t b(t)$ is therefore nondecreasing. Since

$$e^{2t}a(t) = 2e^t b(t) - m(0),$$

the same holds for $e^{2t}a(t)$. This implies, distributionally, $Da + 2a dt \geq 0$. Substitution of the integral identity gives monotonicity of $3m - 2\mu + 2 \int_0^t m$ (up to a constant). Thus for $h > 0$,

$$\mu(t+h) - \mu(t) \leq \frac{3}{2}(m(t+h) - m(t)) + \int_t^{t+h} m(s) ds \leq hm(0).$$

This one-sided slope bound and integrability imply $\mu(t) \rightarrow 0$: a value at least $\varepsilon > 0$ forces an interval immediately before it on which the integral is bounded below by a positive constant depending only on $\varepsilon, m(0)$. If $m(0) = 0$, nonnegativity and mass monotonicity give $m = 0$; the same slope bound excludes positive point values of μ .

The escaping-point ball. Let $a = e^{2/d}$ and $R(s) = R_{\max} e^{2s/d}$. Suppose that containment holds for $0 \leq s \leq T$ but there is a point $p \in \Sigma_{T+1}$ with $|p| \geq aR(T)$. Use the ball

$$\rho = \frac{a-1}{2}R(T), \quad \overline{B_\rho(p)} \subset \{|y| > R(T)\}.$$

Here the balls are Euclidean balls in the original end coordinates. Choose $R_{\max}$ so the indicated exterior avoids the initial obstacle and $\frac{1}{2}\delta \leq g_0 \leq 2\delta$ there.

To estimate the metric on this ball, use its total harmonic conformal factor. Write $g_0 = H^{4/d}\delta$ on the initial end, where $\Delta_\delta H = 0$ and $H \rightarrow 1$. Then Hv_s is Euclidean harmonic outside Σ_s . If $\Sigma_s \subset B_{R(s)}$, its boundary value on $S_{R(s)}$ is nonpositive and its limit at infinity is $-e^{-s}$, so

$$Hv_s(y) \leq e^{-s} \left[\left(\frac{R(s)}{|y|} \right)^d - 1 \right] \quad (|y| \geq R(s)).$$

Using $v_s \leq 0$ on the remaining time interval gives, for $r = |y| \geq R(T)$,

$$Hu_{T+1} \leq H - 1 + e^{-T} + (e^T - 1)R_{\max}^d r^{-d}.$$

Choose $H - 1 \leq C_0 r^{-d}$ and $R_{\max}^d \geq C_0$. The last expression is at most $2e^{-T}$. Taking also $H \geq 1/2$ gives the concrete bounds

$$e^{-T-1} \leq u_{T+1} \leq Ke^{-T}, \quad K = 4, \quad r \geq R(T).$$

Every finite-perimeter variation supported in $B_\rho(p)$ remains an admissible enclosure. Comparison of area elements therefore makes Σ_{T+1} Euclidean perimeter-quasiminimizing there, with ratio

$$\gamma = 2^{n-1}(eK)^q.$$

The isoperimetric proof of the density estimate, using the competitors that fill or delete a ball, applies at every measure-theoretic boundary point, including a singular point. It yields

$$\mathcal{H}_\delta^{n-1}(\Sigma_{T+1} \cap B_\rho(p)) \geq c(n, \gamma)\rho^{n-1}.$$

It follows that the fixed area satisfies

$$\begin{aligned} A &\geq 2^{-(n-1)/2} e^{-q(T+1)} c(n, \gamma) \rho^{n-1} \\ &= 2^{-(n-1)/2} c(n, \gamma) e^{-q} \left(\frac{e^{2/d} - 1}{2} \right)^{n-1} R_{\max}^{n-1}. \end{aligned}$$

All time factors cancel. The constants are independent of T and of further enlargement of $R_{\max}$, giving the required contradiction. To cover all times, enlarge $R_{\max}$ to obtain containment on $[0, 1]$ from the finite-time support bound. If containment holds on $[0, N]$, apply the argument at each $t \in [N, N+1]$ with $T = t - 1$. This proves Equation (10.1) at every real time. $\square$

11 Complete metrics and the auxiliary end factor

Lee's quantitative positive-mass estimate turns small mass into control of a harmonic end factor. To apply it to J_t , we construct complete smooth metrics whose masses tend to zero and whose end factors approach J_t . The construction keeps a fixed coordinate tail while allowing the compact interior to vary. Throughout, $n \geq 8$ and $d = n - 2$.

Proposition 11.1 (Complete approximants and the auxiliary factor). *For the normalized flow of Proposition 9.1 there are fixed radii $R_c < R_1$ and complete smooth connected boundaryless one-ended metrics with nonnegative scalar curvature, harmonic end factors $\mathcal{Z}_t$ on $\{r > R_1\}$, and masses $\mathcal{M}_t$, such that*

$$0 \leq \mathcal{M}_t \longrightarrow 0, \quad \sup_{r \geq 2R_1} r^d |\mathcal{Z}_t - J_t| \longrightarrow 0.$$

The radius R_c encloses every normalized frontier and compact core. For every fixed $s > R_c$,

$$\sup_{r \geq s} r^d |J_t - 1| \longrightarrow 0. \quad (11.1)$$

We first isolate the quantitative input, so that the construction can be read separately from the verification of its positive-mass hypotheses.

Lemma 11.2 (Quantitative positive mass on one harmonic end). *For each $n \geq 8$ there is a fixed constant $\ell_n > 1$ such that, for every $\eta > 0$, there is $\delta(n, \eta) > 0$ with the following property. Let (N, h) be a smooth complete connected boundaryless n -manifold with $R_h \geq 0$. Suppose that outside a compact set it consists of a single coordinate end $\{r > R\}$, where $R > 0$, and that on this end*

$$h = Z^{4/(n-2)} \delta, \quad Z > 0, \quad \Delta_\delta Z = 0, \quad Z \longrightarrow 1.$$

Writing $d = n - 2$ and $m = m_{\text{ADM}}(h)$, we have

$$m < \delta(n, \eta) R^d \implies \sup_{r > \ell_n R} |Z - 1| < \eta.$$

Proof. Rescale the metric by R^{-2} and the coordinates by R^{-1} . The harmonic coordinate radius becomes one, and the mass becomes m/R^d . We apply the argument of [23, Theorem 1.3] at this scale. Its formal hypothesis concerns every complete asymptotically flat metric on the underlying manifold; we verify that the positive-mass statements used in its proof follow here from Lemma 8.1. The one-end assumption also identifies the mass flux in Lee's Lemma 2.1 with the flux in the divergence integral over the entire manifold.

The scalar-flat reduction of [23, Lemma 2.5] gives a metric $k = f^{4/d} h$ with $f > 0$ and $f \rightarrow 1$. The factor is bounded above and below by positive constants, so k remains smooth and complete. Its end factor Zf is Euclidean harmonic. For this scalar-flat reference metric, Lee uses the variations

$$k_s = k + s\zeta \text{Ric}(k), \quad \widehat{k}_s = u_s^{4/d} k_s,$$

where the cutoff ζ is compactly supported and $\widehat{k}_s$ is scalar-flat. Outside the support of ζ , the product of the old end factor and u_s is Euclidean harmonic. The conformal scalar-curvature identity gives

$$c_n = \frac{n-2}{4(n-1)}, \quad L_s = \Delta_{k_s} - c_n R_{k_s}, \quad L_s(u_s - 1) = c_n R_{k_s}.$$

Both scalar-curvature coefficients in the operator definition and inhomogeneous equation preceding Lee's Lemma 2.4 must have this value, as in the earlier conformal equation in that paper. Lemma 2.4 estimates Δ_k , and the ensuing perturbation argument applies with the corrected fixed dimensional coefficient. Its exterior estimates give $u_s - 1 = O(|s|)$ uniformly. On the compact region away from the perturbation, $u_s - 1$ is harmonic, so the maximum principle extends this bound throughout the manifold. For sufficiently small $|s|$, $1/2 \leq u_s \leq 3/2$. The perturbed and conformally corrected metrics are therefore smooth and complete.

The scalar-flat argument uses nonnegativity of the varied mass at a small negative parameter [23, proof following Lemma 2.2]. This follows from Lemma 8.1, since every varied metric has a harmonic end outside a compact set. The same lemma gives $m_{\text{ADM}}(k) \geq 0$ after the initial scalar-flat reduction. Lee's Lemma 2.5 gives $m_{\text{ADM}}(k) \leq m_{\text{ADM}}(h)$ and controls the end-factor difference by the mass difference. Thus all the required positive-mass inputs hold, without a spin or orientability assumption. The constants in Lee's estimates depend only on dimension and the target error at the normalized radius, not on the compact interior. Restoring the scale R gives the asserted implication. $\square$

Proof of Proposition 11.1. We use the following elementary harmonic estimate. If $q \geq 0$ is Euclidean harmonic on $\{r > R\}$, tends to zero, and has expansion $q = br^{-d} + O(r^{-d-1})$, then

$$0 \leq q(x) \leq C(n)br^{-d} \quad (r \geq 2R). \quad (11.2)$$

Indeed, its spherical average is exactly br^{-d} , and the Harnack inequality on scaled annuli bounds its maximum on a sphere by a dimensional constant times this average. In particular, the constant does not depend on the geometry inside B_R .

A smooth reduction from two ends to one. Let (N, k) be a smooth complete connected boundaryless manifold with exactly two harmonically flat ends and compact end complement, and suppose $R_k \geq 0$. Designate one end for retention. There is a harmonic function χ tending to one on that end and zero on the other, with $0 < \chi < 1$. To construct it, choose a smooth function constant with these values outside a compact set and minimize its Dirichlet energy after addition of the homogeneous energy space. The global Sobolev inequality gives the solution; truncation and the strong maximum principle give its bounds. Exterior elliptic estimates give the stated end limits and expansions.

Write $k = Z_i^{4/d} \delta$ on either end. Then $Z_i \chi$ is Euclidean harmonic. On the end to be filled,

$$Z_2 \chi = b_2 r^{-d} + O_j(r^{-d-1}), \quad b_2 > 0.$$

Positivity of b_2 follows by comparison with a positive multiple of r^{-d} from any fixed coordinate sphere. Under inversion $x = y/|y|^2$, the factor

$$|y|^{-d}(Z_2 \chi)(y/|y|^2)$$

extends harmonically across $y = 0$ with value b_2 . Thus $\chi^{4/d} k$ extends to a smooth metric after adding one point. The filled manifold is complete, has one end, and has nonnegative scalar curvature, since $R_{\chi^{4/d} k} = \chi^{-4/d} R_k$ away from the added point and is zero near that point.

At the retained end write $\chi = 1 - Ar^{-d} + O_j(r^{-d-1})$. If $M = m_{\text{ADM}}(k)$, the filled metric has mass $M - 2A$. The function $Z_1(1 - \chi)$ is nonnegative Euclidean harmonic with monopole A . Consequently

$$0 \leq M - 2A \leq M, \quad |Z_1 - Z_1 \chi| \leq C(n)Mr^{-d} \quad (r \geq 2R), \quad (11.3)$$

where the first inequality uses Lemma 8.1 on the smooth filled manifold. The estimate depends only on the retained-end coordinate radius and mass; the second end may vary with the compact interior.

A common coordinate tail. Use the normalized metric G_t and total harmonic factor F_t from Section 9. In this subsection, Σ_t, w_t denote its frontier and capacitary potential in these coordinates; the latter is the pullback previously written $w_t \circ \Phi_t$. A coordinate ball includes the compact core when the whole exterior is under discussion. The mass, capacity, and deficit remain $m(t)$, c_t , and $\mu(t) = m(t) - c_t$. The preceding radius bound and harmonic estimates give $\Sigma_t \subset B_{R_c}$, the factors F_t are uniformly bounded on a fixed sphere outside B_{R_c} , and

$$0 \leq c_t \leq m(t) \leq m(0), \quad \mu(t) \longrightarrow 0.$$

Indeed, take $T = t - 1$ in the radius estimate: $U_t \leq 4e$ on the normalized tail $r \geq R_{\max}$ for $t \geq 1$. The initial bounded time interval is covered by the finite-time flow bounds. Multiplication by the uniformly bounded initial harmonic factor gives the asserted bound for F_t . Increasing R_c once includes that sphere. Let w_t be the capacitary potential equal to one at Σ_t and zero at infinity.

For $0 < \lambda < 1$ put $p = 1 + \lambda w_t$ and $h_{t,\lambda} = p^{4/d} G_t$. The detachment and smoothing construction above provides a smooth enclosing boundary Γ with $H_\nu^\Gamma < 0$. It can be chosen in a ball B_{R_1} whose radius is independent of t and λ . Here is the confinement argument. On the common coordinate end the positive harmonic factor $F_t p$ has a uniform bound on S_{R_c} and tends to one. The maximum principle and scaled gradient estimates give

$$|F_t p - 1| + r |D(F_t p)| \leq C(R_c/r)^d \quad (r \geq 2R_c).$$

Thus all coordinate spheres beyond a fixed R_* have positive outward mean curvature in $h_{t,\lambda}$. Write K_t for the original conductor and L for its detached minimizing enlargement. The functional minimized by L is

$$\mathcal{F}(L) = P_{h_{t,\lambda}}(L) + \theta \int_{L \setminus K_t} h_0 dV_{h_{t,\lambda}}, \quad h_0 \geq 0.$$

Let X be the outward unit normal to the coordinate spheres. For almost every $s > R_*$, the finite-perimeter divergence formula on $L \cap \{r > s\}$ gives

$$P(L \cap \{r < s\}) \leq P(L) - \int_{L \cap \{r > s\}} \operatorname{div} X dV.$$

Both the perimeter and the nonnegative volume penalty decrease under clipping, strictly if the removed tail has positive volume. Minimality therefore confines L to B_{R_*} . The subsequent smooth boundaries lie in shrinking collars, so they lie in B_{R_1} for one fixed $R_1 > R_*$. Fix such a smoothing index before doubling. Thus the coordinate tail is fixed even when the separation distance and smoothing parameters vary.

The capacity comparison and the completed end. Suppress t temporarily and put $h = G_t$, $F = F_t$, $w = w_t$, $c = c_t$, $m = m(t)$, $\mu = \mu(t)$. The potential and capacity of the original conductor for $h_\lambda = p^{4/d} h$ are exactly

$$w_{\Sigma,\lambda} = \frac{(1+\lambda)w}{p}, \quad c_{\Sigma,\lambda} = (1+\lambda)c, \quad m_\lambda = m + 2\lambda c. \quad (11.4)$$

These identities follow from the harmonic conformal transformation formula and the end expansions. If w_Γ is the potential of Γ for h_λ and c_Γ its capacity, enclosure monotonicity and Lemma 8.2 give

$$0 \leq D_\Gamma := c_\Gamma - c_{\Sigma,\lambda} \leq m_\lambda - c_{\Sigma,\lambda} = \mu + \lambda c.$$

On the common tail the function $Fp(w_\Gamma - w_{\Sigma, \lambda})$ is nonnegative Euclidean harmonic with monopole D_Γ . Its sign follows from the maximum principle on the exterior of Γ .

For $0 < a < 1$ the positive two-ended doubling construction in Lemma 8.2 has retained end factor

$$B = Fp\left(1 - \frac{1-a}{2}w_\Gamma\right)$$

and retained end mass

$$m_B = m_\lambda - (1-a)c_\Gamma \leq \mu + \lambda c + am_\lambda.$$

The Miao smoothing and its conformal correction produce smooth complete two-ended metrics with $R \geq 0$ and retained factor Bz_δ on $\{r > R_1\}$. In the notation of that proof, $z_\delta \geq 1$, and

$$e_\delta := B(z_\delta - 1) \geq 0$$

is Euclidean harmonic on the common tail with monopole $A_\delta \rightarrow 0$. Their retained masses are $M_\delta = m_B + 2A_\delta \geq 0$.

The auxiliary Euclidean end factor already defined above is $J = F(1 - w/2)$, with t suppressed. Direct subtraction gives

$$B - J = \frac{\lambda}{2}Fw - \frac{1}{2}Fp(w_\Gamma - w_{\Sigma, \lambda}) + \frac{a}{2}Fpw_\Gamma.$$

Each of the three harmonic functions appearing on the right is nonnegative before its displayed sign, with monopoles respectively c , D_Γ , and c_Γ . Equation (11.2) therefore yields

$$|Bz_\delta - J| \leq C(n)(\mu + \lambda c + am_\lambda + A_\delta)r^{-d}, \quad r \geq 2R_1. \quad (11.5)$$

The capacity difference controls the end-factor error independently of the distance between Γ and Σ_t .

For each t , choose $\lambda_t = a_t = (t+2)^{-1}$; choose the detachment and boundary smoothing parameters next; finally choose δ_t so that $A_{\delta_t} \leq (t+2)^{-1}$. The Sobolev constants used in the seam correction may depend on these fixed choices. This order suffices because Lee's constants do not depend on the compact interior. The preceding bounds imply

$$0 \leq M_{\delta_t} \longrightarrow 0, \quad \sup_{r \geq 2R_1} r^d |B_t z_{\delta_t} - J_t| \longrightarrow 0.$$

Apply the one-end reduction (11.3) to each smooth completed metric. Its underlying double is connected because the original exterior is connected and Γ is nonempty; it is boundaryless with exactly two ends and compact end complement. Denote the resulting masses and harmonic end factors by $\mathcal{M}_t$ and $\mathcal{Z}_t$. These smooth connected one-ended metrics have their harmonic coordinate end on the same $\{r > R_1\}$, and the first two assertions of the proposition follow from Equation (11.3). Lemma 11.2, followed by the end-factor comparison, now gives

$$J_t \longrightarrow 1 \quad \text{uniformly on } \{r > R_2\}$$

for any one fixed $R_2 > \max(2, \ell_n)R_1$. The exterior maximum principle also gives weighted convergence on a slightly larger fixed tail.

The normalization and the inner tail. The initial harmonic factor in the normalized coordinates is $h_t = H \circ \Phi_t$. Thus

$$\bar{G}_t = h_t^{4/d} \delta, \quad F_t = h_t U_t, \quad J_t = h_t W_t, \quad W_t = \frac{1}{2}(U_t - V_t).$$

The initial end expansion gives $h_t - 1 = O_j(e^{-2t}r^{-d})$ on any fixed exterior tail. Consequently the preceding conclusion implies $W_t \rightarrow 1$ there.

It remains to extend the weighted convergence to each fixed radius $s > R_c$. The functions J_t are positive Euclidean harmonic on $\{r > R_c\}$, with spherical averages $1 + \frac{1}{2}\mu(t)r^{-d}$. Their bounded masses therefore give uniform Harnack and elliptic bounds on each compact subannulus. Every subsequential harmonic limit equals one on the distant tail, and hence everywhere on $\{r > R_c\}$ by unique continuation. Thus $J_t \rightarrow 1$ uniformly on S_s . Applying the maximum principle to $J_t - 1$ outside S_s yields

$$\sup_{r \geq s} r^d |J_t - 1| \leq s^d \sup_{S_s} |J_t - 1| \rightarrow 0.$$

The same conclusion holds for W_t , since $h_t \rightarrow 1$ with the stated decay. This proves Equation (11.1). $\square$

12 The end limit and the numerical inequality

The vanishing deficit determines a limiting radial potential. Its zero boundary value identifies the largest limiting frontier radius, and comparison with enclosing spheres gives the sharp full-area bound.

Proposition 12.1 (The inequality for a harmonically flat end). *Let (E, h) satisfy the hypotheses of Theorem 1.1, with $n \geq 8$, nonempty frontier of full area A , and a harmonically flat end. Then*

$$m_{\text{ADM}}(h) \geq \frac{1}{2}(A/\omega)^{d/(n-1)}.$$

Proof. The conformal flow preserves A and has decreasing mass $m(t) \rightarrow M \geq 0$. Proposition 11.1 controls its auxiliary factor on a common tail. Put $q = 2(n-1)/d$.

Harmonic normalization and characteristics. Recall the total harmonic factors $F_t = h_t U_t$, $Z_t = h_t V_t$, and $J_t = h_t W_t$ from Section 9. The flow metric has factor F_t , and the auxiliary metric has factor J_t ; their monopole coefficients are $m(t)/2$ and $\mu(t)/2$. The background factor $h_t = H \circ \Phi_t$ contributes the initial-mass term: the r^{-d} coefficient of U_t is $e^{-t}B(t)$, while that of F_t is $m(t)/2 = e^{-t}B(t) + e^{-2t}m(0)/2$.

Set

$$f_t = F_t - 1 - \frac{m(t)}{2}r^{-d}, \quad j_t = J_t - 1 - \frac{\mu(t)}{2}r^{-d}.$$

The evolution and mass identities give

$$\partial_t f_t = 2(f_t - j_t) + \frac{2}{d}r\partial_r f_t.$$

For $y_s = e^{2(t-s)/d}x$, the chain rule therefore gives

$$\frac{d}{ds}(e^{-2s}f_s(y_s)) = -2e^{-2s}j_s(y_s).$$

Here is the required quantitative bound for j_t . Equation (11.1) gives $\sup_{r \geq s_0} r^d |J_t - 1| \rightarrow 0$ for every fixed $s_0 > R_c$. Choose $s_0 = 2 \max\{R_c, R_{\max}, 1\}$. The Kelvin transform

$$K_t(y) = |y|^{-d}(J_t(y/|y|^2) - 1)$$

extends harmonically across zero, has value $K_t(0) = \mu(t)/2$, and is uniformly small on $|y| \leq s_0^{-1}$. The interior gradient estimate on the half ball therefore gives $|j_t(x)| \leq C\varepsilon r^{-d-1}$ for all sufficiently large t and $r \geq 2s_0$. For any fixed $0 < \beta \leq 1$, this proves, for $s \geq t_0$ and $r \geq 2s_0$,

$$|j_s(x)| \leq C\varepsilon r^{-d-\beta}, \quad |f_{t_0}(x)| \leq C_0 r^{-d-\beta}.$$

Integrating the characteristic equation gives the explicit estimate

$$|f_t(x)| \leq \left[C_0 e^{-2\beta(t-t_0)/d} + \frac{dC}{\beta} \varepsilon (1 - e^{-2\beta(t-t_0)/d}) \right] r^{-d-\beta}.$$

The first term tends to zero with the initial time t_0 fixed. First taking $t \rightarrow \infty$ and then $\varepsilon \downarrow 0$ proves the desired exterior convergence of the total factors. More explicitly, with $M = \lim m(t)$,

$$F_t \rightarrow 1 + \frac{M}{2} r^{-d}, \quad Z_t = F_t - 2J_t \rightarrow -1 + \frac{M}{2} r^{-d}.$$

Since $h_t \rightarrow 1$ uniformly on any fixed exterior region, $V_t = Z_t/h_t$ has the latter limit as well.

Boundary control on fixed annuli. Choose any sequence $t_i \rightarrow \infty$. We obtain a uniform boundary Hölder estimate away from the collapsing compact core; it will transfer the radial potential to a farthest limiting frontier point. Use the direct end coordinates $x = e^{-2t_i/d}y$ on the end chart. In this coordinate region, denote the rescaled time- t_i exterior by E_i and its rescaled frontier by Σ_i . If the original end chart begins at radius R_{chart} , represent the core by the closed ball of radius $R_{\text{chart}}e^{-2t_i/d}$. Adjoin that ball to the rescaled boundary portion in the end and call the resulting compact set $\mathcal{C}_i$. The radius bound puts these compact sets in a fixed ball. They record the frontier and the collapsing core in Euclidean coordinates; enclosure comparisons will still use the original filled regions. On every fixed compact annulus away from the origin, the rescaled upper and lower conformal bounds give uniform comparison of G_{t_i} with δ . More explicitly, for an annulus with inner radius $a_0 > 0$, fix

$$L \geq \max \left\{ 1, \frac{d}{2} \log \frac{R_{\max}}{a_0} \right\}, \quad S = t_i - L \geq 0.$$

If $|x| \geq a_0$, then $|y| = e^{2t_i/d}|x| \geq R(S)$. Since $t_i \geq S + 1$ and u_t is nonincreasing in time, the radius proof directly gives

$$e^{-t_i} \leq u_{t_i}(y) \leq u_{S+1}(y) \leq 4e^{-S}, \quad 1 \leq U_{t_i}(x) \leq 4e^L.$$

Together with $H(e^{2t_i/d}x) \rightarrow 1$, these are the required uniform bounds on the total factor F_{t_i} and metric G_{t_i} . The rescaled initial obstacle is eventually disjoint from that annulus. Thus the enclosure property permits both filling and deleting local balls and gives a uniform Euclidean perimeter-quasiminimality ratio. The isoperimetric argument supplies, with constants independent of i , both-phase density

$$|B_s(p_i) \cap E_i| \geq \eta s^n, \quad |B_s(p_i) \setminus E_i| \geq \eta s^n, \quad p_i \in \Sigma_i, \quad 0 < s < \rho_0.$$

Here p_i lies in a smaller concentric annulus and ρ_0 is chosen so $B_{4\rho_0}(p_i)$ stays in the annulus with the uniform estimates. For example, applying the deleting competitor to a phase of volume $b(s)$ gives $b(s)^{(n-1)/n} \leq Cb'(s)$; integrating this inequality gives the claimed lower volume bound. Apply the same argument to the other phase.

The backgrounds $\bar{G}_{t_i}$ converge smoothly to δ on the annulus. Hence the operators for V_i are uniformly elliptic there. Construct the background equilibrium potential in the original ambient manifold by minimizing Dirichlet energy subject to value one on the filled conductor and zero at infinity. Write $\mathcal{P}_i$ for its pullback to the end chart. The homogeneous energy construction, or smooth conductor exhaustion and weak compactness, extends the potential by one across the conductor. Truncation gives $0 \leq \mathcal{P}_i \leq 1$, and the variational inequality makes $\mathcal{P}_i$ a weak supersolution. Thus $q_i = 1 - \mathcal{P}_i$ has a $W_{\text{loc}}^{1,2}$ zero extension, is a weak subsolution, and satisfies $0 \leq q_i \leq 1$. If $M_i = \sup_{B_{4s}(p_i)} q_i$, the function $M_i - q_i$ is a nonnegative supersolution and equals M_i on a set of volume at least ηs^n . The weak Harnack inequality gives

$$\sup_{B_s(p_i)} q_i \leq (1 - \theta) \sup_{B_{4s}(p_i)} q_i, \quad 0 < \theta < 1.$$

Iteration yields uniform boundary Hölder control

$$|q_i(x)| \leq C \left(\frac{|x - p_i|}{\rho_0} \right)^\kappa \quad (|x - p_i| < \rho_0), \quad \kappa > 0.$$

The variational construction of the flow potential gives $q_i = -V_i$ on the exterior portion of the end chart; see Lemma 4.1 and [6, Lemma 3.2]. Thus the estimate controls $|V_i|$ uniformly in i on each fixed annulus, including at singular frontier points.

The largest limiting radius and the area comparison. Take a Hausdorff limit $\mathcal{C}$ of $\mathcal{C}_i$, and let O be the unbounded component of $\mathbb{R}^n \setminus \mathcal{C}$. Every compact subset of O eventually lies in the actual exterior: join its points to infinity by finitely many paths with positive distance from $\mathcal{C}$, then use Hausdorff convergence and the absence of a boundary crossing along those paths. The uniform annular bounds and interior elliptic estimates give a common subsequence of F_{t_i} and V_i converging on compact subsets of O . Their far-exterior limits identify both functions throughout O by unique continuation:

$$F_\infty(x) = 1 + \frac{M}{2}|x|^{-d}, \quad V_\infty(x) = -1 + \frac{M}{2}|x|^{-d},$$

where $M = \lim_{t \rightarrow \infty} m(t)$. The origin lies in $\mathcal{C}$ and hence outside the continuation domain O . The boundary Hölder estimate implies $V_\infty = 0$ at every point of $\partial O \setminus \{0\}$. Set $R_* = \max\{|x| : x \in \mathcal{C}\}$. If $R_* > 0$, choose $p \in \mathcal{C}$ with $|p| = R_*$. Points on the outward radial ray from p lie in O , so the boundary estimate and the displayed potential give

$$0 = -1 + \frac{M}{2R_*^d}, \quad R_* = r_\infty := (M/2)^{1/d}.$$

In particular $M > 0$, and $\mathcal{C} \subset \bar{B}_{r_\infty}$.

If $R_* = 0$, then $\mathcal{C} = \{0\}$ and the same harmonic limit holds on every fixed annulus. Its sign $V_\infty \leq 0$ forces $M = 0$. For any fixed $R > 0$, the normalized boundaries are eventually inside B_R , while the total factors converge to 1 on S_R by harmonic compactness and unique continuation. The minimizing-enclosure property would give $A \leq \omega R^{n-1}$ for every $R > 0$, contradicting $A > 0$.

Thus $R_* > 0$ and $M > 0$. The enclosing coordinate spheres in this argument also enclose the original compact core.

Finally, for any $R > r_\infty$, outer containment and the minimizing property give, after taking the subsequential limit,

$$A \leq \omega R^{n-1} \left(1 + \frac{M}{2R^d}\right)^q.$$

Letting $R \downarrow r_\infty$ yields

$$A \leq \omega(2M)^{(n-1)/d}, \quad m(0) \geq M \geq \frac{1}{2} \left(\frac{A}{\omega}\right)^{d/(n-1)}.$$

□

13 Proofs of the enclosure lemmas

The harmonic-flat inequality is now established. To pass to general ends, we will first construct a smooth strict inner barrier and then take its minimizing enclosure. We prove here the geometric statements of Section 2, which guarantee that this enclosure has the full area, regularity, and completeness required by the inequality. The arguments themselves use only the fixed-metric hypotheses stated there.

Return to the smooth exterior (X, g) with inner boundary B and the compact filling $O \subset M$ of Section 2. The class $\mathcal{A}$ consists of bounded finite-perimeter sets containing O ; P_g counts their full ambient perimeter, and $a_g(B)$ is the infimum of the areas of full smooth enclosing cuts. The normal at B points into X .

13.1 The full enclosure minimum

Proof of Lemma 2.1. We first establish existence and the equality of infima without using frontier regularity. We then use the local obstacle lemma proved below to identify the exterior.

Confinement and existence. On the coordinate end put $Y = \nabla r / |\nabla r|_g$. Its divergence is the outward mean curvature of a coordinate sphere, so (2.1) gives

$$\operatorname{div}_g Y = \frac{n-1}{r} + O(r^{-1-q}) > 0 \quad (r > R_0) \quad (13.1)$$

for a sufficiently large R_0 . For $E \in \mathcal{A}$ and almost every $R > R_0$, the BV trace on $S_R = \{r = R\}$ is defined and the perimeter measure gives S_R zero mass. Gauss–Green on the bounded region $E \cap \{r > R\}$ yields

$$\int_{E \cap \{r > R\}} \operatorname{div}_g Y \, dV_g = \int_{\partial^* E \cap \{r > R\}} g(Y, \nu_E) \, dA_g - \int_{S_R} \mathbf{1}_E \, dA_g.$$

There is no flux at infinity. The first boundary integral is no larger than the removed perimeter, while the second is the area added by clipping E at S_R . Therefore, writing $E_R = E \setminus \{r > R\}$,

$$P_g(E_R) - P_g(E) \leq - \int_{E \cap \{r > R\}} \operatorname{div}_g Y \, dV_g \leq 0. \quad (13.2)$$

The inequality is strict if the removed region has positive volume.

Fix $R_1 > R_0$. For each competitor choose a good slicing radius in (R_0, R_1) ; the radius need not work for any other competitor. Clipping confines a minimizing sequence to the same compact truncation at R_1 . The competitor O gives a finite perimeter bound. BV compactness and lower semicontinuity on a slightly larger compact set now give a minimizer. Containment of O and absence from the specified exterior tail pass to the L^1 limit. Since clipping did not change the infimum, this is a global minimizer in $\mathcal{A}$, not a minimizer constrained by an artificial outer wall.

For any global minimizer, the strict part of (13.2) excludes positive volume beyond every good slicing radius above R_0 . Choosing one such radius below a fixed $R_2 \in (R_0, R_1)$ confines every minimizer to the truncation at R_2 . The same BV argument shows that any sequence of minimizers has an L^1 -convergent subsequence. Its limit remains admissible, and lower semicontinuity together with minimality makes its perimeter equal to the common minimum. This proves the compactness assertion. The minimum is positive: a zero-perimeter indicator would be constant almost everywhere on connected M , whereas every admissible set contains the nonempty open set O° and misses an end tail.

Recovery by full smooth cuts. A smooth exterior D determines the filled competitor

$$E_D = O \cup (X \setminus \text{int } D), \quad P_g(E_D) = \text{Area}_g(\partial_{\text{man}} D).$$

The identity includes boundary on B . Consequently the perimeter infimum is at most $a_g(B)$.

For the reverse inequality, start with any $E \in \mathcal{A}$. Choose a smooth vector field supported in a collar of B and equal there to the normal pointing from O into X . Its flow Φ_t satisfies $\Phi_t(O) \supset O$ for small positive t , with a positive exterior collar between their boundaries. Hence $E_t = \Phi_t(E)$ contains a neighborhood of O up to null sets. The tangential Jacobian of Φ_t tends uniformly to one on the compact perimeter support, so $P_g(E_t) \rightarrow P_g(E)$.

Fix $t > 0$. Strict BV approximation, with the constant value one retained on a smaller neighborhood of O and zero retained outside a fixed compact set, supplies smooth functions $u_j \in [0, 1]$ such that

$$u_j \rightarrow \mathbf{1}_{E_t} \text{ in } L^1, \quad \int_M |du_j|_g dV_g \rightarrow P_g(E_t).$$

For completeness, smooth in finitely many coordinate charts only between these two constant regions and use a partition of unity; the usual strict approximation proof is local on precisely that region. Choose $\delta_j \downarrow 0$ slowly enough that $\|u_j - \mathbf{1}_{E_t}\|_{L^1}/\delta_j \rightarrow 0$. Coarea and Sard's theorem give a regular value $s_j \in (\delta_j, 1 - \delta_j)$ with

$$P_g(F_j) \leq \frac{\int |du_j|_g dV_g}{1 - 2\delta_j} + o(1), \quad F_j = \{u_j > s_j\}.$$

Also $\|\mathbf{1}_{F_j} - \mathbf{1}_{E_t}\|_{L^1} \leq \|u_j - \mathbf{1}_{E_t}\|_{L^1}/\delta_j$. Lower semicontinuity thus gives $P_g(F_j) \rightarrow P_g(E_t)$. Each F_j contains a neighborhood of O and has compact smooth boundary.

Retain the component of $M \setminus \overline{F_j}$ containing the distant end, and denote its closure by D_j . This amounts to filling the bounded complementary components. The boundary of a compact smooth domain has finitely many components; the operation selects entire boundary components and introduces none. Thus D_j is a connected smooth exterior, closed in X , and its full intrinsic boundary lies in $\text{int } X$, with

$$a_g(B) \leq \text{Area}_g(\partial_{\text{man}} D_j) \leq P_g(F_j).$$

Let first $j \rightarrow \infty$, then $t \downarrow 0$. This proves $a_g(B) \leq P_g(E)$ for every competitor and hence (2.4). Applied to a minimizer, the same argument and the lower bound $a_g(B)$ show convergence of the resulting cut areas to its full perimeter. It also proves independence of the inner metric extension.

The connected exterior. Lemma 2.2, proved in the next subsection, identifies a closed representative with frontier T , whose singular part has zero $\mathcal{H}_g^{n-1}$ measure and whose regular charts have exactly one filled side and one exterior side. The complement has one component containing the distant end. Any other component U is bounded, with $\partial U \subset T$. Since $\mathcal{H}_g^{n-1}(T) = P_g(E) < \infty$, the finite-boundary-measure criterion for perimeter gives U finite perimeter. On its regular boundary charts its outer normal is opposite to ν_E . The singular part has zero perimeter measure, so the perimeter formula for unions gives

$$P_g(E \cup U) = P_g(E) - P_g(U).$$

The open set U has positive volume and is bounded; its perimeter is positive, for otherwise its indicator would be constant on connected M . Filling it contradicts minimality. Thus the exterior is connected. This argument does not require the singular frontier to be a smooth manifold.

If $n \leq 7$, Lemma 2.2 makes T a compact embedded $C^{1,1}$ hypersurface. A smooth field transverse to its continuous outward normal has a flow collar of T . A small positive flow displaces T into $\text{int } X$ and enlarges its filled side. In a smaller collar, smooth a C^1 defining function in C^1 while retaining positive derivative along the transverse field. Its smooth zero set is a small graph over the displaced frontier. The graph displacement extends to a collar isotopy, fixed on O and on the distant end. The exterior remains connected, so these are full cuts. Letting the smoothing error and then the flow time tend to zero gives the asserted C^1 and area convergence.

Finally every bounded finite-perimeter superset of a minimizer still contains O , so global minimality gives the claimed outer minimization directly. $\square$

13.2 Local regularity at the obstacle

We establish the local input used above. Its first step converts the obstacle constraint into a bounded volume error; its second step shows that contact tangent cones are planes.

Proof of Lemma 2.2. The obstacle first converts constrained minimization to an ordinary almost-minimization estimate. Extend the outward unit normal of O to a smooth field Z with $|Z|_g \leq 1$ and bounded divergence. It suffices to work in a collar and to cut off farther away. For a local replacement F , the set $F \cup O$ is admissible. Perimeter submodularity and Gauss–Green give, after canceling unchanged perimeter outside the replacement region V ,

$$\begin{aligned} P_g(E; V) &\leq P_g(F; V) + P_g(O; V) - P_g(F \cap O; V) \\ &\leq P_g(F; V) + \Lambda \text{Vol}_g(E \triangle F), \end{aligned} \tag{13.3}$$

where Λ bounds $|\text{div}_g Z|$ on the region. To check the second inequality without assuming that O has finite total perimeter, take $Z = \nu_O$ on $\partial O \cap V$ and cut it off in a larger neighborhood. Since $F \cap O$ and O agree outside a compact subset of V , Gauss–Green for their indicator difference gives

$$P_g(O; V) - P_g(F \cap O; V) \leq \int_{O \setminus F} \text{div}_g Z \, dV_g \leq \Lambda \text{Vol}_g(E \triangle F).$$

All the perimeters in this calculation are relative to V .

Insertion and deletion of small balls in (13.3), followed by the relative isoperimetric inequality, give the usual two-sided volume-density bounds and upper and lower perimeter-density bounds.

They identify the frontier of the closed representative with the support of its perimeter measure. The codimension-one almost-minimizer regularity theorem gives a $C^{1,\alpha}$ regular part and a singular set of dimension at most $n - 8$; the Riemannian version uses the smooth area integrand in coordinate charts. The error in a radius- r ball is $O(r^n)$, hence vanishes after rescaling perimeter by r^{1-n} . Blowups are therefore genuine Euclidean perimeter-minimizing cones. These are the density and dimension-reduction conclusions of [24, Chapters 21, 26, and 28]; for the smooth minimizing case see [34, Chapter 7, Theorem 5.8].

To see why no contact point is singular, take a tangent filled set at a point of ∂O . It contains the tangent half-space of O , and its boundary cone is supported in the complementary closed half-space. That boundary is a stationary multiplicity-one minimizing cone. Write $k = n - 1$. On its spherical link the nonnegative coordinate height z satisfies, weakly, $\Delta z = -(k - 1)z$. This identity follows by restricting linear coordinate functions to a stationary cone. The link is compact and stationarity includes its singular set, so testing the identity by the constant function gives $0 = -(k - 1) \int z$. Since $k - 1 > 0$, the height vanishes. Thus the boundary cone is the hyperplane bounding the half-space. Constancy and the fact that it is the boundary of a set give multiplicity one; the two density bounds exclude an empty boundary. Density-one regularity for almost-minimizers now gives a single contact graph.

Apply (13.3) to flows of both signs in such a graph cylinder. The resulting first-variation functional has a bounded weak mean curvature. Its graph function v solves a uniformly elliptic prescribed-mean-curvature equation with bounded right-hand side. The coefficients are smooth functions of position, v , and Dv , on the bounded gradient range supplied by $C^{1,\alpha}$ regularity. Difference quotients first yield local $W^{2,2}$ regularity. In nondivergence form the principal coefficients are $C^{0,\alpha}$ and the right-hand side is bounded; the local linear $W^{2,p}$ estimates give $v \in W^{2,p}$ for every finite p . Sobolev embedding then gives $C^{1,\alpha}$ for each $\alpha < 1$. See [16, Chapter 9] for these elliptic estimates. Off the obstacle, all local replacements are admissible, so the weak mean curvature is zero; interior elliptic regularity gives smoothness and minimality on every regular patch.

For the stronger low-dimensional contact conclusion we apply the smooth-obstacle theorem directly: a pure-perimeter minimizer with a C^2 obstacle in ambient dimension below eight has $C^{1,1}$ frontier, smooth away from contact [19, Regularity Theorem 1.3(iii), pp. 368–369]. The smooth metric and obstacle satisfy precisely those local hypotheses. Its graph functions are consequently in $W^{2,\infty}$. Finally the dimension estimate makes the singular set $\mathcal{H}^{n-1}$ -null, while the regular frontier agrees with the reduced boundary up to such a null set. This proves the area identity. $\square$

13.3 Detachment and completeness

It remains to use the strict curvature sign. Adding a thin collar reduces perimeter wherever that collar is not already filled. This comparison proves detachment before requiring smoothness of the minimizing frontier.

Proof of Lemma 2.3. Compactness and strict negativity give a signed-distance collar $0 \leq t \leq 2\varepsilon$ outside B on which $Y = \partial_t$ obeys

$$|Y|_g = 1, \quad \operatorname{div}_g Y \leq -\beta < 0.$$

For a good slicing level $s \in (0, 2\varepsilon)$ set $O_s = O \cup \{0 < t < s\}$ and $W_s = O_s \setminus E$. Gauss–Green,

using a smooth extension of Y across B , gives

$$\int_{W_s} \operatorname{div}_g Y \, dV_g = \int_{\{t=s\}} \mathbf{1}_{E^c} \, dA_g - \int_{\partial^* E \cap \{0 \leq t < s\}} g(Y, \nu_E) \, dA_g.$$

The second integral includes any original frontier at $t = 0$. The perimeter added by adjoining O_s is the first boundary term; the removed perimeter is at least the second. Therefore

$$P_g(E \cup O_s) - P_g(E) \leq \int_{W_s} \operatorname{div}_g Y \, dV_g \leq -\beta \operatorname{Vol}_g(W_s). \quad (13.4)$$

The enlarged set is admissible. Minimality forces W_s to have zero volume. For each minimizer choose one good level $s \in (\varepsilon, 2\varepsilon)$. The common smaller collar $0 \leq t < \varepsilon$ is then filled almost everywhere. The density bounds for the closed representative exclude any frontier point in that collar, so it lies in the interior of E .

The frontier is now separated from the obstacle. Every local perimeter replacement supported sufficiently near it is admissible, so it is locally perimeter minimizing there.

To express this local minimality in the language of integral currents, choose a coordinate ball U with compact closure disjoint from the obstacle, and orient that chart. Since E has finite perimeter, $P_g(E; U) < \infty$. Write $C = \partial \llbracket E \rrbracket$ as a current in U . If Z is an integral $(n-1)$ -cycle compactly supported in U , coning in the ball coordinates fills it by an integral n -current S compactly supported in U . A top-dimensional integral current has the form $S = \llbracket k \rrbracket$, where k is integer-valued and belongs to $BV(U)$; here k has compact support and $\partial S = Z$. Thus $C + Z = \partial \llbracket \mathbf{1}_E + k \rrbracket$. The function $c(t) = \min\{1, \max\{0, t\}\}$ is 1-Lipschitz and maps integers to $\{0, 1\}$. Consequently $c(\mathbf{1}_E + k) = \mathbf{1}_F$ for a finite-perimeter set F agreeing with E outside a compact subset of U . The contraction property of BV variation and set minimality give

$$P_g(E; U) \leq P_g(F; U) \leq |D(\mathbf{1}_E + k)|_g(U) = \mathbf{M}_g(C + Z).$$

Since $P_g(E; U) = \mathbf{M}_g(C)$, this is the local integral-current minimizing property used by interior boundary regularity. The orientation is needed only in this chart; reversing it changes all current signs and leaves the mass comparison unchanged.

Applying ambient flows of both signs gives

$$\int_{\partial^* E} \operatorname{div}_{T_x \partial^* E} V \, dA_g = 0$$

for each smooth vector field V supported near T . Thus stationarity holds across the entire frontier, with no extra first-variation term at the singular set. Lemma 2.2 and interior minimizing-boundary regularity give the stated dimensional conclusions; see [34, Chapter 7, Theorem 5.8]. For the integral-current formulation and its singular-set dimension reduction, see [13, 14]. Detachment removes contact with the obstacle; the dimension restriction is what removes interior singularities.

Lemma 2.1 supplies connectedness and boundedness of the removed region, so the exterior retains exactly the original end. It is closed in the complete ambient manifold. To check completeness also for intrinsic Cauchy sequences, choose a subsequence whose successive intrinsic distances are less than 2^{-j} and join its successive terms by curves in D of lengths less than 2^{1-j} . Their concatenation has finite length and therefore an ambient endpoint. Closedness puts the endpoint in D . Appending it to the concatenated curve gives intrinsic convergence, since the remaining tail lengths tend to zero. The original Cauchy sequence then converges as well. When $n \leq 7$, the exterior is in addition a smooth manifold with boundary T .

Every bounded finite-perimeter enlargement of E remains a competitor containing O , proving outer minimization. In the smooth case any full cut of D determines precisely such an enlargement; all its components and portions on T contribute to its perimeter. The full-area identity follows from (2.4) and the zero $(n-1)$ -dimensional measure of the singular set. $\square$

14 Harmonic-flat approximation after a strict smooth boundary

We pass from a general asymptotically flat end to a harmonically flat one after constructing a strict smooth inner barrier. Uniform metric comparison will control every enclosing area, while an annular flux calculation controls the mass. Set $d = n - 2$, $k = n - 1$, $\omega = |S^{n-1}|$, and $a_n = 4k/d$.

Lemma 14.1 (Smooth exterior harmonic-flat approximation). *Let (N, h) be a smooth connected asymptotically flat exterior, complete including its compact smooth boundary Γ , possibly empty, with one coordinate end and compact end complement. Suppose that*

$$h - \delta = O_2(r^{-\tau}), \quad \tau > d/2, \quad R_h \geq 0, \quad R_h \in L^1, \quad R_h = O(r^{-n-\beta})$$

for some $\beta > 0$, and that its ADM mass exists. There are smooth metrics $\hat{h}_R$, complete with the same boundary, with nonnegative scalar curvature and a harmonically flat end, such that

$$\begin{aligned} (1 - \epsilon_R)h &\leq \hat{h}_R \leq (1 + \epsilon_R)h, & \epsilon_R &\longrightarrow 0, \\ \hat{h}_R &\longrightarrow h \text{ in } C^2 \text{ on every fixed compact set,} \\ m_{\text{ADM}}(\hat{h}_R) &\longrightarrow m_{\text{ADM}}(h). \end{aligned}$$

In particular, if $H_\nu(\Gamma, h) < 0$ for the normal toward the end, the same strict sign holds for $\hat{h}_R$ when R is large.

Proof. We cut off the metric on a distant annulus and remove the resulting scalar-curvature error by a Dirichlet conformal correction. The annular flux then recovers the original ADM mass.

Step 1: the cutoff and Dirichlet correction. Choose a smooth radial cutoff $\chi_R = \chi(r/R)$, with $\chi = 1$ on $(-\infty, 1]$, $\chi = 0$ on $[2, \infty)$, and $0 \leq \chi \leq 1$. On the end put

$$h_R = \delta + \chi_R(h - \delta),$$

and set $h_R = h$ on the compact core. For large R this is a smooth positive metric, equal to h on $r \leq R$ and Euclidean on $r \geq 2R$. Extend $\chi_R = 1$ on the compact core and define

$$Q_R = R_{h_R} - \chi_R R_h.$$

Thus Q_R is supported in the annulus $\mathcal{A}_R = \{R < r < 2R\}$. The differentiated metric bounds and the scalar-curvature formula give

$$|Q_R| \leq CR^{-\tau-2}, \quad \|Q_R\|_{L^{n/2}(h_R)} \leq CR^{-\tau}, \quad \|Q_R\|_{L^{2n/(n+2)}(h_R)} \leq CR^{d/2-\tau}.$$

The metrics h_R are uniformly comparable with h , so the Sobolev inequality

$$\|v\|_{L^{2^*}(h_R)}^2 \leq C \int_N |dv|_{h_R}^2 dV_{h_R}, \quad 2^* = \frac{2n}{n-2},$$

has a constant independent of large R on the homogeneous completion of $C_c^\infty(\text{int } N)$. This is the Euclidean-end Sobolev inequality combined with a compact-core Poincaré inequality with trace on one fixed end annulus. The completion imposes zero trace at Γ .

Since $\|Q_R\|_{n/2} \rightarrow 0$, the bilinear form

$$\mathcal{B}_R(v, \varphi) = a_n \int \langle dv, d\varphi \rangle_{h_R} + \int Q_R v \varphi$$

is coercive, with $\mathcal{B}_R(v, v) \geq (a_n/2) \int |dv|^2$. Solve

$$\mathcal{B}_R(v_R, \varphi) = - \int Q_R \varphi, \quad z_R = 1 + v_R.$$

Then

$$(-a_n \Delta_{h_R} + Q_R) z_R = 0, \quad z_R|_\Gamma = 1, \quad \|dv_R\|_2 + \|v_R\|_{2^*} \leq CR^{d/2-\tau}.$$

All integrals in these estimates use h_R . Local elliptic regularity makes v_R smooth up to Γ . On the Euclidean end it is harmonic and belongs to L^{2^*} ; the scaled mean-value estimate implies $v_R \rightarrow 0$ at infinity.

Step 2: uniform and compact convergence. We first prove a uniform supremum estimate by annular rescaling. Cover $\mathcal{A}_R$ by a fixed number of balls of radius comparable to R in $\{R/2 < r < 4R\}$, and rescale each by R . The rescaled metrics have uniform ellipticity and bounded coefficients through order two; the rescaled zeroth-order term $R^2 Q_R$ is $O(R^{-\tau})$. The scalar interior boundedness estimate for this equation gives

$$\sup_{\mathcal{A}_R} |v_R| \leq C \left[R^{-d/2} \|v_R\|_{2^*} + R^2 \|Q_R\|_\infty \right] \leq CR^{-\tau}.$$

The same bound holds on the closed source annulus by using these slightly larger balls. Off that annulus v_R is harmonic, has zero trace on Γ , and tends to zero on the end. The maximum principle on the inner and outer complementary regions therefore gives $\|v_R\|_\infty \leq CR^{-\tau}$ on all of N . In particular $z_R > 0$ for large R . On each fixed compact region the equation for v_R is $\Delta_h v_R = 0$ once R is large. Interior and fixed smooth boundary estimates, with $v_R = 0$ on Γ , give $v_R \rightarrow 0$ in C^2 there, and in every fixed higher norm.

Put $\hat{h}_R = z_R^{4/d} h_R$. The conformal scalar formula now gives exactly

$$R_{\hat{h}_R} = z_R^{-(n+2)/d} (-a_n \Delta_{h_R} z_R + R_{h_R} z_R) = z_R^{-4/d} \chi_R R_h \geq 0.$$

The global supremum estimate and $\sup_N |h_R - h|_h = O(R^{-\tau})$ give the stated metric comparison and completeness. The end is $z_R^{4/d} \delta$, with z_R positive Euclidean harmonic and tending to one, hence is harmonically flat to every order.

Step 3: the ADM limit and the boundary flux. Write

$$z_R = 1 + A_R r^{-d} + O_j(r^{-d-1})$$

on that Euclidean end. Let ν denote the h -unit normal at Γ pointing into N , and put $I_R = \int_\Gamma \partial_\nu z_R dA_h$. The boundary C^2 convergence gives $I_R = o(1)$. Since $h_R = h$ near Γ , integrating $a_n \Delta_{h_R} z_R = Q_R z_R$ over a large truncation and then letting its radius tend to infinity gives, with the outward normal of the exterior domain equal to $-\nu$ at Γ ,

$$-a_n d\omega A_R - a_n I_R = \int_N Q_R z_R dV_{h_R}.$$

The nonlinear factor is harmless:

$$\left| \int_N Q_R v_R dV_{h_R} \right| \leq \|Q_R\|_{2n/(n+2)} \|v_R\|_{2^*} \leq CR^{d-2\tau} = o(1).$$

For the remaining integral, write $b = h - \delta$ and

$$\mathcal{L}(b) = \partial_i \partial_j b_{ij} - \Delta_\delta \operatorname{tr}_\delta b.$$

On $\mathcal{A}_R$,

$$R_{h_R} dV_{h_R} = \left[\mathcal{L}(\chi_R b) + O(R^{-2\tau-2}) \right] dx.$$

Indeed all nonlinear scalar terms and the change of volume density are bounded by $C(|b_R| |D^2 b_R| + |Db_R|^2)$, where $b_R = \chi_R b$. Their annular integral is $O(R^{d-2\tau})$. The outer linear flux is zero since $b_R = 0$ there; its inner flux is the negative of the original ADM flux on S_R . Consequently

$$\int_N Q_R dV_{h_R} = - \int_{S_R} (\partial_j h_{ij} - \partial_i h_{jj}) n_\delta^i dA_\delta + O(R^{d-2\tau}) - \int_{\mathcal{A}_R} \chi_R R_h dV_{h_R}.$$

The last term tends to zero, by the scalar tail assumption (or by integrability and uniform metric comparison). Thus

$$\int_N Q_R dV_{h_R} = -2k\omega m_{\text{ADM}}(h) + o(1).$$

As $a_n d = 4k$, the flux identity yields

$$A_R \longrightarrow \frac{1}{2} m_{\text{ADM}}(h), \quad m_{\text{ADM}}(\hat{h}_R) = 2A_R \longrightarrow m_{\text{ADM}}(h).$$

Finally $H_\nu(\Gamma, \hat{h}_R) \rightarrow H_\nu(\Gamma, h)$ uniformly by the compact C^2 convergence. Strict negativity therefore persists on the compact boundary. $\square$

Proof of Theorem 1.1. The case $3 \leq n \leq 7$ is the numerical Bray–Lee theorem described in Section 1, with the classical positive mass theorem when the frontier is empty. Assume $n \geq 8$. Proposition 12.1 has proved the nonempty-frontier case for harmonically flat ends. We now let (E, h) have the general asymptotics of the theorem and full perimeter $A > 0$.

The capacitary coefficient at infinity. Let w be its capacitary potential and c its capacity. The local minimizing density bounds and the weak Harnack contraction for the equilibrium potential, as in the proof of Lemma 4.1, give its constant extension the continuous boundary value one on the full frontier. For any $0 < \alpha < \min\{\tau, 1\}$, the positive function $r^{-d}(1 - r^{-\alpha})$ is h -superharmonic for all sufficiently large r : its Euclidean Laplacian is $-\alpha(d + \alpha)r^{-n-\alpha}$, while the metric error is $O(r^{-n-\tau})$. Comparison with a fixed positive multiple on a large coordinate sphere, followed by the maximum principle on the end, gives $w = O(r^{-d})$. Scaled interior estimates give $Dw = O(r^{-d-1})$, $D^2w = O(r^{-d-2})$. Consequently

$$\Delta_\delta w = (\delta^{ij} - h^{ij}) \partial_{ij} w + h^{ij} \Gamma_{ij}^\ell \partial_\ell w = O(r^{-n-\tau}).$$

Extend a cutoff of w , supported in its end chart, smoothly by zero to $\mathbb{R}^n$. Its Euclidean Laplacian has a finite α -moment. The Newton potential, split into inner, comparable and outer annuli, is therefore a monopole plus $O_1(r^{-d-\alpha})$. Its difference from the extended function is a decaying harmonic function on all of $\mathbb{R}^n$, hence zero. The cutoff has reduced this uniqueness statement to

Euclidean space. If the monopole coefficient is c_0 , flux integration on $\{w < s\}$, for regular values $s \uparrow 1$, gives

$$\int_{\{w < s\}} |dw|_h^2 dV_h = s d\omega c_0.$$

The flux at infinity is $-d\omega c_0$; the other boundary is the regular level $w = s$, and the buried boundary is absent. Monotone convergence and the capacity normalization identify $c_0 = c$. Thus the finite-energy construction and this end argument give

$$0 < w < 1 \text{ in the open exterior,} \quad w = cr^{-d} + O_1(r^{-d-\alpha}), \quad 0 < \alpha < \min\{\tau, 1\}.$$

In particular the conformal mass formula applies.

A strict smooth inner boundary. Fix a small $\lambda > 0$ and set

$$p = 1 + \lambda w, \quad h_\lambda = p^{4/d} h.$$

Then $h_\lambda \geq h$, its asymptotically flat rate is $\min\{\tau, d\} > d/2$, and

$$R_{h_\lambda} = p^{-4/d} R_h \geq 0, \quad R_{h_\lambda} dV_{h_\lambda} = p^2 R_h dV_h.$$

Since $1 \leq p \leq 1 + \lambda$, scalar integrability and the stated scalar decay persist. The end expansion gives

$$m_{\text{ADM}}(h_\lambda) = m_{\text{ADM}}(h) + 2\lambda c.$$

To apply the static part of Corollary 6.2, write this perturbation as

$$h_\lambda^{\text{base}} = (1 + \lambda)^{4/d} h, \quad \hat{u}_\lambda = \frac{1 + \lambda w}{1 + \lambda}, \quad h_\lambda = \hat{u}_\lambda^{4/d} h_\lambda^{\text{base}}.$$

The normalized harmonic factor equals one on the frontier and lies strictly between zero and one on the connected open exterior. Constant scaling preserves local perimeter minimality of the original filled set. Its smooth ambient metric supplies the minimizing-boundary density and regular-patch estimates, and the harmonic factor has the constant boundary data required by the separation argument. The asymptotically flat end supplies the positive-mean-curvature coordinate spheres used to confine the prescribed-curvature minimizer. Thus the static detachment conclusion applies to the original conductor. Proposition 7.2 then gives a smooth compact enclosing boundary Γ with $H_\nu(\Gamma, h_\lambda) < 0$. Retain its connected end exterior. The metric h_λ is smooth on a neighborhood of its closure. All parameters producing Γ are fixed before approximation.

Comparison of all enclosing areas and passage to the limit. Apply Lemma 14.1 on this smooth exterior. Let k_j be its approximating metrics, and choose $\epsilon_j \downarrow 0$ so that

$$k_j \geq (1 - \epsilon_j) h_\lambda, \quad H_\nu(\Gamma, k_j) < 0, \quad m_{\text{ADM}}(k_j) \longrightarrow m_{\text{ADM}}(h_\lambda).$$

Take a minimizing enclosure of the region removed by Γ in k_j . Lemma 2.1 gives compact existence, and Lemma 2.3 puts the entire minimizing frontier a positive distance from the strict smooth obstacle. The resulting frontier Σ_j is detached and locally perimeter minimizing, and is outer minimizing with all components counted. Lemma 2.1 gives its connected end exterior. It has precisely the boundary class of Theorem 1.1, a harmonically flat end and scalar curvature zero on that distant end.

Every such enclosure contains the original conductor. Its full perimeter in h is therefore at least A , by the original outer-minimizing property. This comparison applies directly to finite-perimeter enclosures, or follows from strict smooth approximation keeping a fixed neighborhood of the conductor. Tangent-plane metric comparison gives

$$\mathcal{H}_{k_j}^k(\Sigma_j) \geq (1 - \epsilon_j)^{k/2} \mathcal{H}_{h_\lambda}^k(\Sigma_j) \geq (1 - \epsilon_j)^{k/2} A.$$

Proposition 12.1 consequently gives

$$m_{\text{ADM}}(k_j) \geq \frac{1}{2} \left(\frac{(1 - \epsilon_j)^{k/2} A}{\omega} \right)^{d/k}.$$

First let $j \rightarrow \infty$ at fixed λ , and then $\lambda \downarrow 0$. The mass formula proves

$$m_{\text{ADM}}(h) \geq \frac{1}{2} (A/\omega)^{d/k}.$$

The strict boundary and elliptic constants may depend on the fixed λ and on Γ ; this is consistent with the stated order of limits. The area estimate came from metric comparison and conductor inclusion.

The empty frontier. A nonempty locally perimeter-minimizing frontier has positive perimeter by the density lower bound. If $A = 0$, the frontier is therefore empty. On that complete boundaryless manifold apply the same cutoff-density proof with $\Gamma = \emptyset$; there is then no compact-boundary flux. Applying Lemma 8.1 to the harmonic-flat approximants and passing their masses to the limit gives $m_{\text{ADM}}(h) \geq 0$. $\square$

A A quantitative singular-set bound for weighted minimizers

The smooth positive mass theorem used in Section 8 is [10, Corollary 1.6]. In its dimension-descent proof, a weighted perimeter minimizer with a volume forcing is completed across its singular set. This appendix verifies the quantitative covering estimate needed for that variational class. It supplies that input to the cited proof; the positive mass conclusion remains the external theorem.

The quantitative input here is [27, Theorem 1.3], with the Riemannian bounded-mean-curvature hypotheses (1.14)–(1.15) of preprint version 4. Bounded mean curvature alone does not place a singular set in a quantitative stratum of codimension eight. We use local minimization to prove that inclusion at every sufficiently small scale, then apply the quantitative theorem.

Lemma A.1 (A quantitative estimate for weighted minimizers). *Let $n \geq 8$, let g be smooth on an open set U , and let $\hat{\rho} > 0$ and Φ be smooth there. Put*

$$g_* = \hat{\rho}^{2/(n-1)} g, \quad f = \Phi \hat{\rho}^{-1/(n-1)}.$$

Suppose that a locally finite-perimeter set E locally minimizes

$$\mathcal{F}(E; W) = P_{g_*}(E; W) - \int_{E \cap W} f dV_{g_*}$$

against changes compactly supported in $W \Subset U$. Let $\mathcal{S}$ be a compact subset of the singular set of its perimeter support in U . There are constants $C, t_0 > 0$ such that

$$\text{Vol}_g(B_t^g(\mathcal{S})) \leq C t^8, \quad 0 < t < t_0. \tag{A.1}$$

The constants are local: they may depend on the minimizer and on a buffered compact neighborhood of $\mathcal{S}$.

Proof. The proof has three parts. Absorbing the density into the metric gives bounded generalized mean curvature and strict blow-up compactness. Compactness and low-dimensional regularity then exclude cones with too many translation symmetries at every small scale. The quantitative stratification theorem converts this exclusion into the tube estimate.

The conformal metric absorbs the positive perimeter density: $d\mathcal{H}_{g_*}^{n-1} = \hat{\rho} d\mathcal{H}_g^{n-1}$. Also $f dV_{g_*} = \Phi \hat{\rho} dV_g$, so the displayed functional is exactly the corresponding weighted perimeter minus weighted volume functional. All estimates below are first made in g_* . Choose neighborhoods

$$\mathcal{S} \subset U_0 \Subset U_1 \Subset U.$$

Smoothness and positivity make their metric and forcing bounds uniform after working on a slightly larger compact neighborhood of $\overline{U}_1$.

First variation and local estimates. Let $V = |\partial^* E|$ be the multiplicity-one boundary varifold, computed in g_* . Domain variation by any compactly supported smooth vector field Y gives

$$\delta_{g_*} V(Y) = \int_{\partial^* E} f \langle Y, \nu_E \rangle_{g_*} d\mathcal{H}_{g_*}^{n-1}.$$

Here the divergence theorem for finite-perimeter sets is applied to fY . Thus V has bounded generalized mean curvature, with no additional first-variation measure on its singular set. In particular

$$|\delta_{g_*} V(Y)| \leq \|f\|_{L^\infty(U_1)} \int |Y|_{g_*} d\|V\|.$$

The minimizing comparison also gives

$$P_{g_*}(E; W) \leq P_{g_*}(F; W) + \|f\|_{L^\infty(U_1)} \text{Vol}_{g_*}(E \Delta F) \quad (\text{A.2})$$

for changes supported in $W \Subset U_1$. The usual ball comparisons and relative isoperimetric inequality for this bounded-volume-forcing functional give uniform local perimeter upper bounds and two-phase density lower bounds. In particular, for x in the perimeter support in $\overline{U}_0$ and all sufficiently small s , both phases occupy at least cs^n of $B_s^{g_*}(x)$, and

$$cs^{n-1} \leq P_{g_*}(E; B_s^{g_*}(x)) \leq Cs^{n-1}.$$

These are the local estimates for almost-minimizing boundaries; they do not say that the boundary is Euclidean area minimizing in a fixed coordinate chart.

Strict blow-up compactness. Consider singular points $x_j \in \overline{U}_0$ and radii $r_j \downarrow 0$. Choose g_* -orthonormal frames at x_j and write

$$\Psi_j(y) = \exp_{x_j}^{g_*}(r_j y), \quad g_j = r_j^{-2} \Psi_j^* g_*, \quad f_j(y) = r_j f(\Psi_j(y)).$$

The rescaled sets $E_j = \Psi_j^{-1}(E)$ minimize $P_{g_j} - \int f_j dV_{g_j}$ on domains exhausting $\mathbb{R}^n$. On every fixed ball $g_j \rightarrow \delta$ and $f_j \rightarrow 0$ smoothly. The preceding perimeter bounds give, after passage to a subsequence,

$$\chi_{E_j} \longrightarrow \chi_{E_\infty} \quad \text{in } L_{\text{loc}}^1(\mathbb{R}^n).$$

We need strict perimeter convergence as well, since L^1 convergence by itself could lose cancelling sheets.

Fix $0 < a < b$ in a bounded coordinate ball. Coarea permits radii $s_j \in (a, b)$ for which the traces exist, neither perimeter measure charges ∂B_{s_j} , and

$$\int_{\partial B_{s_j}} |\chi_{E_j} - \chi_{E_\infty}| d\mathcal{H}^{n-1} \longrightarrow 0.$$

Glue E_∞ inside B_{s_j} to E_j outside. The modification is compactly supported in a larger fixed ball. The BV gluing formula and minimality give

$$\begin{aligned} P_{g_j}(E_j; B_{s_j}) &\leq P_{g_j}(E_\infty; B_{s_j}) + C \int_{\partial B_{s_j}} |\chi_{E_j} - \chi_{E_\infty}| d\mathcal{H}^{n-1} \\ &\quad + \|f_j\|_{L^\infty(B_b)} \text{Vol}_{g_j}((E_j \triangle E_\infty) \cap B_{s_j}). \end{aligned}$$

The last two terms tend to zero. Uniform convergence of the metrics compares their perimeters with Euclidean perimeter by factors $1 + o(1)$, so

$$\limsup_j P_\delta(E_j; B_a) \leq P_\delta(E_\infty; B_b).$$

Let $b \downarrow a$ at a radius with $|D\chi_{E_\infty}|(\partial B_a) = 0$, and use lower semicontinuity. It follows that

$$P_\delta(E_j; B_a) \longrightarrow P_\delta(E_\infty; B_a). \quad (\text{A.3})$$

The same argument on other relatively compact balls gives local perimeter-measure convergence. Together with $D\chi_{E_j} \rightharpoonup D\chi_{E_\infty}$, Reshetnyak continuity now gives boundary-varifold convergence: the unoriented tangent plane is a continuous even function of the measure-theoretic normal. Changing between the g_j and Euclidean varifold conventions introduces only vanishing metric errors. In particular there is no residual varifold mass from disappearing or cancelling sheets.

The limit is a Euclidean locally perimeter-minimizing boundary. Indeed, for $F \triangle E_\infty \Subset B_a$, repeat the same gluing with F inside B_{s_j} . On the gluing annulus $F = E_\infty$, so the trace error is unchanged. Letting $j \rightarrow \infty$ and then $b \downarrow a$ proves $P(E_\infty; B_a) \leq P(F; B_a)$ at continuity radii. Such a ball can contain any compact modification. Finally, the two-phase density bounds at x_j pass to every fixed rescaled ball centered at zero. Both phases of E_∞ have positive density there; in particular its boundary varifold is nonzero.

Uniform exclusion of highly symmetric cones. We claim that there are single constants $\epsilon, R_0 > 0$ such that, at every singular point in $\bar{U}_0$ and every physical scale $0 < s < R_0$, the rescaled varifold is not ϵ -close to an $(n-7)$ -symmetric comparison cone. Comparison is in the weak-varifold metric on B_1 used in quantitative stratification, with balls contained in the buffered neighborhood. Such a cone is dilation invariant about zero and translation invariant along an $(n-7)$ -plane; it is not assumed to be a minimizing boundary.

If the claim fails, choose x_j, r_j as above and comparison cones C_j with distance less than $1/j$ on B_1 . Denote their invariant $(n-7)$ -planes by K_j and pass to $K_j \rightarrow K$ in the Grassmannian. Closeness on the open ball gives a uniform mass bound for C_j on $B_{1/2}$, by using a cutoff equal to one there and supported in $B_{3/4}$. No mass bound at the boundary of the comparison ball is needed. Cone homogeneity propagates this interior bound to every fixed B_R . Radon compactness on the Grassmann bundle therefore gives a global varifold limit C , homogeneous about zero and translation invariant along K . The strict compactness just proved identifies it on the comparison ball:

$$C \llcorner B_1 = |\partial^* E_\infty| \llcorner B_1. \quad (\text{A.4})$$

There was no first-variation bound for the arbitrary C_j . Consequently it is this identification with the minimizing-boundary limit, not their mass bounds alone, that gives rectifiability, integrality and multiplicity one in B_1 . Homogeneity extends these properties to C globally.

We next construct a boundary-set cone; we do not assert that the moving-center limit E_∞ is conical outside B_1 . The radial direction is tangent almost everywhere to a rectifiable cone. Equation (A.4) therefore implies, on $B_1 \setminus \{0\}$,

$$x \cdot D\chi_{E_\infty} = 0, \quad \operatorname{div}(x\chi_{E_\infty}) = n\chi_{E_\infty} dx.$$

The second expression includes the dimensional term. In polar coordinates the first identity says that the phase is constant along almost every radial segment. Extend those phase values radially to a conical set $E_0 \subset \mathbb{R}^n$. It agrees with E_∞ almost everywhere in B_1 . The homogeneous perimeter bound gives locally finite perimeter also at the origin: cutting out a ball of radius a introduces at most $O(a^{n-1})$ additional perimeter, which vanishes as $a \downarrow 0$. Hence

$$|\partial^* E_0| = C \quad \text{on } \mathbb{R}^n,$$

first by agreement inside B_1 , and then by homogeneity.

The conical set E_0 is globally locally perimeter minimizing. Given a compact modification, homothetically contract a ball containing its support into B_1 . In that ball E_0 equals E_∞ , so the minimizing comparison is available. Undoing the contraction and using the $(n-1)$ -degree scaling of perimeter proves the comparison at the original scale. Only the phase extension E_0 , not the earlier limit outside B_1 , is used in this scaling argument.

Splitting as a minimizing boundary. Translation invariance of the rectifiable varifold C along K makes every $v \in K$ tangent almost everywhere. Thus

$$v \cdot D\chi_{E_0} = 0 \quad (v \in K).$$

Distributional constancy in those directions and Fubini yield $E_0 = K \times F$ up to a null set, for a locally finite-perimeter conical set $F \subset K^\perp$. This preserves the phase information that would be absent from a statement about splitting stationary varifolds alone.

The factor F minimizes perimeter. To see this directly, let $\ell = \dim K$ and $G \triangle F \Subset D \subset K^\perp$ be a bounded compact modification. Over a cube $Q_L \subset K$ of side L , replace F by G , retaining E_0 elsewhere. The product gluing formula and the minimality of E_0 give

$$0 \leq |Q_L| [P(G; D) - P(F; D)] + P(Q_L) |G \triangle F|.$$

There is no interface contribution at ∂D because the phases agree near it. The displayed lateral cost at ∂Q_L is essential. Divide by $|Q_L|$ and let $L \rightarrow \infty$; its coefficient tends to zero. Thus $P(F; D) \leq P(G; D)$. Here $\ell = n - 7 \geq 1$; if there are more invariant directions the residual dimension only decreases.

The residual ambient dimension is at most seven. The classical regularity theorem for minimizing boundaries therefore makes the minimizing cone F regular at its vertex [34]. A nontrivial boundary cone smooth at its vertex is a hyperplane, so F is a halfspace. Nontriviality follows from the two-phase density bounds. Consequently C is a multiplicity-one hyperplane through zero.

The strict boundary-varifold convergence to this plane, $g_j \rightarrow \delta$ smoothly, and $f_j \rightarrow 0$ now give flat-boundary regularity in a fixed smaller ball. One can use the metric-uniform almost-minimizing-boundary theorem or Riemannian Allard regularity; the prescribed-mean-curvature

equation then bootstraps the graph to smoothness. This contradicts singularity of E_j at zero. The argument uses Riemannian regularity with converging smooth metrics, not an unsupported bound for Euclidean mean curvature obtained by changing coordinates. It proves the claimed uniform ϵ, R_0 , at every singular center in the patch and every scale below R_0 .

The complete quantitative scale interval. Choose finitely many balls $B_{\rho_i/2}^{g*}(p_i)$ covering $\mathcal{S}$, with

$$B_{4\rho_i}^{g*}(p_i) \Subset U_0, \quad 4\rho_i < R_0.$$

They may be shrunk so that the rescaled ambient geometry has the sectional-curvature and injectivity-radius bounds in the cited Riemannian quantitative theorem. The first-variation bound and local perimeter bound already give its bounded generalized mean curvature and finite mass assumptions. The constants are finite on this fixed finite cover; no uniformity over unrelated metrics is asserted.

Apply [27, Theorem 1.3] to $B_{2\rho_i}^{g*}(p_i)$ after rescaling lengths by ρ_i^{-1} . Its target ball B_1 is $B_{\rho_i}^{g*}(p_i)$. For every singular point in that target ball and every rescaled scale $0 < s < 1$, the comparison ball lies in $B_{2\rho_i}^{g*}(p_i)$ and has physical radius $\rho_i s < R_0$. Thus the same symmetry exclusion holds on the entire interval $[r, 1)$, for each $0 < r < 1$. In the notation of quantitative stratification, the rescaled singular set in the target ball lies in

$$S_{\epsilon, r}^{n-8} \quad \text{for every } 0 < r < 1.$$

The stratum index is $n-8$ because the excluded cones have $n-7$ invariant directions. In particular this statement is stronger than a statement merely about tangent cones or a sequence of scales.

The quantitative theorem gives rescaled tube volume at most $C_i r^8$. Returning to physical variables, for $0 < t < \rho_i/2$ we obtain

$$\text{Vol}_{g*} \left(B_t^{g*}(\mathcal{S} \cap B_{\rho_i/2}^{g*}(p_i)) \right) \leq C_i \rho_i^{n-8} t^8.$$

The entire displayed tube lies in $B_{\rho_i}^{g*}(p_i)$, which is the target ball where the theorem controls its volume. For $t < \min_i \rho_i/2$, these finitely many tubes cover $B_t^{g*}(\mathcal{S})$. Summing gives its Ct^8 bound. Finally, smooth metric equivalence on the buffered compact neighborhood compares small tubes and volume elements for g and g_* , proving Equation (A.1). Ambient dimension eight corresponds to stratum index zero and is included. $\square$

The estimate supplies the covering bound used in [10, Theorem 3.34] for the weighted minimizers of its Definition 3.19 and Lemma 3.20. Indeed, disjoint equal-radius balls of radius t centered on a fixed compact singular set number at most Ct^{8-n} : each has volume at least ct^n , and their union lies in the tube just estimated. A maximal disjoint family has doubled balls covering the singular set. The resulting bound also implies every estimate with an arbitrarily small loss in its exponent, as required in that dimension-descent construction.

References

- [1] Luigi Ambrosio, Nicola Fusco, and Diego Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Mathematical Monographs. Oxford University Press, Oxford, 2000.
- [2] Victor Bangert. Sets with positive reach. *Archiv der Mathematik*, 38:54–57, 1982.

- [3] Robert Bartnik. The mass of an asymptotically flat manifold. *Communications on Pure and Applied Mathematics*, 39(5):661–693, 1986.
- [4] Yuchen Bi and Jintian Zhu. Curvature-free effects from volume growth and ends-counting and their applications. <https://arxiv.org/abs/2605.12403v2>, 2026. arXiv:2605.12403v2, 15 July 2026.
- [5] Yuchen Bi and Jintian Zhu. Riemannian Penrose inequality in all dimensions. <https://arxiv.org/abs/2605.00680v2>, 2026. arXiv:2605.00680v2, 15 September 2026.
- [6] Yuchen Bi and Jintian Zhu. Riemannian Penrose inequality in all dimensions. <https://arxiv.org/abs/2605.00680v1>, 2026. arXiv:2605.00680v1, 1 May 2026.
- [7] E. Bombieri and E. Giusti. Harnack’s inequality for elliptic differential equations on minimal surfaces. *Inventiones Mathematicae*, 15:24–46, 1972.
- [8] Hubert L. Bray. Proof of the Riemannian Penrose inequality using the positive mass theorem. *Journal of Differential Geometry*, 59(2):177–267, 2001.
- [9] Hubert L. Bray and Dan A. Lee. On the Riemannian Penrose inequality in dimensions less than eight. *Duke Mathematical Journal*, 148(1):81–106, 2009.
- [10] Simon Brendle and Yipeng Wang. A dimension descent scheme for the positive mass theorem in arbitrary dimension. <https://arxiv.org/abs/2604.08473v2>, 2026. arXiv:2604.08473v2, 20 April 2026.
- [11] Levi Lopes de Lima and Frederico Girão. The ADM mass of asymptotically flat hypersurfaces. *Transactions of the American Mathematical Society*, 367(9):6247–6266, 2015. Theorem locators refer to arXiv:1108.5474v3, 3 July 2012.
- [12] Daniela De Silva and David Jerison. A gradient bound for free boundary graphs. *Communications on Pure and Applied Mathematics*, 64(4):538–555, 2011. Theorem and page locators refer to the author manuscript, <https://math.mit.edu/~jerison/gbf.finaldraft.pdf>.
- [13] Herbert Federer. *Geometric Measure Theory*, volume 153 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, New York, 1969.
- [14] Herbert Federer. The singular sets of area minimizing rectifiable currents with codimension one and of area minimizing flat chains modulo two with arbitrary codimension. *Bulletin of the American Mathematical Society*, 76(4):767–771, 1970.
- [15] Robert Geroch. Energy extraction. *Annals of the New York Academy of Sciences*, 224(1):108–117, 1973.
- [16] David Gilbarg and Neil S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. Classics in Mathematics. Springer, Berlin, second edition, 2001. Reprint of the 1998 edition.
- [17] S. W. Hawking. Gravitational radiation in an expanding universe. *Journal of Mathematical Physics*, 9(4):598–604, 1968.
- [18] Marc Herzlich. A Penrose-like inequality for the mass of Riemannian asymptotically flat manifolds. *Communications in Mathematical Physics*, 188:121–133, 1997.

- [19] Gerhard Huisken and Tom Ilmanen. The inverse mean curvature flow and the Riemannian Penrose inequality. *Journal of Differential Geometry*, 59(3):353–437, 2001.
- [20] Tom Ilmanen. A strong maximum principle for singular minimal hypersurfaces. *Calculus of Variations and Partial Differential Equations*, 4:443–467, 1996.
- [21] Pong Soo Jang and Robert M. Wald. The positive energy conjecture and the cosmic censor hypothesis. *Journal of Mathematical Physics*, 18(1):41–44, 1977.
- [22] Mau-Kwong George Lam. The graph cases of the Riemannian Positive Mass and Penrose Inequalities in all dimensions. <https://arxiv.org/abs/1010.4256v1>, 2010. arXiv:1010.4256v1, 20 October 2010.
- [23] Dan A. Lee. On the near-equality case of the positive mass theorem. <https://arxiv.org/abs/0705.0677v1>, 2007. arXiv:0705.0677v1, 4 May 2007. Theorem 1.3 is the fixed-radius-factor version used here.
- [24] Francesco Maggi. *Sets of Finite Perimeter and Geometric Variational Problems: An Introduction to Geometric Measure Theory*, volume 135 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2012.
- [25] Edward Malec and Niall Ó Murchadha. Trapped surfaces and the Penrose inequality in spherically symmetric geometries. *Physical Review D*, 49(12):6931–6934, 1994.
- [26] Pengzi Miao. Positive mass theorem on manifolds admitting corners along a hypersurface. *Advances in Theoretical and Mathematical Physics*, 6(6):1163–1182, 2002. <https://arxiv.org/abs/math-ph/0212025v3>.
- [27] Aaron Naber and Daniele Valtorta. The singular structure and regularity of stationary varifolds. *Journal of the European Mathematical Society*, 22(10):3305–3382, 2020. Theorem and page locators refer to arXiv:1505.03428v4, 28 July 2017.
- [28] OpenAI. Boundary graph deformations for the spacetime Penrose inequality. OpenAI Math Release preprint [OAI:Boundary-graph-deformations-for-the-spacetime-Penrose-inequality-September-27-2026](#), 2026.
- [29] OpenAI. Equality and rigidity in the spacetime Penrose inequality. OpenAI Math Release preprint [OAI:Equality-and-rigidity-in-the-spacetime-Penrose-inequality-September-27-2026](#), 2026.
- [30] OpenAI. The spacetime Penrose inequality and enclosing area. OpenAI Math Release preprint [OAI:The-spacetime-Penrose-inequality-and-enclosing-area-September-27-2026](#), 2026.
- [31] Roger Penrose. Naked singularities. *Annals of the New York Academy of Sciences*, 224(1):125–134, 1973.
- [32] Ludovic Rifford. A Morse–Sard theorem for the distance function on Riemannian manifolds. *Manuscripta Mathematica*, 113:251–265, 2004. Author manuscript: https://math.univ-cotedazur.fr/~rifford/Papiers_en_ligne/RiffordSard.pdf.
- [33] Leon Simon. A strict maximum principle for area minimizing hypersurfaces. *Journal of Differential Geometry*, 26(2):327–335, 1987.

- [34] Leon Simon. Introduction to geometric measure theory. <https://math.stanford.edu/~lms/ntu-gmt-text.pdf>, 2018. NTU lecture notes, version updated 7 March 2018.
- [35] F. R. Tangherlini. Schwarzschild field in n dimensions and the dimensionality of space problem. *Il Nuovo Cimento*, 27:636–651, 1963.
- [36] Neshan Wickramasekera. A sharp strong maximum principle and a sharp unique continuation theorem for singular minimal hypersurfaces. *Calculus of Variations and Partial Differential Equations*, 51:799–812, 2014. Theorem and page locators refer to arXiv:1310.4031v1.